Refresh and Extract Operators Manual

BASE24®

Jan-2010, Release 6.0 Version 10 Publication Number: BA-AE000-04

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ACI Worldwide, Inc.

Preface

The ***BASE24 Refresh and Extract Operators Manual*** contains operator procedures for performing refreshes and extracts using the BASE24 Refresh and Super Extract processes. It also provides an overview of processing for these processes.

Audience

This manual is intended for those operators responsible for performing file refreshes and extracts.

Prerequisites

The screens used to control BASE24 refresh and extract processing are part of the BASE24 CRT access system. This manual assumes the reader is familiar with the BASE24 CRT access system.

Additional Documentation

The BASE24 documentation set is arranged so that each manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following publications:

- The ***BASE24 Logical Network Configuration File Manual*** discusses in detail the assigns and params used in configuring the Refresh and Super Extract processes.

- The ***BASE24 CRT Access Manual*** provides an introduction to the CRT access system, including instructions for logging on to the system and for accessing screens in the system.
- The ***BASE24 Base Files Maintenance Manual*** documents BASE24 file maintenance screen fields, including those for the Extract Configuration File (ECF) and the Token File (TKN). These files are used in configuring extracts.
- The ***BASE24 Text Command Reference Manual*** documents in detail the text commands that can be sent to the Refresh or Super Extract processes.
- The ***BASE24 Tokens Manual*** describes how to use the TKN to configure the tokens extracted from transaction log files by the Super Extract process.
- The ***BASE24 Event Message Reference Manual*** describes how to access online event message documentation and documents the file maintenance screen fields for the Log Message Configuration File (LMCF). The online documentation includes causes and remedies for event messages generated by Refresh and Super Extract processes. The LMCF can be used to modify the event messages generated by the Refresh or Super Extract process.
- The ***BASE24 Files and Record Structures Reference Manual*** provides general information about the structure of BASE24 files and records, details the location of the compiled Data Definition Language (DDL) dictionary for each file or record structure, and provides instructions on how to use the DDLDOC program to access file and record structure information.
- The ***BASE24 Remote Banking Transaction Processing Manual*** describes the mandatory and optional Internal Transaction Data (ITD) fields that determine the variable-length records in the ITS Transaction Log File (ITLF).
- The ***BASE24-pos Stored Value Support Manual*** describes the BASE24-pos Stored Value add-on product in detail.
- The ***BASE24-atm Multiple Currency Support Manual*** and the ***BASE24-pos Multiple Currency Support Manual*** describe multiple currency impacts for refreshing the Positive Balance File (PBF) and Cardholder Authorization File (CAF).
- The ***NET24-XPNET Network Command Reference Manual*** describes the procedures involved in sending DELIVER PROCESS commands and setting the STARTUP attribute.
- The ***BASE24 ITS Kernel Files Maintenance Manual*** describes the Process File (PRO) screens, the Server Class Type File (SCT) screen, and the Server Class/Process Relationship File (SPRF) screen.
- The ***ACI MAKE Release 5 User Guide*** contains detailed information on the ACI MAKE utility.

This manual also references the following HP NonStop publications:

- The ***ENSCRIBE Programmer's Guide*** and the ***File Utility Program (FUP) Reference Manual*** provide detailed information on Enscribe files, partitioned files, and FUP commands.
- The ***NonStop SQL/MP Reference Manual*** provides detailed information on SQL tables, table partitions, and SQLCI commands.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

"Conventions Used in this Manual" follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, "Introduction," provides an introduction to the BASE24 Refresh and BASE24 Super Extract processes, identifies the Enscribe files and SQL tables that can be refreshed or extracted, and describes refresh and extract processing options.

Section 2, "Configuration and Control," provides information about the configuration and control of refreshes and extracts.

Section 3, "Refresh Operator Procedures," provides an overview of the network control text commands that can be sent to the Refresh process, and provides procedures for using the Refresh screens.

Section 4, "Extract Operator Procedures," provides an overview of the network control text commands that can be sent to the Super Extract process, and provides procedures for using the Extract screens.

Appendix A, "Refresh Disk and Tape Specifications," provides specifications for refresh disks and tapes. It also documents the refresh file headers and trailers.

Appendix B, "Extract Disk and Tape Specifications," provides specifications for extract disks and tapes. It also documents the extract file headers and trailers.

Appendix C, "Initialization and Processing," describes the processing performed by the Refresh and Super Extract processes.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, BA-AE000-04 for the ***BASE24 Refresh and Extract Operators Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Jan-2010 for January, 2010) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in this Manual

This section explains how syntax notation, field descriptions, blank characters, and unlabeled fields are documented in this manual.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that must be entered exactly as shown.
lowercase italics	Identify variables that you must provide.
Brackets [ ]	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar (|).
Vertical bar |	Separates alternatives in a list of items.
Ellipsis ...	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.

Notation	Meaning
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. Additional spaces can be inserted between any adjacent words or symbols, but cannot be inserted within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Field Descriptions

Each field appearing on an operations screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from Refresh (REFR) screen 1.

USER — The name that uniquely identifies you to the BASE24 CRT access system. This name must match your alias in the Security File (SEC) or access to the refresh screens is denied.

Normally, your supervisor gives you your user name. You must have security clearance under an identical user name for all logical networks you want to access, if you want to access a different logical network without logging on again.

Field Length:	1–16 alphanumeric characters
Required Field:	Yes, if network control security is enabled for your installation (i.e., the ENABLE-SECURITY param for the NCPI Server process is set to a value of ON).
Default Value:	None

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term *alphanumeric* includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term *hexadecimal* includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is *System Protected*.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the **F8** key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

Blank Characters

Field descriptions for internal and external messages, tokens, files maintenance screens, and reports often include lists of valid values. When the value contains a blank character, this manual uses the symbol ƀ to indicate the blank character.

Unlabeled Fields on Screens

Angle brackets (< >) indicate an unlabeled field on a screen. An unlabeled field is a field that is present on a screen but is not preceded by an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen, the unlabeled field appears according to its place on the screen and is identified by the same label.

Unlabeled fields are not included in the index by field name; they appear by subject in the main index.

1: Introduction

File refresh is a method by which BASE24 authorization files and tables can be refreshed, that is, updated, with transaction information processed by host computers. The following Refresh processes are documented in this manual:

- The Base National Negative Refresh process is an Enscribe process that is used to refresh the Base national negative files used by the BASE24-pos product. The Base national negative files are refreshed with information provided by Visa.
- The INAS National Negative Refresh process is an Enscribe process that is used to refresh the INAS national negative files used by the BASE24-pos product. The INAS national negative files are refreshed with information provided by MasterCard.
- The Enscribe Refresh process is used to refresh Enscribe files other than national negative files.
- The SQL Refresh process is used to refresh SQL tables.

Note: Some interchanges use interchange-specific Refresh processes to refresh Interchange Prefix Files (IPFs). These Refresh processes are documented in interchange-specific documentation.

File extract is a method by which records can be extracted, that is, copied, from BASE24 files and placed on tape or disk for processing by the host. File extracts are performed by the Super Extract process.

The Refresh and Super Extract processes keep information in the BASE24 database current with host processing activity. This section provides an overview of the Refresh and Super Extract processes. For each process, this section includes the following topics:

- An overview of the process
- The files or tables that can be refreshed or extracted
- Processing options

Refresh Overview

Using disk files or computer tapes generated by a host computer, Refresh processes update specific BASE24 files and tables. (A host computer is any computer except the HP NonStop processor on which the BASE24 product is installed.) The refresh tape or disk input file can contain records for one or more BASE24 files or tables.

Files and tables are normally refreshed at the beginning of an institution's business day, after the accounts on the host system have been impacted with the previous day's business. Transaction processing can take place while refreshing any of the BASE24 files and tables.

Refresh processes are controlled by Refresh screens. Operators can also control Refresh processes with text commands entered from a network control facility.

Refer to section 3 for information on how to access the Refresh screens. Section 3 also provides an overview of the text commands that can be used to control Refresh processes.

ACI Proactive Risk Manager Files

BASE24 Refresh processes can be used to refresh the Scoring Engine Master File (SEMF), which is used by the ACI Proactive Risk Manager product.

BASE24 Shared Files

BASE24 Refresh processes can be used to refresh the following files shared by multiple BASE24 products:

- Cardholder Authorization File (CAF)
- Negative Card File (NEG)
- Positive Balance File (PBF)
- Stop Payment File (SPF)
- Switch Dispute File (SDF)

BASE24-atm self-service banking Files

BASE24 Refresh processes can be used to refresh the following files used by the BASE24-atm self-service banking (SSB) product:

- Corporate Check File (CCF)
- Check Status File (CSF)

BASE24 Customer Service

BASE24 Refresh processes can be used to refresh the following files used by the BASE24 Customer Service product:

- Customer/Card Information File (CCIF)
- Customer/Card Memo File (CCMF)

BASE24-mail Files

BASE24 Refresh processes can be used to refresh the Mail Box File (MBF), which is used by the BASE24-mail product.

BASE24-pos CRT Authorization Files

BASE24 Refresh processes can be used to refresh the Card History File (CHF), which is used by the BASE24-pos CRT Authorization product.

BASE24-pos National Negative Files

BASE24 Refresh processes can be used to refresh the following files used by the BASE24-pos product:

- Base National Negative File (BNEG)
- Base National Range File (BNR)
- INAS National Negative File (INEG)
- INAS National Range File (INR)

BASE24 Remote Banking Files and Tables

BASE24 Refresh processes can be used to refresh the following files and tables used by the BASE24 Remote Banking products:

- Customer Table (CSTT)
- Customer/Account Relation Table (CACT)
- Customer/Personal ID Relation Table (CPIT)
- Personal Information Table (PIT)
- Transaction History File (THF)

BASE24 Remote Banking products include the BASE24-telebanking and BASE24-billpay products.

BASE24-teller Files

BASE24 Refresh processes can be used to refresh the following files used by the BASE24-teller product:

- No Book File (NBF)
- Warning/Hold/Float File (WHFF)

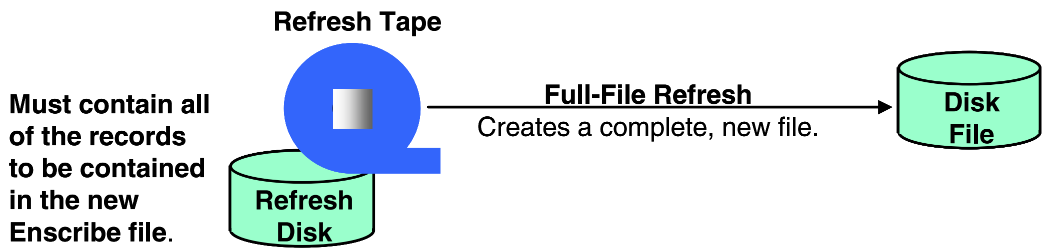
Refresh Methods

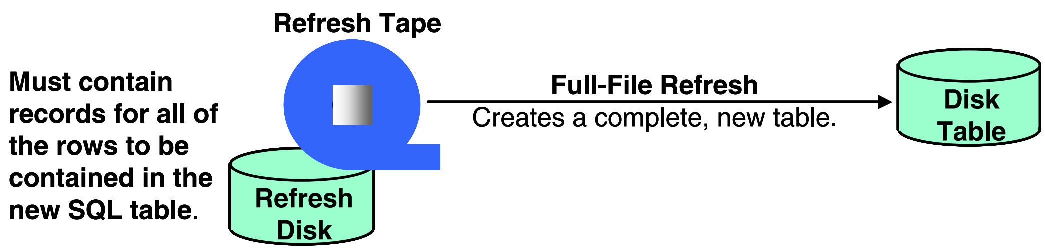
The two basic methods of refreshing BASE24 files are full-file refreshes and partial-file refreshes.

Full-File Refreshes

A full-file refresh creates an entirely new file or table from the refresh input file. The records in the host-generated refresh input file are used to create a new Enscribe file or SQL table, as shown below.

Note: Full-file refresh does not support input file records with a record type of D (Delete). If the Refresh process encounters an input file record with a record type of D during a full-file refresh, it generates an event message and stops running.





The Refresh process never accesses the existing file or table during a full-file refresh. Once a refresh has completed, the new file is available for system use and the old file remains as a backup file until the next full-file refresh.

Because the Refresh process builds the new file on the same volume as the old file, the volume must contain enough space for the current file and the new file.

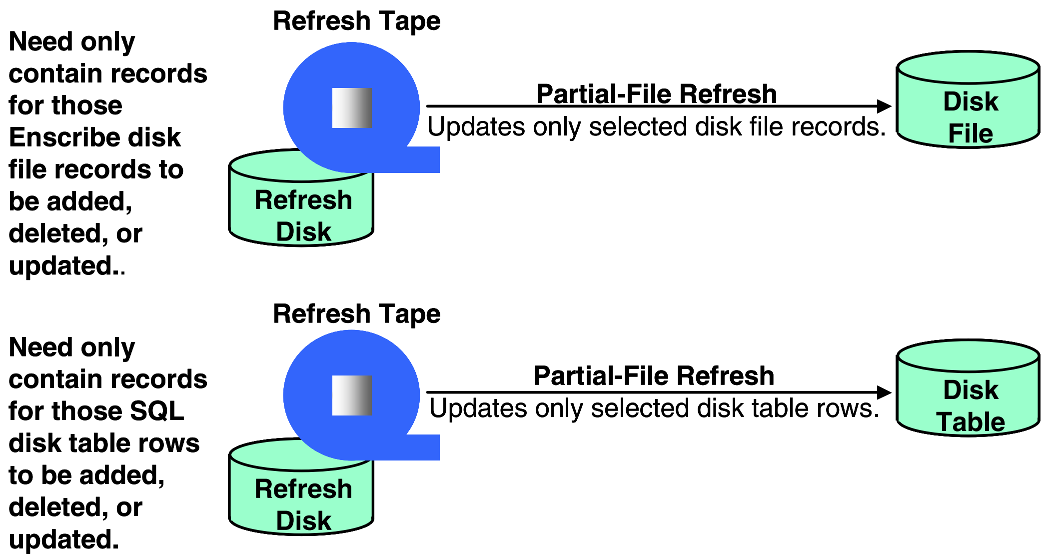
Following the refresh of most files, the existing file is renamed and the new file is given the name of the existing file. For some files and tables, however, the existing file or table cannot be renamed because it is audited by the HP NonStop Transaction Monitoring Facility (TMF). For files and tables audited by TMF, the Refresh process gives each file or table a unique name by changing the last character of the file or table name each time a full-file refresh occurs. This character, known as a version indicator, is stored in the File Control File (FCF). The Transaction History File (THF) currently uses a version indicator.

Caution: BASE24 Remote Banking applications require system downtime to perform the full-file refresh of an SQL table. This is because the name of an SQL table cannot be changed without recompiling the processes that use the table. A full-file refresh is recommended only at system installation for the following tables:

- Customer Table (CSTT)
- Customer/Account Relation Table (CACT)
- Customer/Personal ID Relation Table (CPIT)
- Personal Information Table (PIT)

Partial-File Refreshes

A partial-file refresh, illustrated below, updates records in an Enscribe disk file or rows in an SQL disk table. Unlike a full-file refresh, a partial-file refresh modifies the existing file or table. Each record on the refresh input file indicates whether the record must be added to, deleted from, or used to update a record in the existing Enscribe file or a row in the existing SQL table. The refresh tape can contain records for multiple Enscribe files or multiple SQL tables. However, records for Enscribe files and SQL tables cannot be combined on the same refresh tape.



Record Formats

The format of the refresh input file records does not match, field for field, the format of the application file records or table rows being refreshed. Some fields in the application file record or table row cannot be refreshed. Therefore, for most refreshable files and tables, the refresh input file records contain only selected fields or rows that can be refreshed.

When records in a file or rows in a table are updated during a partial-file refresh, the Refresh process replaces values for the fields in the input file record format only. Fields not present in the input file record format remain unchanged, unless they are initialized by the Refresh process (e.g., the Refresh process resets the CUR-FLOAT field in the PBF to zero).

When records in a file or rows in a table are added during a partial- or full-file refresh, the Refresh process creates the new records or rows using only those fields present in the input file record format. Fields not present in the input file record format are set to default values.

Refresh record structures are documented online by comments in the Data Definition Language (DDL) source schema files that define the record's structure. The DDLs are located on the same subvolume as the corresponding file record structure or table row structure. The table below identifies the standard subvolume and DDL definition name of the refresh format for each file or table that can be refreshed, where *xx* is the number of the current release. Refresh record structures can be retrieved from the data dictionaries on the identified subvolume and printed using DDLDOC. For more information on using DDLDOC, and on ACI-supplied input files to print refresh record formats, refer to the ***BASE24 Files and Record Structures Reference Manual***. The printed output for refresh record formats is also provided on the BASE24 Product Documentation CD-ROM.

File or Table Acronym	**Subvolume**	**Definition Name**
CACT	OC*xx*CUST	RFCACTDS
CAF	BA*xx*DDL	CAFXD
CCF	AT*xx*CDDL	CCFX
CCIF	OC*xx*CS	CCIFXDS
CCMF	OC*xx*CS	CCMFXDS
CHF	BP*xx*CRTA	CHX
CPIT	OC*xx*PSNL	RFCPITDS
CSF	AT*xx*CDDL	CSFX
CSTT	OC*xx*CUST	RFCSTTDS
MBF	MA*xx*DDL	MBFX*
NBF	TR*xx*DDL	NBFX-DISPLAY
NEG	BA*xx*DDL	NEGXD
PBF	BA*xx*DDL	PBFXD
PIT	OC*xx*PSNL	RFPITDS
SDF	SW*xx*SDF	SDFX
SEMF	PI*xx*DDL	SEMFXD
SPF	BA*xx*DDL	SPFX-DISPLAY
THF	OC*xx*HIST	THFXDS
WHFF	TR*xx*DDL	WHFX-DISPLAY

* The Refresh process sends BASE24-mail request messages to the Mail process, which creates and stores the MBF record.

Refresh Options

A refresh can be accomplished in a number of different ways, depending on the institution's needs and the files or tables the institution is using. The different refresh options are explained on the following pages. Some of these options apply to Enscribe files and SQL tables and others apply to Enscribe files only.

Refreshing by Group

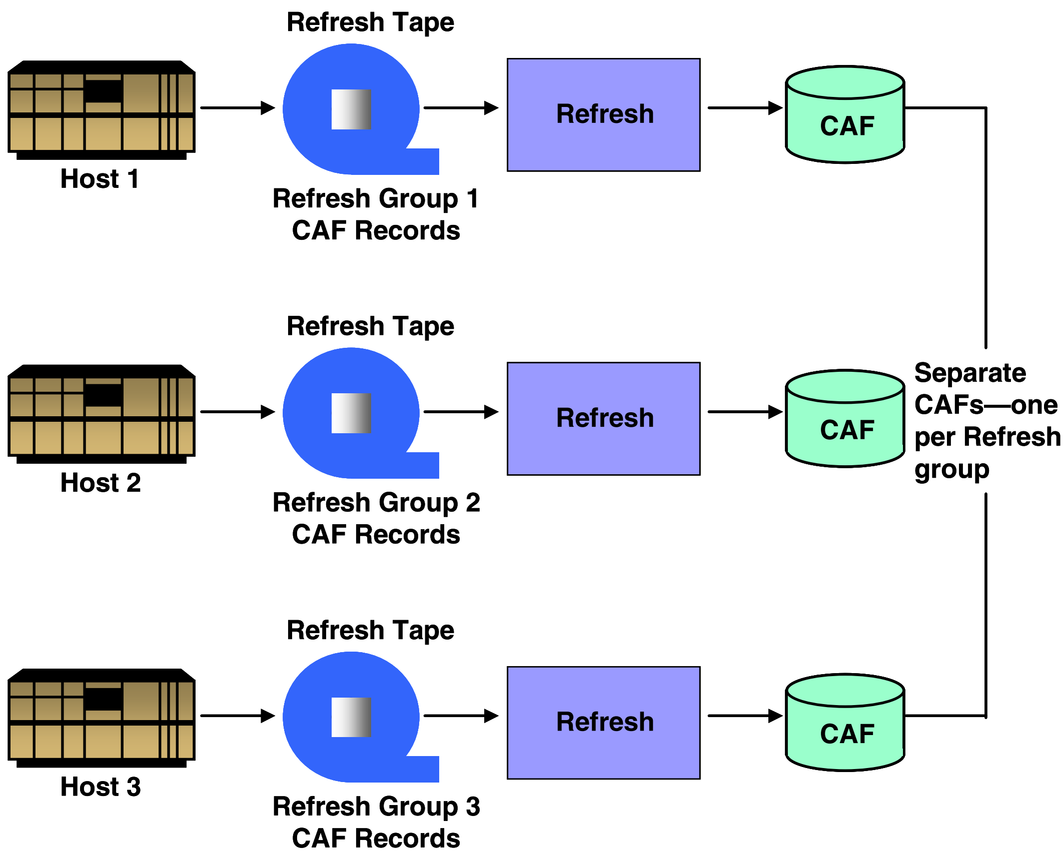
The Enscribe Refresh process can refresh a file for several institutions at once if those institutions use the same physical disk files. These institutions form a refresh group. If institutions use different physical disk files, they must belong to different refresh groups.

Refreshing by group applies only to Enscribe files because a logical network can have more than one Enscribe file, but can have only one SQL table. For example, a logical network can have more than one CAF, which is an Enscribe file, but can have only one CACT, which is an SQL table.

BASE24 requires that a refresh group be specified in the refresh file header. Institutions are assigned to their respective refresh groups through a field in the Institution Definition File (IDF).

This one-to-one correspondence between refresh groups and groups of disk files allows institutions that use different host processors to be divided and grouped. For example, if a number of institutions in a logical network use the CAF, and those institutions are processed by three different host processing centers, three different refresh groups could be set up—each refresh group corresponding to one host processing center. Likewise, three physical CAF disk files must exist on the system—one corresponding to each refresh group. Each CAF would be used by only those institutions that are a part of the corresponding refresh group. In such a situation, CAF refreshes would be performed by refresh group, as shown on the following page, using separate refresh input files from each host processing center.

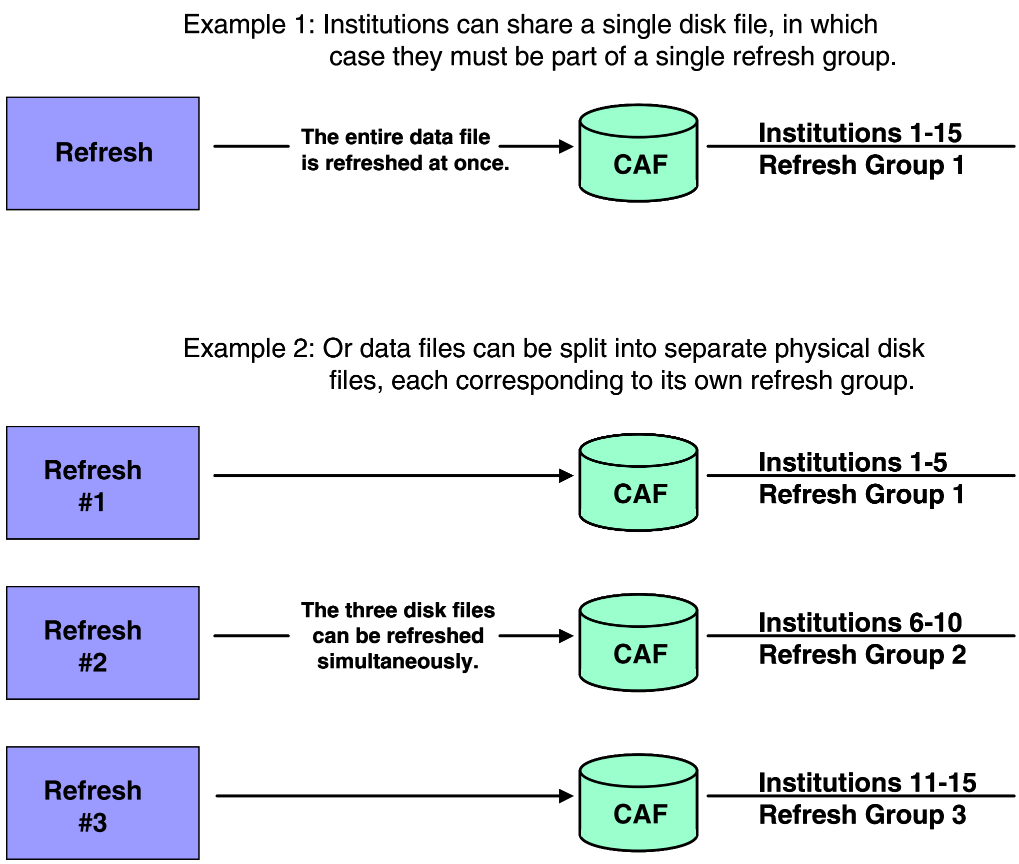
CAF Refresh


Refresh Tape
Host 1
Refresh
CAF
Refresh Group 1
CAF Records
Refresh Tape
Host 2
Refresh
CAF
Separate
CAFs—one
per Refresh
group
Refresh Group 2
CAF Records
Refresh Tape
Host 3
Refresh
CAF
Refresh Group 3
CAF Records

Refreshing by group also can be used to make the refresh more efficient. Instead of refreshing one large disk file, several refreshes can be performed simultaneously on several smaller files. This results in faster refresh processing. In the first example below, 15 institutions use the same disk file. In the second example, 15 institutions are divided into three refresh groups, allowing three separate refreshes to be run.

Note: Refresh groups also can be used to perform transaction log file extracts.

Refresh Groups



Bypassing Records with Parity Errors

Occasionally, tape parity errors occur during refreshes. Parity errors denote inconsistent data in the records. Unless otherwise specified, the first record with a parity error causes the Refresh process to abend.

Enscribe and SQL Refresh processes handle parity errors in the same manner.

The REF-BYPASS-PARITY-ERRORS param in the Logical Network Configuration File (LCONF) specifies the maximum number of parity errors that the Refresh process can encounter without abending. An operator can override the value in this param when starting the Refresh process through the Refresh screens or a network control facility. If the REF-BYPASS-PARITY-ERRORS param does not exist or is set to a value of 0, and a corresponding value was not entered on the Refresh screen or network control facility at startup, the Refresh process abends when it encounters the first parity error.

When the REF-BYPASS-PARITY-ERRORS param is set to a value other than 0, the Refresh process bypasses records having parity errors; therefore, the file or table is not updated with those records. Either a separate partial-file refresh tape must be created for the records that were bypassed, or a BASE24 operator must manually update the bypassed records or rows.

The Refresh process logs event messages to the network logging facility and to the refresh report as parity errors occur. The messages indicate the last record in the refresh input file successfully applied before the parity error and the first record applied after the parity error. It also generates warning messages at the end of the refresh, indicating that the record counts are incorrect. By issuing a STATUS command from the network control facility to a running Refresh process, an operator can determine if parity errors have been encountered, the number of parity errors that have been encountered, and the maximum number of parity errors that can be encountered.

Refreshing Partitioned Files or Tables

A partitioned file or table is logically considered to be one file or table, although an Enscribe file can be physically spread across as many as 16 different disk volumes and an SQL table can be spread across as many as 99 different disk volumes. A partitioned Enscribe file is created using the HP NonStop Kernel File Utility Program (FUP). A partitioned SQL table is created using the HP NonStop Kernel SQL conversational interface (SQLCI).

The Refresh process can refresh individual partitions of a single file or table. Refreshing an individual partition can be quicker than either a partial- or full-file refresh of the entire partitioned file or table.

To start a full-file refresh on a single partition, the number of the partition to be refreshed is specified in the refresh file header or in the REFRESH command. The REFRESH command setting overrides the value specified in the refresh file header.

Because the Refresh process builds new partitions on the same disk volume as the old partitions, the volume must contain enough space for the current partition and the new partition.

Partitions of a file or table that is audited by TMF cannot be refreshed individually because they cannot be renamed and the Refresh process maintains only one version indicator per file. This restriction applies to the THF.

Note: Because of the eight-character length limitation on file names and the way the Refresh process handles renaming of files during processing, partitions for different files of the same type should not be placed on the same subvolumes. For example, the primary or secondary partitions of one Cardholder Authorization File (CAF) should not be placed on the same subvolumes as the primary or secondary partitions of any other CAF. During refresh processing, this could lead to the corruption of the backup file (the renamed old CAF), or it could cause the Refresh process to abend because of not being able to properly distinguish and rename the appropriate files. This only affects multiple partitioned files of the same type. Partitions of different file types can share the same subvolumes without causing the problem noted above. For example, a CAF partition could be placed on the same subvolume as a Positive Balance File (PBF) partition.

For detailed information on partitioned files, how to use them, and how to create them, refer to the HP NonStop ***ENSCRIBE Programmer's Manual*** and the ***File Utility Program (FUP) Reference Manual*** concerning the FUP SET and FUP CREATE commands. For more information on performing a refresh on a single partition, refer to section 3.

For detailed information on partitioned tables, how to use them, and how to create them, refer to the HP ***NonStop SQL/MP Reference Manual*** concerning the SQLCI CREATE CATALOG, CREATE INDEX, and CREATE TABLE commands.

Performing Multiple Concurrent Refreshes

A single Refresh process can refresh only one file or table at a time. However, it is possible to perform multiple concurrent refreshes by defining multiple copies of the Refresh process, and then using each separate Refresh process to refresh a separate file or table.

Although it is possible for multiple copies of the Refresh process to perform a partial-file refresh of the same file or table concurrently, caution should be taken. If records in both refresh input files update a single database record or row, it is impossible to know which refresh record will be applied first. This can result in database corruption, because the record or row is updated incorrectly.

Transaction Impacting

Impacting refers to the ability of the Refresh process to update the newly refreshed records in a file with the transaction activity that has occurred since the last extract. Refresh input files are generated using host files; therefore, the refresh input file is only as up-to-date as the last extract. Impacting updates the refreshed file or table with the transactions that have occurred since the last extract. The Refresh process can impact records in the CAF, PBF, SPF, NBF, and WHFF. The Refresh process determines whether to impact the file based on a setting in the refresh file header. Impacting for each of these files is described below.

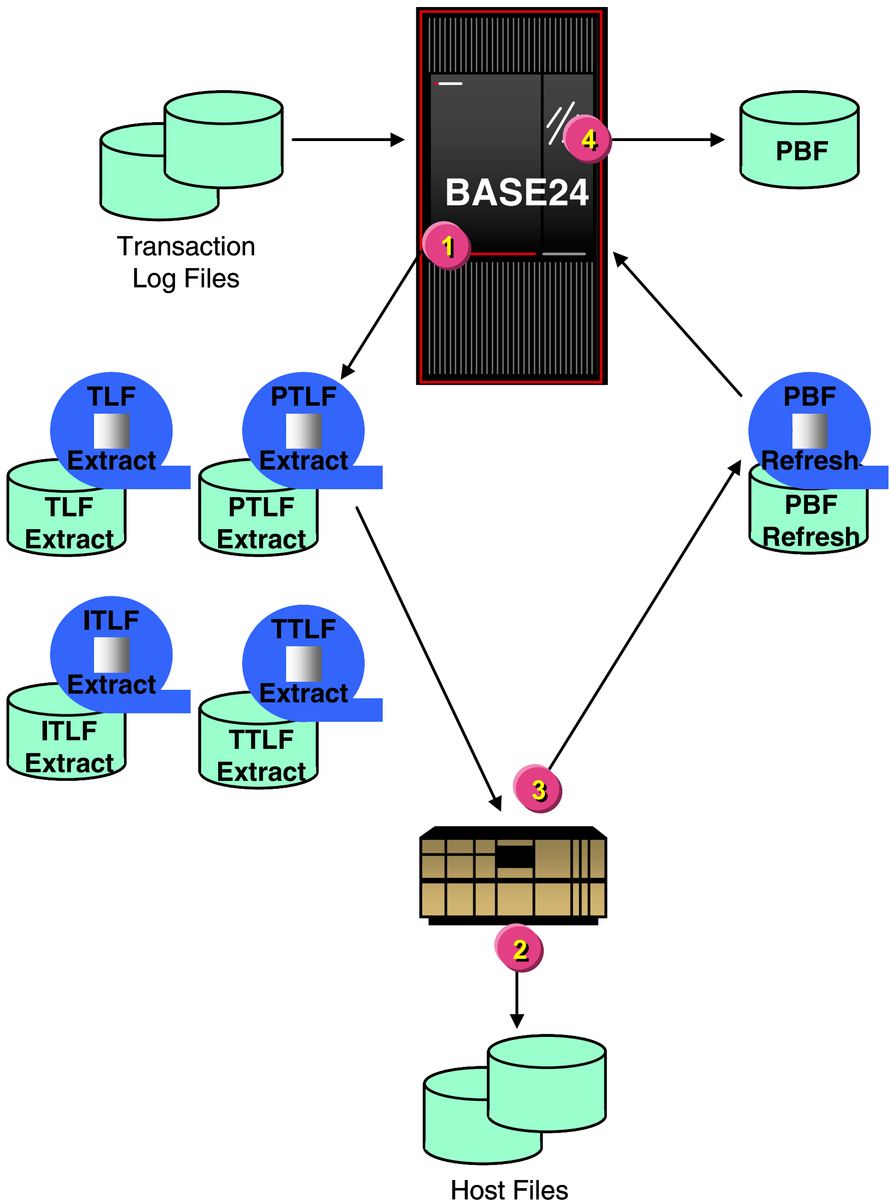
CAF Impacting

To update the CAF, the Refresh process impacts the newly refreshed CAF with records from the BASE24-atm Transaction Log File (TLF). TLFs are created daily; they contain information on each transaction processed by the BASE24-atm system. The Refresh process searches through the TLF for PIN Change transactions that have occurred since the last TLF extract. PIN Change transactions are identified by a transaction code of 81. When the Refresh process finds a PIN Change transaction in the TLF, the Refresh process uses the information in the record to update PIN offset information in the CAF.

PBF Impacting

To update the PBF, the Refresh process impacts the newly refreshed PBF with records from the appropriate transaction log files (i.e., BASE24-atm Transaction Log File (TLF), ITS Transaction Log File (ITLF), POS Transaction Log File (PTLF), and Teller Transaction Log File (TTLF)). Transaction log files are created daily; they contain information on each transaction processed by the BASE24 system. BASE24-telebanking and BASE24-billpay are the only products that currently use the ITLF. To impact the PBF, the Refresh process updates the account balance fields in the PBF with transactions that have occurred since the last extract. PBF impacting is illustrated below. The processing steps are explained following the illustration.

PBF Impacting



1. The transaction log files are extracted to tape or disk to update host files.
2. Information from the extracts is applied to the host files.
3. The host files are used to create the PBF refresh tape or disk file.
4. The PBF is impacted with transaction log file records that have been created since the last extract.

SPF Impacting

To update the SPF, the Refresh process impacts the newly refreshed SPF with records from the TTLF. TTLFs are created daily; they contain information on each transaction processed by the BASE24-teller system. The Refresh process searches through the TTLF for SPF Update transactions that have occurred since the last TTLF extract. SPF Update transactions are identified by a transaction code of 80 or 81. When the Refresh process finds an SPF Update transaction in the TTLF, the Refresh process uses the information in the record to update the SPF.

Note: Although the SPF is a shared file, only BASE24-teller transactions add, update, or delete records in the SPF. BASE24-atm SPF information is added, updated, or deleted from the files maintenance system. Because BASE24-atm transactions do not impact the records in the SPF (i.e., do not add, update, or delete SPF records), the TLF is not used for SPF impacting.

NBF Impacting

To update the NBF, the Refresh process impacts the newly refreshed NBF with records from the TTLF. The Refresh process impacts the NBF with all passbook transactions in the TTLF that have occurred since the last TTLF extract. A single TTLF record can be used to impact multiple NBF records, depending on the passbook accounts used in the transaction. The Refresh process updates the NBF records for each passbook account involved in the transaction.

WHFF Impacting

To update the WHFF, the Refresh process impacts the newly refreshed WHFF with records from the TTLF. The Refresh process searches through the TTLF for WHFF Update transactions that have occurred since the last TTLF extract. WHFF Update transactions are identified by a transaction code of 84, 85, 86, 87, 88, or 89. When the Refresh process finds a WHFF Update transaction in the TTLF, the Refresh process uses the information in the record to update the WHFF.

Determining Which Transaction Log Files to Use for Impacting

Impacting is not necessarily limited to the current day's transaction activity. Because extracts are normally done on a daily basis, the CAF, PBF, SPF, NBF, and WHFF are normally impacted with records from the current and next day's transaction log files. However, the Refresh process can impact the file with previous day's transaction log files.

The Refresh process identifies the transaction log files to use for impacting using the dates specified in the LST-EXTR-DAT fields in the refresh file header. There is one LST-EXTR-DAT field for BASE24-atm, one for BASE24-pos, one for BASE24-telebanking, and one for BASE24-teller. Since the LST-EXTR-DAT fields contain the date of the last transaction log file fully extracted, the Refresh process adds one to the date in the LST-EXTR-DAT field to determine the first transaction log file to use for impacting for each product. If the calculated date is not the current date, each time the Refresh process reaches the end of a transaction log file, the Refresh process increments the date of the file name, opens the transaction log file, and impacts the newly refreshed file records with the records from the transaction log file. This continues until the Refresh process finishes processing the transaction log files through the current date. For example, if the current date is 12/22/01 and the ATM.LST-EXTR-DAT field in the refresh file header is 011220, the Refresh process uses the TLFs named TL011221, TL011222, and TL011223 for impacting. The Refresh process impacts the PBF with transactions from log files for each product that accesses the PBF. The Refresh process impacts the SPF, NBF, and WHFF with transactions from the TTLF only, and the CAF with transactions from the TLF only.

Determining the Impact Start Date and Time

Normally, the Refresh process uses all records in the transaction log files for impacting. When institutions perform multiple extracts of the transaction log files during the course of one day, however, there may be records in the log file that should not be used for impacting. These records can be bypassed by specifying an impact start date and time in the refresh file header or on Refresh screen 1. There is a separate impact start date and time for each product. If no impact start date and time is set in the refresh file header or on Refresh screen 1 and impacting is specified, the Refresh process impacts the newly refreshed file (i.e., CAF, NBF, PBF, SPF, or WHFF) with all of the records in the selected transaction log files.

If a start date and time are specified, the Refresh process locates the first transaction log file record that has a timestamp greater than the specified date and start time. This timestamp is the time the record was logged to the transaction log file. This record and all subsequently logged records within the log file are used to impact the file. If more than one day's log files are being used for impacting, all transactions from subsequent day's log files are used for impacting.

Note: When performing a refresh with impacting using a previous release refresh format, the Refresh process locates the first transaction log file record that has a timestamp greater than or equal to the specified date and start time. This record and all subsequent records within the log file are used to impact the file.

When the Super Extract process extracts a log file, the Super Extract process places the timestamp of the last record extracted in the first 20 bytes of the USER-FLD2 field of the file trailer. The date is in the format YYYYMMDDhhmmssmmmmmm (year, month, day, hour, minute, second, millisecond, and microsecond). This information can be echoed in the refresh file header as the impact start date and time.

Naming the Cardholder Authorization File

The name assigned to the Cardholder Authorization File (CAF) on Institution Definition File (IDF) screen 1 depends on whether the Card segment is used and whether alternate key files are used.

- If alternate key files are not present, the standard eight-character limit applies to the name assigned to the CAF.
- If alternate key files are present, the name assigned to the CAF is limited to six characters. This limitation applies when the Card segment is present because it uses alternate key files.

Multiple Currency Support

The Enscribe Refresh process provides multiple currency support for the Cardholder Authorization File (CAF) and Positive Balance File (PBF) when the Multiple Currency add-on product is used with the BASE24-atm or BASE24-pos products. For more information on multiple currency CAF and PBF refresh impacts, refer to the ***BASE24-atm Multiple Currency Support Manual*** and the ***BASE24-pos Multiple Currency Support Manual***.

Previous Release Support

The Enscribe Refresh process supports previous release (i.e., BASE24-atm release 5.*x*, BASE24-mail release 5.*x*, BASE24-pos release 5.*x*, BASE24-telebanking release 1.*x*, and BASE24-teller release 5.*x*) file formats for all files that can be refreshed.

Refresh Early Notification

The Refresh process receives the PBF refresh input tape or disk from the Host and loads the new data into the HP NonStop files. Refresh early notification processing allows the Refresh process to notify BASE24-atm, BASE24-pos, BASE24-teller, and BASE24-telebanking Authorization processes immediately after data is loaded in the new PBF without waiting for impacting to be finished. The Refresh process begins impacting after notification to Authorization processes is complete. Authorization processes authorize against the new PBF file after this notification message is received.

Refresh early notification processing is used where the Host program or batch system suffers from load fluctuations (typically at month-end) and the Refresh input file arrives to the HP NonStop system late. In this instance, the BASE24 Refresh process loads the new PBF file and the impacting phase starts. If the refresh has started really late, the Authorization process logs online transactions into the transaction log files at a rate much faster than Refresh can impact those transactions from transaction log files. If the impacting rate is significantly slower than the log rate by Authorization, the Refresh process cannot release the new PBF file to the Authorization processes. Therefore, Authorization processes use an outdated PBF file for the whole business day.

Refresh early notification processing allows the new PBF file to be available at the earliest opportunity and therefore reduces the possibility of the Authorization processes working with an outdated PBF file. Refresh Early Notification processing also improves the performance of refresh processing by reducing the number of records to be impacted under these circumstances. All transactions authorized after the new PBF delivery are marked as not requiring impact.

Processing

If a PBF is being refreshed and the REFR-EARLY-NOTIFICATION param in the LCONF is set to a value of Y, the Refresh process updates the IDF with a PBF early refresh indicator, sets itself to refuse to be suspended, and notifies the authorization processes that the new PBF is available. The Refresh process sets a global value to the current time when the first Authorization process acknowledges it has started processing using the new PBF and impacting the PBF with the new transactions being processed. When PBF Refresh early notification is used, Authorization processes do not impact the PBF for transaction log file records with timestamps that exceed the time when the first Authorization process acknowledges it has started processing using the new PBF. A message is logged that indicates the impact stop time placed in the global memory used by the Refresh process.

Broadcast Notification

Broadcast notification enables BASE24 users to effectively minimize Refresh process and Settlement Initiator process notify lists in the LCONF, replacing multiple params with a single service name. Without broadcast notification, BASE24 users must configure a separate param for each process to be notified.

Services can be configured in the following LCONF params used by the Enscribe Refresh process: POS-PREF-NOTIFY, REF-NOTIFY, REF-SWI-NOTIFY, REF-TLR-NOTIFY, and SEMF-NOTIFY. When broadcast notification is not in use, these params contain an individual process to be notified instead of a service.

If broadcast notification is used with both the Refresh process and the Settlement Initiator process, the same service name can be used in the refresh and settlement notification params. However, if broadcast notification is used with the Refresh process or the Settlement Initiator process individually, one service must be defined for the refresh notification params and a different service must be defined for the settlement notification params.

For broadcast notification, the BROADCAST-NOTIFY param in the LCONF must be set to a value of 2 and all processes requiring a refresh notify message must have a service configured that matches the entry in the appropriate refresh notification param. A second LCONF param associated with broadcast notification, BROADCAST-DELAY, defaults to 2, if not present or invalid.

If the broadcast notification is enabled, the Refresh process continues cycling until the number of seconds specified in the BROADCAST-DELAY param have elapsed. Then, the Refresh process completes processing and stops, regardless if all the authorization processes have acknowledged. If all the authorization processes have responded in a timely manner, the Refresh process does not start cycling until all PAUSEACK messages have been processed. If the PAUSEACK messages are received after the Refresh process has stopped, an event message is issued indicating the BROADCAST-DELAY time should be increased to allow the authorization processes sufficient time to respond.

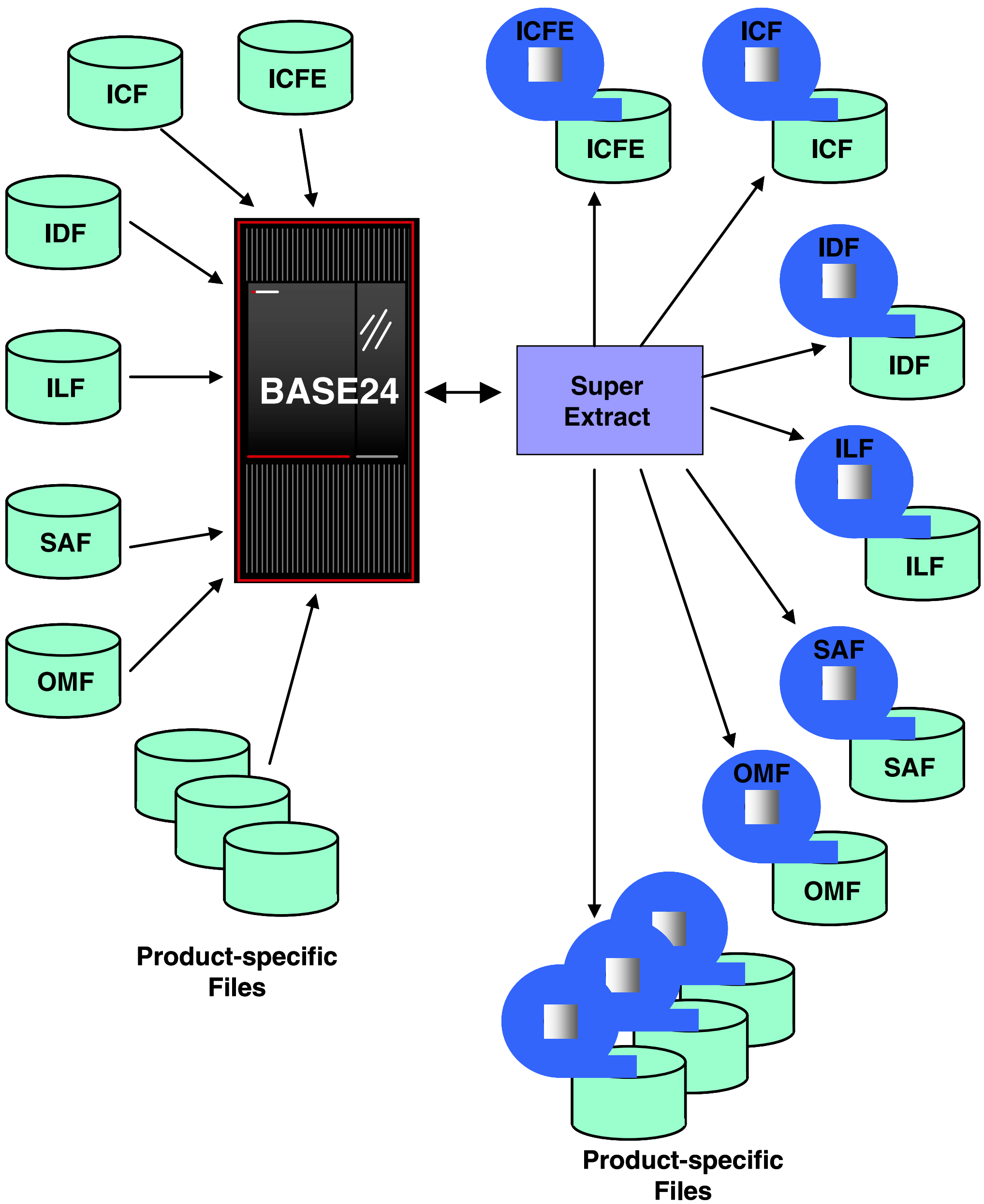
When the broadcast notification is enabled, the delay functionality is used for acknowledgement processing, regardless if one notify param is configured to use service-based routing and another notify param is configured to use individual process routing.

If the BROADCAST-NOTIFY param is missing or contains an invalid value, the Refresh process treats information defined in the POS-PREF-NOTIFY, REF-NOTIFY, REF-SWI-NOTIFY, REF-TLR-NOTIFY, and SEMF-NOTIFY params as individual process names instead of service names.

For more information about service-based routing, including broadcast messages, see the ***NET24-XPNET Network Control Operations Guide.***

Super Extract Overview

The Super Extract process allows files on the HP NonStop processor to be placed on tapes or on disk so that hosts can use that information for processing. The process of placing files on tape or disk, as shown in the illustration below, is known as extraction. One or more files can be extracted to a single tape or disk file.



Extracting can be performed anytime during normal online transaction processing, since the Super Extract process runs continuously. The user can set a time for the extract to run, ensure that a tape is in place on the drive or that disk space is available, and let the extract complete automatically. The user can also start and control extracts using the Extract screens or network control text commands. In both situations, records must be set up in the Extract Configuration File (ECF) in order for the extract to take place. For more information on how to start an extract using the Extract screens, refer to section 4. Section 4 also provides an overview of the network control text commands used to start and control the Super Extract process. For more detailed information about network control text commands, refer to the ***BASE24 Text Command Reference Manual***.

Extracts can be performed both on files shared among the current BASE24 release compatible products and on product-specific files. The shared files are shown in the illustration on the previous page. The following table shows the BASE24 files that can be extracted.

BASE24 Product	File Name
Shared	Enhanced Interchange Configuration File (ICFE)
	Institution Definition File (IDF)
	Interchange Configuration File (ICF)
	Interchange Log File (ILF)
	Online Maintenance File (OMF)
	Store-and-Forward File (SAF)
BASE24-atm	Hardware Status File (HSF)
	Transaction Log File (TLF)
BASE24-from host maintenance	Update Log File (ULF)
BASE24-mail	Host Mail Box File (HMBF)
	Mail Box File (MBF)

BASE24 Product	File Name
BASE24-pos	POS Retailer Definition File (PRDF)
	POS Transaction Log File (PTLF)
	POS Stored Value History File (SVHF)
BASE24-telebanking	ITS Transaction Log File (ITLF)
BASE24-teller	Teller Transaction File (TTF)
	Teller Transaction Log File (TTLF)

Automatic and Manual Extracts

File extract sessions can be performed either automatically or manually. Automatic file extracts are performed at set times controlled by the ECF. Manual file extracts are performed using the BASE24 Extract screens or a network control facility.

Both automatic and manual extracts require the use of ECF records. Each ECF record defines the general characteristics of a particular extract session, such as whether the extract is to tape or to disk, the time the extract is to start, the type of tape labels being used, the character set being used, and the BASE24 files that are included in the extract session. This provides all of the information required to perform the automatic extract. The ECF parameters are also used for manual extracts. However, manual extracts can override some file settings in the ECF record. For more information on configuring extracts using the ECF, refer to section 2.

Using Extract Position Markers

File reextracts can be initiated only after an extract has been completed. The processing performed for a reextract depends on the type of file being reextracted. For most files, a reextract is performed in the same manner as an extract—the Super Extract process begins extracting records at the beginning of the file and continues to extract each record until the end of the file is reached. For transaction log files, however, reextract works differently from extract.

The files for which reextract produces different results from extract are as follows:

- Transaction Log File (TLF)
- POS Transaction Log File (PTLF)
- Teller Transaction Log File (TTLF)
- ITS Transaction Log File (ITLF)

Note: The extract and reextract functionality described in the following paragraphs is performed only when the value ALL is specified in the appropriate GROUP NAME field on ECF screen 5, 7, 9, or 23. Refer to the topic “Extracting by Group” for more information on what happens when any other group is named in a GROUP NAME field.

In the first record (i.e., record 0) of each of the transaction log files are 100 sets of fields used to store extract positions. Each set of fields includes a relative byte address field and a block record count field. Before an extract has been performed, the relative byte address fields are set to -1, and the block record count fields are set to 0.

Extract Position Markers and Extracts

When the first extract is performed, the Super Extract process begins at the second record (record 1) and continues to the end of file. At the end of the extract, the Super Extract process sets the first relative byte address field to the relative byte address of the beginning of the last block that it processed in the transaction log file. The Super Extract process sets the block record count to the number of the record within the last block that it processed.

Record 0, which contains the extract position markers, is never extracted from the file. The extract always begins with record 1, which is the first transaction record logged to the file.

When a second extract is performed, the Super Extract process uses the information in the first set of extract position fields to position in the log file at the relative byte address of the last block processed, and within that block at the last record processed. The Super Extract process then begins extracting at the next record, and continues to the end of file. When the extract is complete, the Super Extract process sets the second set of extract position fields with the relative byte address and block record number information from the extract.

After the first extract, subsequent extracts always begin where the previous extract finished plus one record. The extract continues to the current end of the file. After the extract completes, the Super Extract process updates the appropriate set of extract position fields. This allows the financial institution to perform multiple file extracts of transaction log files over the course of a single day without extracting all of the records in the file every time. Basically, for the transaction log files, each extract after the first begins extracting where the last extract left off.

If more than 100 extracts are performed in one day, the Super Extract process updates the last set (i.e., the one hundredth set) of extract position fields after each extract over 100.

Extract Position Markers and Reextracts

When the appropriate GROUP NAME field contains the value ALL, reextract allows the financial institution to repeat one or more previous extracts. When a reextract is requested, the Super Extract process reextracts the records identified by the values in the start-partial and end-partial fields in the reextract message, as described below:

- If the start-partial and end-partial fields are both set to 0 or blanks (i.e., they were not included in the message), the Super Extract process extracts all records that were extracted in any previous extract of the file.
- If a value is specified in the start-partial field only, or if the start-partial and end-partial fields both contain the same value, the Super Extract process extracts the records for the single specified extract session. For example, if the start-partial field contains the value 3 and the end-partial field contains blanks, the Super Extract process reextracts only those records that were included in the third partial extract.
- If values are specified in both fields and the value in the start-partial field is less than the value in the end-partial field, the Super Extract process reextracts the records for the extract sessions beginning with the session identified in the start-partial field through the session identified in the end-partial field. For example, if the start-partial field contains the value 2 and the end-partial field contains the value 4, the Super Extract process reextracts those records that were included in the second, third, and fourth partial extracts.
- If values are specified in both fields and the value in the start-partial field is greater than the value in the end-partial field, the Super Extract process logs an error message indicating that the start-partial field contained a value larger than the value in the end-partial field. No records are reextracted from the file.

For a reextract when the appropriate GROUP NAME field contains the value ALL, the Super Extract process reextracts only those records that were included in the original extract (or extracts, if the start-partial and end-partial values indicate several partial extracts). If new records have been added to the file, they are not included in the reextract.

Extracting by Group

The Super Extract process can extract records from the transaction log files by refresh group. This option is selected by entering a refresh group name in the GROUP NAME field on ECF screen 5, 7, 9, or 23.

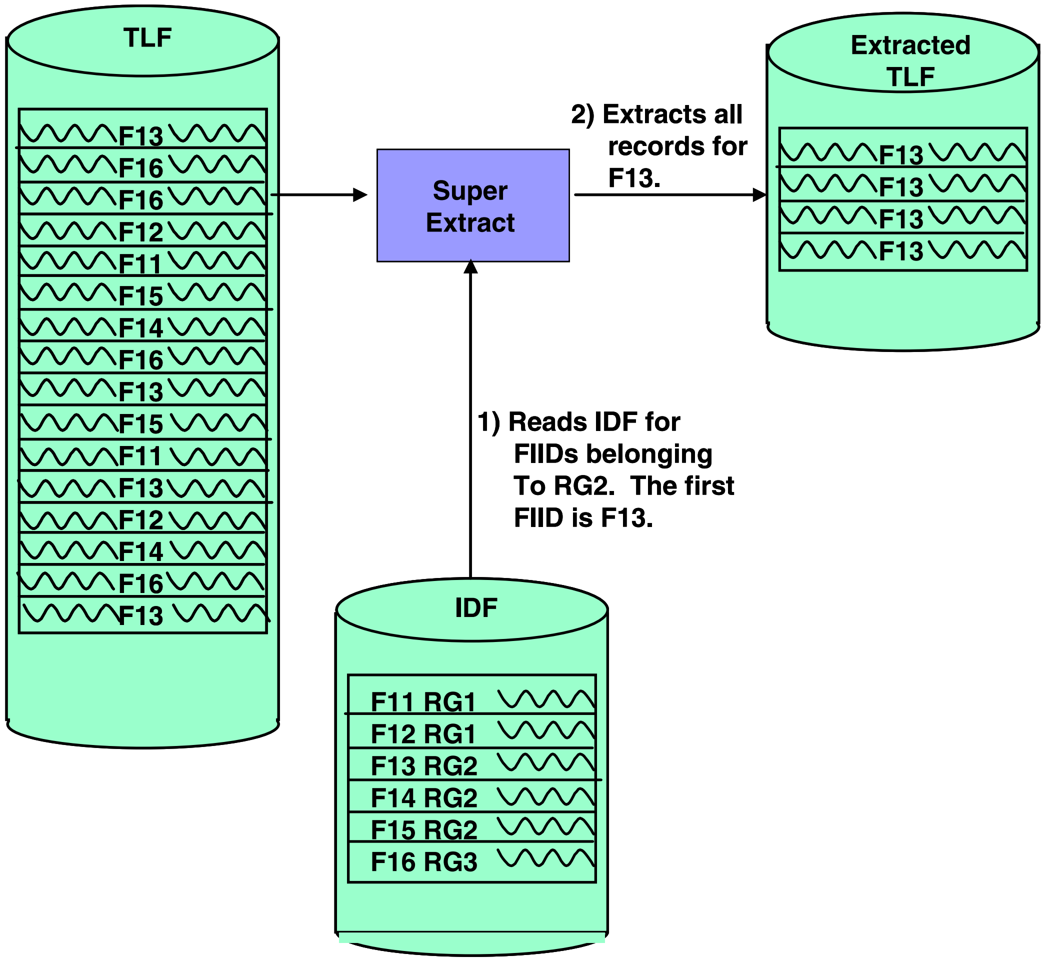
When the Super Extract process extracts records by refresh group, the Super Extract process extracts only those records associated with financial institutions in the refresh group. Not all of the records in the file are extracted, and as a result, the extract positions stored in record 0 are not updated.

When a transaction log file extract is requested for a specific refresh group, the Super Extract process reads the IDF for a financial institution that belongs to that group. The Super Extract process then uses the FIID from the IDF to read the transaction log file by alternate key, looking for log file records where the card-issuing (or account-owning) institution matches the IDF FIID. The Super Extract process extracts all of the log file records for that FIID. The Super Extract process then reads the IDF for the next financial institution in the refresh group, and extracts all log file records for that financial institution.

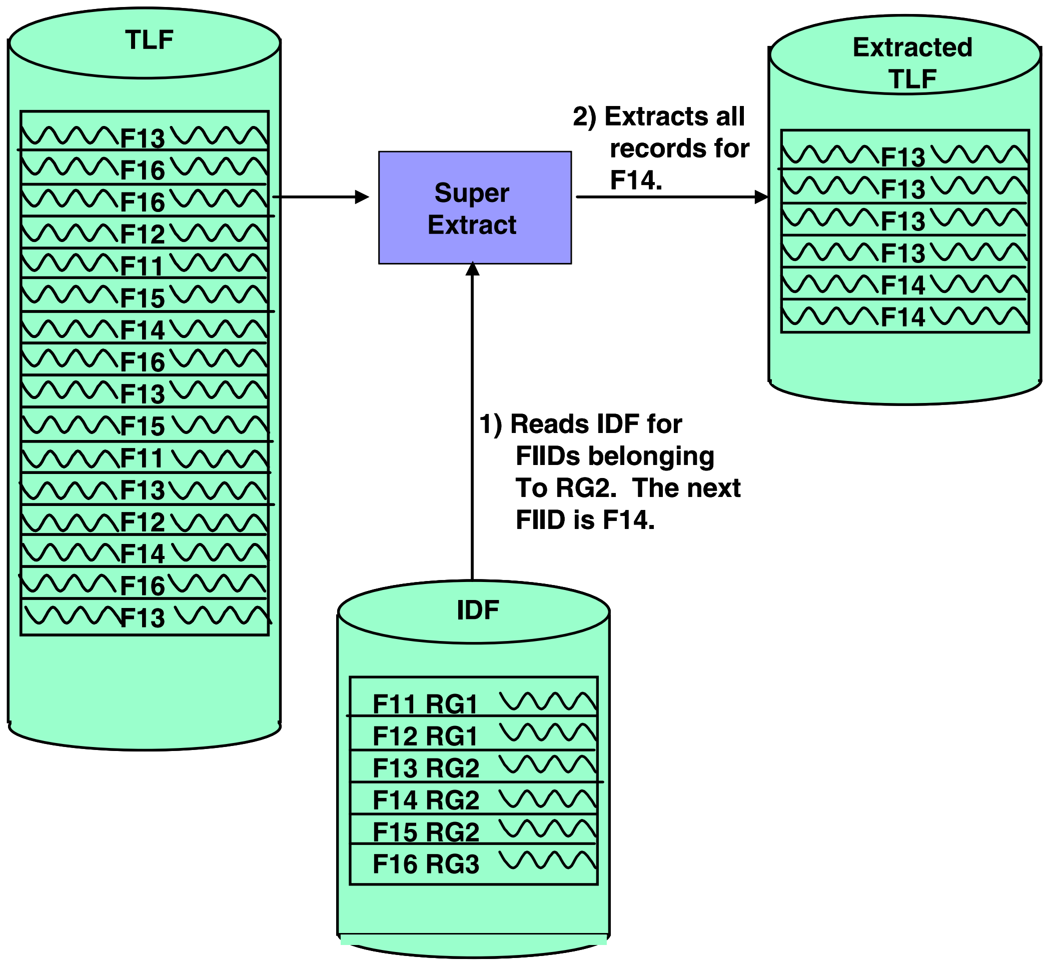
This process continues until the Super Extract process has extracted the records for all financial institutions in the refresh group. For example, assume that a transaction log file contains records for six different financial institutions. These financial institutions are divided into refresh groups as follows:

- Financial institutions 1 and 2 (FIIDs FI1 and FI2) are in refresh group 1 (RG1)
- Financial institutions 3 through 5 (FIIDs FI3, FI4, and FI5) are in refresh group 2 (RG2)
- Financial institution 6 (FIID FI6) is in refresh group 3 (RG3)

If an extract for refresh group 2 is requested, the Super Extract process locates the first IDF record with the value RG2 in the REFR-GRP field. Assuming that this record is for FI3, the Super Extract process extracts all records for FI3, as illustrated below.



After the Super Extract process extracts all of the records for FI3, the Super Extract process locates the next IDF record with the value RG2 in the REFR-GRP field. Assuming this record is for FI4, the Super Extract process extracts all records for FI4, placing them after the records for FI3 in the extract file. The graphic below illustrates this processing.



The Super Extract process continues reading an IDF record for the specified refresh group and extracting the log file records for the identified financial institution until all of the log file records for all institutions in the refresh group have been extracted.

If a subsequent extract or reextract is requested when extracting by group (i.e., when the appropriate GROUP NAME field contains a value other than ALL), the same processing is performed, and the same records are extracted. In addition, the Super Extract process extracts any new records for financial institutions in the refresh group logged to the file since the first extract was performed.

Determining the Date of the File to Extract or Reextract

Many of the BASE24 files that can be extracted are log files. These files record activity or events that occurred over the course of a particular business day. Each day, the processes that write to these log files use a different file. The file name includes the date of the information stored in the files. For example, the BASE24-atm Transaction Log File for August 12, 2001 would be named TL010812. In this way, the activity occurring on a specific date is easily located.

Because multiple copies of these files exist, the Super Extract process must determine which day's log file to extract or reextract. The information the Super Extract process uses to determine the file date depends on whether an extract or reextract has been requested.

For extracts, the Super Extract process determines the date of the file to extract using the current system date and the DATE OFFSET fields in the ECF. There is a separate DATE OFFSET field for each dated file that can be extracted. Each DATE OFFSET field contains the time difference, in number of days, to add to or subtract from the current system date to obtain the date of the file to extract. For example, if the DATE OFFSET field contains a value of -2 and the current system date is August 12, the date of the file to extract is August 10 (two days prior to the current system date).

For reextracts of most files, the Super Extract process determines the date of the file to reextract using the LAST EXTRACT DATE field and the DATE OFFSET field in the ECF. In this case, the value in the DATE OFFSET field is added to or subtracted from the value in the LAST EXTRACT DATE field on ECF screen 1 to obtain the date of the file to reextract. For example, if the DATE OFFSET field contains the value -5 and the value in the LAST EXTRACT DATE field is August 10, the date of the file to reextract is August 5 (five days prior to the date of the last extract).

For reextracts of the ILF, MBF, or HMBF, the Super Extract process determines the date of the file to reextract in the same manner as for an extract, using the current system date and the DATE OFFSET field in the ECF.

Using the Extract Screens

The Extract screens are used to initiate manual extracts and reextracts only. They allow extracts to be performed at a different time on all or a subset of the files indicated in the ECF record used to control the extract. It is important to note, however, that more files cannot be included in the manual extract than are specified in the ECF record being used. For example, suppose that the ECF record has flags set so that the ILF, OMF, and SAF are the only files extracted. A manual extract using that ECF record could only extract those three files or any combination thereof. It could not extract any other files, regardless of how flags on the Extract screens are set.

On reextracts of transaction log files, the Extract screens allow the user to set the start partial and end partial extract numbers, as well as to override the volume IDs of the tapes used in the reextract.

Fixed-Format Extracts

The Super Extract process supports fixed-format extracts of the following transaction log files:

- Interchange Log File (ILF)
- BASE24-atm Transaction Log File (TLF)
- POS Transaction Log File (PTLF)
- Teller Transaction Log File (TTLF)

Fixed-format extracts of these files are specified by individual flags in the ECF record.

In the transaction log file, each record is composed of a log file record and, optionally, token data that was logged to the file. The log file record has a standard structure, based on the record type and subtype. The tokens that are logged depend on the tokens present in the internal message and on the tokens configured to be logged for the record type and subtype. Tokens are configured in the Token File (TKN).

The fixed-format option affects the way the Super Extract process handles both the basic log file record and the token data. Each extract record is made up of a basic log file record, plus any tokens associated with that record that are configured to be extracted. Fixed-format processing makes the basic log file record portion of each extract record the same length for all types of records, and it makes the token data portion of the extract record the same length for all records of the same type.

For example, TLF extracts can include four different types of records: financial transaction records (485 bytes), terminal balancing records (856 bytes), terminal cash adjustment records (150 bytes), and terminal settlement records (182 bytes). When the FORMAT field on ECF screen 5 is set to 00 (fixed), all basic log file records are extracted for the length of the longest record type—in this case, for 856 bytes, the length of terminal balancing records. Other record types are padded with blanks so that the total record length of basic log file record portion of the extract record is 856 bytes.

The token data portion of the extract records is handled similarly but separately based on the record type. All extract records of a given type will be created with the same amount of token data; however, the amount of token data included will vary by type of record. When the format is fixed, the Super Extract process reads the TKN to determine what tokens should be included in the extract for the record type and subtype. For each token configured to be extracted, the Super Extract process checks the transaction log file record for the token. Depending on whether the token is found in the log file record, and whether the token itself is fixed length or variable length, the Super Extract process performs as follows:

- If the token is present in the transaction log file record and the token is fixed length, the Super Extract process extracts the token from the transaction log file record.
- If the token is present in the transaction log file record and the token is variable length, the Super Extract process determines whether the token includes enough data to be the maximum length for the token. If the token includes enough data to be the maximum length for the token, the Super Extract process extracts the token. If the token data does not fill the token to its maximum length, the Super Extract process pads the unused positions with spaces (alphanumeric fields) and zeros (numeric fields) before extracting the token.
- If the token is not present in the transaction log file record, the Super Extract process creates an *empty* token for the maximum size of the token, and extracts the empty token. To create the empty token, the Super Extract process fills the token with spaces (alphanumeric fields) and zeros (numeric fields) before extracting the token.

In this way, the length of the token data is made uniform for each type of record extracted. Thus, using the fixed-format option does not necessarily mean that all extract records will be the same length. Although this would be the case if no tokens were extracted, if tokens were extracted, different types of extract records would differ in length—based on the tokens configured to be extracted for each type of record. For example, all financial transaction extract records would be the same length, and all terminal settlement extract records would be the same length; however, these two types of records would differ in length because of the different token data being extracted.

For more information on configuring the tokens to be extracted from transaction log file records, and for information on record types and subtypes, refer to the ***BASE24 Tokens Manual***.

ACI's DDLDOC program can be used to print log file DDL record structures to identify their lengths. The DDLDOC program is documented in the ***BASE24 Files and Record Structures Reference Manual***.

Variable-Format Extracts

The Super Extract process supports only a variable-format extract of the ITS Transaction Log File (ITLF), due to the variable length of Internal Transaction Data (ITD) logged to the ITLF. Because some ITD fields are mandatory and some are optional, the length of ITLF records can vary. The ***BASE24 Remote Banking Transaction Processing Manual*** describes each ITD field and its contents, a tag number that is used to arrange optional fields in an ITLF record, the type of ITD field for ITLF logging purposes (i.e., mandatory, optional, or not logged), and the data name for the ITD field when it appears in an ITLF record.

The Super Extract process adds a binary 16 (two-character) field containing the tag number to the beginning of each optional ITD field in an ITLF record. Host programs must use these tag numbers to determine which optional ITD fields an ITLF record contains. Optional ITD fields that contain data are arranged in ascending order by tag number. There are no eye-catcher characters or other special characters separating ITD fields.

Previous Release Support

The Super Extract process supports extracts in previous release (i.e., BASE24-atm release 5.*x*, BASE24-from host maintenance release 5.*x*, BASE24-mail release 5.*x*, BASE24-pos release 5.*x*, BASE24-telebanking release 1.*x*, and BASE24-teller release 5.*x*) file formats for all files that can be extracted. Note that in some cases, the file formats have not changed from the previous release.

Since the release number flags in the ECF record are at the product level, all files being extracted in a single extract session for a given product will be in the same format. For example, assume an ECF record specifies to extract the IDF, ILF, TLF, and TTLF. Since the IDF and ILF are both base files, the extracts of these two files will both be in the same release format (i.e., either both in the format for releases 6.0 and newer or both in the format for release 5.*x*). The TLF and TTLF are product-specific files, and each will be in the format specified by the product-specific flag.

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2: Configuration and Control

This section provides information about the configuration and control of refreshes and extracts. It identifies the files that contain configuration and control information and identifies the commands that can be issued to Refresh and Super Extract processes to control processing.

Refer to appendix A for refresh input disk and tape specifications and refer to appendix B for extract output disk and tape specifications.

Refresh Process Configuration and Control

Refresh processes are configured and controlled by information in several locations. The following Refresh processes use this information:

- The Base National Negative Refresh process is an Enscribe process that is used to refresh the Base national negative files used by the BASE24-pos product.
- The INAS National Negative Refresh process is an Enscribe process that is used to refresh the INAS national negative files used by the BASE24-pos product.
- The Enscribe Refresh process is used to refresh Enscribe files other than national negative files.
- The SQL Refresh process is used to refresh SQL tables.

The following table identifies which Refresh processes use the information kept in each location.

Location	Base	INAS	Enscribe	SQL
Refresh File Header (RFH)	Yes	Yes	Yes	Yes
Refresh Group Control File (RGCF)	No	No	Yes	Yes
Logical Network Configuration File (LCONF)	Yes	Yes	Yes	Yes
Log Message Configuration File (LMCF)	Yes	Yes	Yes	No
Institution Definition File (IDF)	No	No	Yes	No
Transaction History Configuration File (THCF)	No	No	Yes	No
Process Type File (PTF)	No	No	Yes	No
Process File (PRO)	No	No	Yes	No
Server Class Type File (SCT)	No	No	Yes	No

Location	Base	INAS	Enscribe	SQL
Server Class/Process Relationship File (SPRF)	No	No	Yes	No
File Control File (FCF)	No	No	Yes	No

The network control operator can also issue commands that control refresh processing. The following items address configuration and control issues related to the Refresh processes.

Refresh File Header

The Refresh File Header (RFH) identifies the file or table to be refreshed and the type of refresh to be performed. It also contains various processing parameters required by the Refresh processes. There is one RFH in the refresh input file for each application file or table to be refreshed.

For more information on the RFH, refer to appendix A.

Refresh Group Control File

The Refresh Group Control File (RGCF) contains one record for each refresh performed. The Refresh processes use the RGCF to store information about each refresh. It contains information from the RFH and LCONF, as well as information about the progress of a specific refresh.

When a Refresh process processes an RFH, it checks for an RGCF record for the file or table to be refreshed. If a record for the file or table being refreshed does not exist, the Refresh process creates an RGCF record and adds it to the file. The Refresh process updates the RGCF with current information at the following points in processing:

- When the Refresh process receives a STATUS command from the network operator.
- When the Refresh process receives a SUSPEND command from the network operator. The SQL Refresh process does not support the SUSPEND command at the present time.
- When the Refresh process receives a STOP command from the network operator.

- When the Refresh process abends for any reason.
- When the Refresh process completes normal refresh processing for an individual file or table.

If a refresh is suspended, the information in the RGCF allows the Refresh process to resume processing of the input file at the point where the refresh left off.

REFRESH Command

The Refresh processes are set up in the XPNET system to start on demand, which means that the processes must be started with the REFRESH command. The REFRESH command can be generated from the Refresh screens, or it can be entered by an operator through a network control facility. For directions on using the Refresh screens to start the Refresh processes, refer to section 3. For information on starting the Refresh processes with a text command from a network control facility, refer to the ***BASE24 Text Command Reference Manual***.

Multiple copies of the Refresh processes can be used to initiate more than one refresh at a time.

Logical Network Configuration File

The Refresh processes use a number of Logical Network Configuration File (LCONF) assigns and params in their processing. These assigns and params are identified on the following pages. The assign and param descriptions use the following codes to identify which Refresh processes use the assign or param.

Base	=	Enscribe Base National Negative Refresh process
INAS	=	Enscribe INAS National Negative Refresh process
Enscribe	=	Enscribe Refresh process for files other than national negative files
SQL	=	SQL Refresh process used to refresh SQL tables

For more information on configuring the Refresh processes with these assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual***.

General Assigns

The following assigns identify the names of files that contain Refresh process configuration and control information. The type of Refresh process identified in the Process Use field reads these assigns for all refreshes.

ALT-COLL — The PPD name of the alternate EMS Collector process. If this assign is invalid or missing, it defaults to a value of $0.

Process Use: SQL

BP-RPTR — Identifies the spooler location for refresh reports.

Process Use: Base, INAS

IDF — Identifies the name of the Institution Definition File (IDF).

Process Use: Enscribe

FUP-FILE-NAME — The fully qualified name of the File Utility Program (FUP) process if it is to be run from a location other than $SYSTEM.SYSTEM.FUP.

Process Use: Enscribe

FUP-OUTFILE-NAME — The output location for the File Utility Program (FUP) process started by the Enscribe Refresh process if it is to be sent to a location other than $0.

Process Use: Enscribe

LMCF — Identifies the name of the Log Message Configuration File (LMCF).

Process Use: Base, INAS, Enscribe

PRI-COLL — The PPD name of the primary EMS Collector process. If this assign is invalid or missing, it defaults to a value of $0.

Process Use: SQL

REF-RPT-SQL — Identifies the spooler location for refresh reports.

Process Use: SQL

REF-SOURCEDICT — Identifies the fully qualified volume and subvolume location of the source record dictionary used in full-file refreshes. This assign must contain a file name, such as DICTOBL. However, the Refresh process uses only the volume and subvolume names.

Process Use: SQL

RGCF — Identifies the name of the Refresh Group Control File (RGCF).

Process Use: Enscribe, SQL

RPTR — Identifies the spooler location for refresh reports.

Process Use: Enscribe

File-Specific Assigns

The following assigns are used for a specific type of refresh. For each assign, the description indicates how the assign is used and the type of refresh for which the assign is retrieved from the LCONF.

BP-BNEG — Identifies the template file name for the Base National Negative File (BNEG). This assign is retrieved for full-file refreshes of the BNEG.

Process Use: Base

BP-BNR — Identifies the template file name for the Base National Range File (BNR). This assign is retrieved for full-file refreshes of the BNR.

Process Use: Base

BP-INEG — Identifies the template file name for the INAS National Negative File (INEG). This assign is retrieved for full-file refreshes of the INEG.

Process Use: INAS

BP-INR — Identifies the template file name for the INAS National Range File (INR). This assign is retrieved for full-file refreshes of the INR.

Process Use: INAS

CACT-PART*xx* — Identifies the fully qualified table location for a partition of the Customer/Account Relation Table (CACT), where *xx* is the partition number. Each partition must be placed on a different volume. This assign is retrieved for full-file refreshes of the CACT. Only the CACT-PART00 assign is used if the CACT is not partitioned or if the CACT is partitioned and all partitions are being refreshed.

Process Use: SQL

CAF*identifier* — Identifies the template file name for the Cardholder Authorization File (CAF). This assign can include an optional identifier to specify assigns used for different CAFs in the same logical network. This assign is retrieved for full-file refreshes of the CAF.

Process Use: Enscribe

CCIF — Identifies the fully qualified file name of the default Customer/Cardholder Information File (CCIF). This assign is used for refreshes of the CCIF when an attempt to use the CCIF identified by a CCIF*identifier* assign is unsuccessful, where *identifier* specifies a particular refresh group configured in the Institution Definition File (IDF).

Process Use: Enscribe

CCIF*identifier* — Identifies the fully qualified file name of the Customer/Cardholder Information File (CCIF) for a particular refresh group. This assign includes an *identifier* used for different CCIFs in the same logical network, where *identifier* specifies a particular refresh group. Refresh groups are established by institution in the Institution Data File (IDF). The BASE24 Customer Service product uses the CCIF. This assign is used for refreshes of the CCIF.

Process Use: Enscribe

CCMF — The fully qualified file name of the default Customer/Cardholder Memo File (CCMF). This assign is used for refreshes of the CCMF when an attempt to use the CCMF identified by a CCMF*identifier* assign is unsuccessful, where *identifier* specifies a particular refresh group configured in the Institution Definition File (IDF).

Process Use: Enscribe

CCMF*identifier* — The fully qualified file name of the Customer/Cardholder Memo File (CCMF) for a particular refresh group. This assign includes an *identifier* used for different CCMFs in the same logical network, where identifier specifies a particular refresh group. Refresh groups are established by institution in the Institution Data File (IDF). The BASE24 Customer Service product uses the CCMF. This assign is used for refreshes of the CCMF.

Process Use: Enscribe

CPIT-PART*xx* — Identifies the fully qualified table location for a partition of the Customer/Personal Information Relation Table (CPIT), where *xx* is the partition number. Each partition must be placed on a different volume. This assign is retrieved for full-file refreshes of the CPIT. Only the CPIT-PART00 assign is used if the CPIT is not partitioned or if the CPIT is partitioned and all partitions are being refreshed.

Process Use: SQL

CRT-AUTH-CHF — Identifies the template file name for the Card History File (CHF). The CHF is used with BASE24-pos CRT Authorization. This assign is retrieved for full-file refreshes of the CHF.

Process Use: Enscribe

CSTT-PART*xx* — Identifies the fully qualified table location for a partition of the Customer Table (CSTT), where *xx* is the partition number. Each partition must be placed on a different volume. This assign is retrieved for full-file refreshes of the CSTT. Only the CSTT-PART00 assign is used if the CSTT is not partitioned or if the CSTT is partitioned and all partitions are being refreshed.

Process Use: SQL

FCF — Identifies the name of the File Control File (FCF). This assign is retrieved for refreshes of the Transaction History File (THF).

Process Use: Enscribe

ISCF-*group* — Identifies the name of the Integrated Server Control File (ISCF) and the template file name for the ISCF. Refreshes of the Positive Balance File (PBF) and Transaction History File (THF) can use this assign. This assign is retrieved for full-file or single-partition refreshes when object-oriented processes must be notified. The name of this assign, including the *group* variable, must match the value in the CONTROL FILE ASSIGN field on Server Class Type File (SCT) screen 1.

Process Use: Enscribe

LCF — Identifies the file name of the Log Configuration File (LCF). This assign is retrieved for PBF refreshes with impacting when the BASE24-telebanking product is part of the logical network.

Process Use: Enscribe

NBF*identifier* — Identifies the template file name for the NBF. This assign is retrieved for full-file refreshes of the NBF. This assign can include an optional identifier to specify assigns used for different NBFs in the same logical network.

Process Use: Enscribe

NEG*identifier* — Identifies the template file name for the Negative Card File (NEG). This assign is retrieved for full-file refreshes of the NEG. This assign can include an optional identifier to specify assigns used for different NEGs in the same logical network.

Process Use: Enscribe

OBJ — Identifies the name of the Object File (OBJ). Refreshes of the Positive Balance File (PBF) and Transaction History File (THF) can use this assign. This assign is retrieved for full-file or single-partition refreshes when object-oriented processes must be notified.

Process Use: Enscribe

PBF*identifier* — Identifies the template file name for the Positive Balance File (PBF). This assign is retrieved for full-file refreshes of the PBF. This assign can include an optional identifier to specify assigns used for different PBFs in the same logical network.

Process Use: Enscribe

PIT-PART*xx* — Identifies the fully qualified table location for a partition of the Personal Information Table (PIT), where *xx* is the partition number. Each partition must be placed on a different volume. This assign is retrieved for full-file refreshes of the PIT. Only the PIT-PART00 assign is used if the PIT is not partitioned or if the PIT is partitioned and all partitions are being refreshed.

Process Use: SQL

PORF — Identifies the name of the Process/Object Relationship File (PORF). Refreshes of the Positive Balance File (PBF) and Transaction History File (THF) can use this assign. This assign is retrieved for full-file or single-partition refreshes when object-oriented processes must be notified.

Process Use: Enscribe

POS-PRDF — The fully qualified file name of the POS Retailer Definition File (PRDF).

Process Use: Enscribe

POS-PTLF — Identifies the file name of the POS Transaction Log File (PTLF). This assign is retrieved for PBF refreshes with impacting when the BASE24-pos product is part of the logical network.

Process Use: Enscribe

SCT — Identifies the name of the Server Class Type File (SCT). Refreshes of the Positive Balance File (PBF) and Transaction History File (THF) can use this assign. This assign is retrieved for full-file or single-partition refreshes when object-oriented processes must be notified.

Process Use: Enscribe

SDF — Identifies the template file name for the Switch Dispute File (SDF). This assign is retrieved for refreshes of the SDF.

Process Use: Enscribe

SEMF — Identifies the name of the Scoring Engine Master File (SEMF). The SEMF is used with the ACI Proactive Risk Manager product and is retrieved for refreshes of the SEMF.

Process Use: Enscribe

SPF*identifier* — Identifies the template file name for the Stop Payment File (SPF). This assign is retrieved for full-file refreshes of the SPF. This assign can include an optional identifier to specify assigns used for different SPFs in the same logical network.

Process Use: Enscribe

SPRF — Identifies the name of the Server Class/Process Relationship File (SPRF). Refreshes of the Positive Balance File (PBF) and Transaction History File (THF) can use this assign. This assign is retrieved for full-file or single-partition refreshes when object-oriented processes must be notified.

Process Use: Enscribe

SSB-CCF — Identifies the template file name for the Corporate Check File (CCF). The CCF is used with the BASE24-atm self-service banking Enhanced Check Application. This assign is retrieved for full-file refreshes of the CCF.

Process Use: Enscribe

SSB-CSF — Identifies the template file name for the Check Status File (CSF). The CSF is used with the BASE24-atm self-service banking Enhanced Check Application. This assign is retrieved for full-file refreshes of the CSF.

Process Use: Enscribe

THCF — Identifies the name of the Transaction History Configuration File (THCF). This assign is retrieved for refreshes of the Transaction History File (THF).

Process Use: Enscribe

THF-*group* — Identifies the name of the Transaction History File (THF) and the template file name for the THF. This assign is retrieved for refreshes of the THF. The name of this assign, including the *group* variable, must match the value in the ASSIGN field on Transaction History Configuration File (THCF) screen 1.

Process Use: Enscribe

TLF — Identifies the file name of the BASE24-atm Transaction Log File (TLF). This assign is retrieved for CAF or PBF refreshes with impacting when the BASE24-atm product is part of the logical network.

Process Use: Enscribe

TLF-*group* — Identifies the file name of the ITS Transaction Log File (ITLF). This assign is retrieved for PBF refreshes with impacting when the BASE24-telebanking product is part of the logical network. The name of this assign, including the *group* variable, must match the value in the ASSIGN field on Log Configuration File (LCF) screen 1.

Process Use: Enscribe

TLR-TTLF — Identifies the file name of the TTLF. This assign is retrieved for PBF, SPF, No Book File (NBF), or Warning/Hold/Float File (WHFF) refreshes with impacting when the BASE24-teller product is part of the logical network.

Process Use: Enscribe

TLR-WHFF*identifier* — Identifies the template file name for the WHFF. This assign is retrieved for full-file refreshes of the WHFF. This assign includes an optional identifier to specify assigns used for different WHFFs in the same logical network.

Process Use: Enscribe

General Params

The following params contain configuration information used for all refreshes. The type of Refresh process identified in the Process Use field reads these params for all refreshes.

ALT-TIM-LMT — Specifies the maximum time the application waits for the call to the WRITE procedure to complete. This param is used when logging to the alternate EMS Collector process.

Process Use: SQL

BASE24-TAPE-LBL-PROC-USED — Indicates whether BASE24 or the HP NonStop Kernel TAPECOM product performs tape label processing.

Process Use: Base, INAS, Enscribe, SQL

CONSOLE-PRT — Indicates whether EMS event messages are printed on the operator console if the EMS compatibility distributor is used.

Process Use: SQL

PRI-TIM-LMT — Specifies the maximum time the application waits for the call to the WRITE procedure to complete. This param is used when logging to the primary EMS Collector process.

Process Use: SQL

REF-INPUT-BLK-DISK — Specifies the size of block segments on the disk used as input for refreshes.

Process Use: SQL

REF-INPUT-BLK-TAPE — Specifies the size of block segments on the tape used as input for refreshes.

Process Use: SQL

REFR-INPUT-DATA-UNSORTED — Specifies whether the Refresh process allows unsorted input data records. If the flag is set to no, the Refresh process expects input data records to be sorted.

Process Use: Enscribe, SQL

REFR-MAX-RECS-UNSORTED — Ondicates the number of records in the refresh input file. The File Utility Program (FUP) uses this value to determine the size of the scratch file used by the SORT process. This value should be equal to or greater than the actual number of records in the refresh input file.

Process Use: Enscribe, SQL

REF-MONITOR-COMMAND — Specifies the number of records to process between checking for a command from a network control operator.

Process Use: Enscribe, SQL

REF-NOTIFY*identifier* — Specifies the name of a process to be notified after a new file is generated by a full-file refresh.

Process Use: Enscribe

REFR-EARLY-NOTIFICATION — Specifies whether refresh early notification processing is enabled. If the flag is set to yes, the Enscribe Refresh process notifies the Authorization processes immediately after data is loaded on the new PBF without waiting for impacting to be completed.

Process Use: Enscribe

REF-RGCF-RETN-PRD — Specifies the number of days to retain RGCF records.

Process Use: Enscribe, SQL

REF-SQLCI-CPU — Identifies the CPU in which the NonStop SQL conversational interface (SQLCI) process runs for a full-file refresh.

Process Use: SQL

REF-SQLCI-DELAY — Specifies the number of seconds to delay before the SQL Refresh process attempts to stop the NonStop SQL conversational interface (SQLCI) process. This delay allows the SQLCI process to stop on its own before being stopped by the SQL Refresh process.

Process Use: SQL

File-Specific Params

The following params are used for a specific type of refresh. For each param, the description indicates how the param is used and the type of refresh for which the param is retrieved from the LCONF.

ATM-LN-CUTOVER — Specifies the time of day that the entire logical network is cut over for BASE24-atm. This param is only read for refreshes with impacting.

Process Use: Enscribe

CACT-PRI-PARTONLY — Indicates whether all partitions of the Customer/ Account Relation Table (CACT) or only the primary partition of the CACT are refreshed during a full-file refresh.

Process Use: SQL

CACT-SOURCEREC — Identifies the record name within the Data Definition Language (DDL) dictionary for the CACT-LOAD definition. This param is used in conjunction with the REF-SOURCEDICT assign to obtain the correct DDL record when performing a full-file refresh of the CACT.

Process Use: SQL

CPIT-PRI-PARTONLY — Indicates whether all partitions of the Customer/ Personal Information Relation Table (CPIT) or only the primary partition of the CPIT are refreshed during a full-file refresh.

Process Use: SQL

CPIT-SOURCEREC — Identifies the record name within the Data Definition Language (DDL) dictionary for the CPIT-LOAD definition. This param is used in conjunction with the REF-SOURCEDICT assign to obtain the correct DDL record when performing a full-file refresh of the CPIT.

Process Use: SQL

CSTT-PRI-PARTONLY — Indicates whether all partitions of the Customer Table (CSTT) or only the primary partition of the CSTT are refreshed during a full-file refresh.

Process Use: SQL

CSTT-SOURCEREC — Identifies the record name within the Data Definition Language (DDL) dictionary for the CSTT-LOAD definition. This param is used in conjunction with the REF-SOURCEDICT assign to obtain the correct DDL record when performing a full-file refresh of the CSTT.

Process Use: SQL

LGNT-CUTOVER — Specifies the logical network cutover time. This param is retrieved for PBF refreshes with impacting when the BASE24-telebanking product is part of the logical network.

Process Use: Enscribe

LOGICAL-NET — Specifies the name of the logical network of which the Refresh process is a part. This param is read for all refreshes.

Process Use: Base, INAS, Enscribe, SQL

PIT-NAM-ALGO — Indicates whether the BASE24-telebanking and BASE24-billpay products support a name token generation algorithm, such as Soundex. This param is read for refreshes of the Personal Information Table (PIT).

Process Use: SQL

PIT-PRI-PARTONLY — Indicates whether all partitions of the Personal Information Table (PIT) or only the primary partition of the PIT are refreshed during a full-file refresh.

Process Use: SQL

PIT-SOURCEREC — Specifies the record name within the Data Definition Language (DDL) dictionary for the PIT-LOAD definition. This param is used in conjunction with the REF-SOURCEDICT assign to obtain the correct DDL record when performing a full-file refresh of the Personal Information Table (PIT).

Process Use: SQL

POS-LN-CUTOVER — Specifies the time of day that the entire logical network is cut over for BASE24-pos. This param is read for refreshes with impacting.

Process Use: Enscribe

POS-PREF-NOTIFY*identifier* — Specifies the symbolic name of a process that the national negative refresh process should notify after new files are generated in a full-file refresh.

Process Use: Base, INAS

REF-BYPASS-PARITY-ERRORS — Specifies the maximum number of tape parity errors that can occur during a refresh before the Refresh process abends. This param is read for all SQL and Enscribe refreshes.

Process Use: Enscribe, SQL

REF-CAF-ACCT-CNT — Specifies the maximum number of accounts that can be associated with each fixed-length CAF record in the refresh input file. This param is used for previous-release format CAF refreshes.

Process Use: Enscribe

REF-FUP-CPU — Specifies the CPU where the File Utility Program (FUP) is to execute for full-file refreshes.

Process Use: Enscribe

REF-IMPACT-TABLE-SIZE — Specifies the number of bytes that the Refresh process must reserve in memory to build an internal table for use during PBF partial-file refreshes with impacting.

Process Use: Enscribe

REF-PBF-LOCKOUT-FACTOR — Indicates the number of PBF records the Refresh process can lock on disk and update at one time during PBF partial-file refreshes with impacting.

Process Use: Enscribe

REF-PRE-ALLOCATE-EXTENTS — Specifies the number of extents that the Refresh process must preallocate for creating a new file during a full-file refresh.

Process Use: Enscribe

REF-SWI-NOTIFY*identifier* — Specifies the name of a process to be notified after a partial-file refresh of the SDF.

Process Use: Enscribe

SEMF-NOTIFY*identifier* — Specifies the symbolic name of a Scoring Engine Interface process that the Enscribe Refresh process should notify after a full-file refresh of the Scoring Engine Master File (SEMF).

Process Use: Enscribe

TLR-LN-CUTOVER — Specifies the time of day that the entire logical network is cut over for BASE24-teller. This param is only read for refreshes with impacting.

Process Use: Enscribe

TLR-WRITE-HLDS-DEP-FLOATS — Specifies whether warning and hold transactions are written to the WHFF. This param is only read for WHFF refreshes with impacting.

Process Use: Enscribe

Log Message Configuration File

The Log Message Configuration File (LMCF) contains event message information, such as text, routing code, and severity, that has been modified by the user. At initialization, the Base National Negative, INAS National Negative, and Enscribe Refresh processes build a table in memory from the LMCF records. When an event message is generated, the Refresh processes access the table to verify whether that event message has been modified. If the event message has been modified, the Refresh processes log the modified information instead of the standard event message.

Operator functions are used to access the LMCF. The routing code, severity, and text can be modified for each event message through the LMCF. Because LMCF records are read into memory by the processes using them, updates to the LMCF do not take effect until the processes are reinitialized. For more information on accessing the LMCF, refer to the ***BASE24 Event Message Reference Manual***.

Event Messages

All types of Refresh processes generate event messages to assist network operators in performing their daily tasks. Event messages provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Event messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

ACI has developed a method that allows customers to access event message documentation electronically. Event message descriptions are contained in files on a designated subvolume. Each file contains event messages for a specific BASE24 process. The files are organized according to product module ID (PMID). One PMID exists for each BASE24 process. PMIDs identify a specific process in the BASE24 system (e.g., BAREFR identifies the Enscribe Refresh process and ISCOREFR identifies the SQL Refresh process). Providing event message documentation in files on the HP NonStop computer gives customers greater flexibility by allowing them to view and print event message documentation as needed at their location.

Documentation is available in file form for the event messages generated by the Refresh processes. These files are located on the BA*xx*LOGM or OK*xx*LOGM subvolume, where *xx* is the number of the current release, and are named according to PMID. For a complete explanation of event message documentation, refer to the ***BASE24 Event Message Reference Manual***.

The Enscribe Refresh process generates event messages using a standard routing code of 0, unless the file being refreshed is tied to only one product. In that case, the Enscribe Refresh process uses the product's routing code (e.g., CHF is a BASE24-pos file, and the Enscribe Refresh process uses the standard BASE24-pos routing code of 2 when logging messages during a CHF refresh).

Institution Definition File

The Enscribe Refresh process uses the Institution Definition File (IDF) to obtain the fully qualified name of the current file to be refreshed and to identify the institutions in the refresh group. The IDF also identifies the file segments supported by each financial institution. File names appear on IDF screens 1, 16, and 25, the refresh group for each institution appears on IDF screen 1, and the supported file segments appear on IDF screens 5 and 6.

The Enscribe Refresh process also uses refresh indicators in the Base segment of the IDF during CAF, NBF, PBF, SPF, and WHFF refreshes with impacting. These refresh indicators are increased and checked against the refresh indicators in the transaction log files. These indicators do not appear on file maintenance screens.

Refer to the ***BASE24 Base Files Maintenance Manual*** for the IDF screen illustrations and field descriptions.

Transaction History Configuration File

The Enscribe Refresh process uses the Transaction History Configuration File (THCF) to obtain the LCONF assign that identifies the proper Transaction History File (THF). The BASE24-telebanking product can use multiple THFs, and the THCF identifies the THF according to refresh group, financial institution, and account number range. At least one THCF record must exist for each THF.

Refer to the ***BASE24 Core Files and Tables Maintenance Manual*** for the THCF screen illustrations and field descriptions.

Process Type File

The Process Type File (PTF) must be configured when the Enscribe Refresh process is used with object-oriented applications such as BASE24-telebanking. The Enscribe Refresh process uses the PTF indirectly because a process type must be defined for Enscribe Refresh processes in the PTF before records for the individual Enscribe Refresh processes can be defined in the Process File (PRO).

Refer to the ***BASE24 ITS Kernel Files Maintenance Manual*** for the PTF screen illustration and field descriptions. The following table contains suggested values for fields on the PTF screen.

Field	Value
PROCESS TYPE	REFRESH
PROCESS TYPE DESCRIPTION	REFRESH

Process File

The Process File (PRO) must be configured when the Enscribe Refresh process is used with object-oriented applications such as BASE24-telebanking. The Enscribe Refresh process must use process control commands called FILE CLOSE OPEN requests to notify BASE24-telebanking Integrated Authorization Server processes each time it completes a full-file refresh. The PRO contains a record defining each process that can receive process control commands. The PRO already contains a record for each BASE24-telebanking Integrated Authorization Server process; however, a record must be added to the PRO for each Enscribe Refresh process. The Integrated Authorization Server processes then use information from the PRO records to respond to the Enscribe Refresh processes.

Refer to the ***BASE24 ITS Kernel Files Maintenance Manual*** for the PRO screen illustration and field descriptions. The following table contains suggested values for fields on the PRO screen.

Field	Value
PROCESS NAME	The process name assigned to the Enscribe Refresh process. For example, P1A^REFR1. A separate PRO record is required for each Enscribe Refresh process.

Field	Value
PPD NAME	The HP NonStop Process Pair Directory (PPD) name assigned to the Enscribe Refresh process. For example, $H5AR1.
PROCESS TYPE	The process type assigned to all Enscribe Refresh processes in the PTF. For example, REFRESH. The process type must be defined in the PTF before it can be used in the PRO.
QUEUE FLAG	N

Server Class Type File

The Server Class Type File (SCT) must be configured when the Enscribe Refresh process is used with object-oriented applications such as BASE24-telebanking. The Enscribe Refresh process must read an Integrated Server Control File (ISCF) to receive process control commands and responses from Integrated Authorization Server processes. The SCT contains a record defining each ISCF that can contain process control commands.

The SCT must contain a record for the ISCF being read by Enscribe Refresh processes before records for the individual Enscribe Refresh processes can be defined in the Server Class/Process Relationship File (SPRF). However, you have the option of adding a new SCT record or using an existing SCT record, as follows:

- If process control commands for the Enscribe Refresh process are being placed in an existing ISCF, an additional SCT record is not required.
- If process control commands for the Enscribe Refresh process are being placed in a separate ISCF, a record must be added to the SCT to define that ISCF.

Refer to the ***BASE24 ITS Kernel Files Maintenance Manual*** for the SCT screen illustration and field descriptions. The following table contains suggested values for fields on the SCT screen when a separate ISCF is used for Enscribe Refresh processes.

Field	Value
SERVER CLASS TYPE	PROREFR

Field	Value
SERVER CLASS TYPE DESCRIPTION	REFRESH
CONTROL FILE ASSIGN NAME	ISCF-PROREFR **Note:** An assign for this file must be defined in the LCONF.
CONTROL FILE NUMBER	0

Server Class/Process Relationship File

The Server Class/Process Relationship File (SPRF) must be configured when the Enscribe Refresh process is used with object-oriented applications such as BASE24-telebanking. Each BASE24-telebanking Integrated Authorization Server process and Enscribe Refresh process reads only one ISCF for its process control commands. Records in the SPRF identify which ISCF contains process control commands for each process. A record must be added to the SPRF to identify which ISCF contains process control commands that are read by each Enscribe Refresh process. The ISCF must be defined in the SCT and the Enscribe Refresh process must be defined in the PRO before they can be identified in an SPRF record.

Refer to the ***BASE24 ITS Kernel Files Maintenance Manual*** for the SPRF screen illustration and field descriptions. The following table contains suggested values for fields on the SPRF screen when a separate ISCF is used for Enscribe Refresh processes.

Field	Value
SERVER CLASS TYPE	The server class type assigned to all Enscribe Refresh processes in the SCT. For example, PROREFR. The server class type must be defined in the SCT before it can be used in the SPRF.

Field	Value
PROCESS NAME	The process name assigned to the Enscribe Refresh process in the PRO. For example, P1A^ REFR1. The process name must be defined in the PRO before it can be used in the SPRF. A separate SPRF record is required for each Enscribe Refresh process.

File Control File

The File Control File (FCF) must contain records for the Transaction History File (THF) prior to updating it with a full-file refresh.

Each FCF record contains a version indicator that is used to identify the newest version of the file. For most BASE24 files, the Enscribe Refresh process renames the existing file and builds a new file with the name originally assigned to the old file during a full-file refresh. The THF cannot be renamed because it is audited by the HP NonStop Kernel Transaction Monitoring Facility (TMF), and a file that is audited by TMF cannot be renamed. A record must be added to the FCF for each file that is audited by TMF so the Enscribe Refresh process can change the version indicator during a full-file refresh.

Network Control Commands

All Refresh processes are started by a command issued by a network operator. The REFRESH command can be issued either from the Refresh screens or from a network control facility. The startup message provides information about the refresh input file and includes some configuration fields that can be used to override information in the RFH and LCONF.

In addition to the startup message, the network operator can issue the following commands to all Refresh processes:

- STATUS
- CONTINUE

The network operator can also issue the following commands to the Base National Negative, INAS National Negative, and Enscribe Refresh processes, but not the SQL Refresh process:

- SUSPEND
- RESUME

For more information on issuing these commands to the Refresh processes, refer to the procedures in section 3. For more information on the processing that a Refresh process performs when it receives one of these commands, refer to the ***BASE24 Text Command Reference Manual***.

Super Extract Process Configuration and Control

The Super Extract process is configured and controlled by information in the following locations:

- XPNET configuration
- Logical Network Configuration File (LCONF)
- Log Message Configuration File (LMCF)
- Extract Configuration File (ECF)
- Token File (TKN)

The network control operator can also issue commands that control extract processing. The following items address configuration and control issues related to the Super Extract process.

XPNET Configuration

Super Extract processes must be set up in the XPNET system with the value of the process STARTUP attribute set to AUTOMATIC, which indicates that the Super Extract process is started automatically when the XPNET node is initiated. This allows the Super Extract process to read the ECF and set timers to expire when automatic extracts are to occur. Refer to the ***NET24-XPNET Network Control Command Reference Manual*** for additional information about the STARTUP attribute.

Multiple copies of a Super Extract process can be defined in the system to increase the number of extract timers that can be set or to run different extracts to separate locations at the same time.

Logical Network Configuration File

The Super Extract process uses a number of Logical Network Configuration File (LCONF) assigns and params in its processing. These assigns and params are identified on the following pages. For more information on configuring the Super Extract process with these assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual***.

General Assigns

The following assigns identify the file names of files that contain Super Extract process configuration and control information. The Super Extract process reads these assigns for all extracts.

ECF — Specifies the name of the Extract Configuration File (ECF).

LMCF — Specifies the name of the Log Message Configuration File (LMCF).

TKN — Specifies the name of the Token File (TKN).

File-Specific Assigns

The following assigns are read for a specific type of extract. For each assign, the description indicates how the assign is used and the type of extract for which the assign is retrieved from the LCONF.

HMBF — Specifies the file name of the Host Mail Box File (HMBF). This assign is retrieved for HMBF extracts.

HMTF — Specifies the file name of the Host Message Text File (HMTF). This assign is retrieved for HMBF extracts.

HSF — Specifies the file name of the Hardware Status File (HSF). This assign is retrieved for HSF extracts.

ICF — Specifies the file name of the Interchange Configuration File (ICF). This assign is retrieved for Interchange Log File (ILF) and ICF extracts.

ICFE — Specifies the file name of the Enhanced Interchange Configuration File (ICFE). This assign is retrieved for Interchange Log File (ILF) and ICFE extracts.

IDF — Specifies the file name of the IDF. This assign is retrieved for IDF, ITS Transaction Log File (ITLF), Mail Box File (MBF), PTLF, TLF, TTLF, and Update Log File (ULF) extracts.

ILF — Specifies the file name of the ILF. This assign is retrieved for ILF extracts.

LCF — Identifies the file name of the Log Configuration File (LCF). This assign is retrieved for ITLF extracts.

MBF — Specifies the file name of the Mail Box File (MBF). This assign is retrieved for MBF extracts.

MEF — Specifies the file name of the Message Explanation File (MEF). This assign is retrieved for MEF extracts.

MTF — Specifies the file name of the Message Text File (MTF). This assign is retrieved for MBF extracts.

OMF — Specifies the file name of the Online Maintenance File (OMF). This assign is retrieved for OMF extracts.

POS-PRDF — Specifies the file name of the POS Retailer Definition File (PRDF). This assign is retrieved for PRDF extracts.

POS-PTLF — Specifies the file name of the POS Transaction Log File (PTLF). This assign is retrieved for PTLF extracts.

POS-SVHF — Specifies the file name of the POS Stored Value History File (SVHF). This assign is retrieved for SVHF extracts.

SAF — Specifies the file name of the Store-and-Forward File (SAF). This assign is retrieved for SAF extracts.

TLF — Specifies the file name of the TLF. This assign is retrieved for TLF extracts.

TLF-*group* — Identifies the file name of the ITS Transaction Log File (ITLF). This assign is retrieved for ITLF extracts. The name of this assign, including the *group* variable, must match the value in the ASSIGN field on Log Configuration File (LCF) screen 1.

TLR-TTF — Specifies the file name of the Teller Transaction File (TTF). This assign is retrieved for TTF extracts.

TLR-TTLF — Specifies the file name of the TTLF. This assign is retrieved for TTLF extracts.

ULF — Specifies the file name of the ULF. This assign is retrieved for ULF extracts.

General Params

The following params contain configuration information used for all extracts.

ACCT-NUM-TYP — Indicates whether account numbers are carried in the system in decimal or binary format.

BASE24-TAPE-LBL-PROC-USED — Indicates whether BASE24 or the HP NonStop Kernel TAPECOM product performs tape label processing.

DATE-DISPLAY-TYP — Indicates the format in which dates are displayed in reports.

EXTR-DISK-EXTENT-SIZE-MULT — The disk extract extent size multiplier. This is an integer value that is multiplied by the primary and secondary extent size values in the Extract Configuration File (ECF) to obtain the actual size of the extents used when creating an extract disk file. This param applies to the ECF primary and secondary extent sizes for both partitioned and unpartitioned files and is needed only when extracting extremely large files (e.g., 2 gigabytes or more) to disk and the maximum extent size value in the ECF (i.e., 9999) is insufficient.

EXTR-DISPOSITION — Specifies how the Super Extract process handles oversized records.

EXTR-TAPE-BUFFERING — Indicates whether record buffering should be used when extracting records to tape drives configured for streaming mode.

LOGICAL-NET — Specifies the name of the logical network of which the Super Extract process is a part.

File-Specific Params

The following params contain configuration information used for extracting a specific file. For each param, the description indicates how the param is used and the type of extract for which the param is retrieved from the LCONF.

ATM-LN-CUTOVER — Specifies the time of day that the entire logical network is cut over for BASE24-atm. This param is used for SAF extracts.

BP-DATE-FORMAT — Specifies the format in which dates are displayed or entered for the BASE24-billpay product. This param is used for Online Maintenance File (OMF) extracts.

FHM-LN-CUTOVER — Specifies the time of day that the entire logical network is cut over for BASE24-from host maintenance. This param is used for SAF extracts.

LGNT-CUTOVER — Specifies the logical network cutover time. This param is retrieved for SAF extracts when the BASE24-telebanking product is part of the logical network.

MAIL-LN-CUTOVER — Specifies the time of day that the entire logical network is cut over for BASE24-mail. This param is used for SAF extracts.

OMF-RETENTION-PERIOD — Specifies the number of days to retain an OMF. This param is used for OMF extracts.

POS-LN-CUTOVER — Specifies the time of day that the entire logical network is cut over for BASE24-pos. This param is used for SAF extracts.

TLR-LN-CUTOVER — Specifies the time of day that the entire logical network is cut over for BASE24-teller. This param is used for SAF extracts.

TLR-TTLFX-NOTIFY*identifier* — Identifies a process to be notified that the PBF, SPF, NBF, and WHFF are not current after a TTLF extract. When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the logical network.

TLR-WRITE-HLDS-DEP-FLOATS — A flag indicating whether to write a record to the Warning/Hold/Float File (WHFF) when processing a hold or deposit float on a Positive Balance File (PBF) record.

Log Message Configuration File

The Log Message Configuration File (LMCF) contains event message information, such as text, routing code, and severity, that has been modified by the user. At initialization, the Super Extract process builds a table in memory from the LMCF records. When an event message is generated, the Super Extract process accesses the table to verify whether that event message has been modified. If the event message has been modified, the Super Extract process logs the modified information instead of the standard event message.

Operator functions are used to access the LMCF. The routing code, severity, and text can be modified for each event message through the LMCF. Because LMCF records are read into memory by the processes using them, updates to the LMCF do not take effect until the processes are reinitialized. For more information on accessing the LMCF, refer to the ***BASE24 Event Message Reference Manual***.

Event Messages

The Super Extract process generates event messages to assist network operators in performing their daily tasks. Event messages provide information regarding the internal condition of the system. They are a network operator’s primary link to what is actually occurring in the system. Event messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

ACI has developed a method that allows customers to access event message documentation electronically. Event message descriptions are contained in files on a designated subvolume. Each file contains event messages for a specific BASE24 process. The files are organized according to product module ID (PMID). One PMID exists for each BASE24 process. PMIDs identify a specific process in the BASE24 system (e.g., BASUPERX identifies the Super Extract process). Providing event message documentation in files on the HP NonStop computer gives customers greater flexibility by allowing them to view and print event message documentation as needed at their location.

Documentation is available in file form for the event messages generated by the Super Extract process. It is located on the BA*xx*LOGM subvolume in a file named BASUPERX, where *xx* is the number of the current release. For a complete explanation of event message documentation, refer to the ***BASE24 Event Message Reference Manual***.

The Super Extract process generates event messages using a standard routing code of 0.

Extract Configuration File

The Extract Configuration File (ECF) defines processing parameters for each type of extract an institution might perform in a particular logical network. The ECF provides users with the capability to extract a single file or multiple files to a tape or disk file. In addition, users have the option to set parameters that make it possible to perform extracts automatically on a timed basis each day.

One ECF record must exist for each extract session to be performed. For example, an institution may choose to extract the Transaction Log File (TLF) to one tape; the Interchange Log File (ILF) to another tape; the ITS Transaction Log File (ITLF) to another tape; and the POS Transaction Log File (PTLF) and POS Retailer Definition File (PRDF) to yet another tape. Each one of these extracts must have a separate record defined in the ECF.

Each Super Extract process is limited to setting 20 extract timers. Each ECF record is tied to one Super Extract process. However, more than one ECF record can be tied to a single Super Extract process.

Manual extracts require the entry of a tag on the Extract screen or from a network control facility, which identifies the ECF record to use to control processing for the extract.

Using the ECF to Configure Extracts to Tape

The following are operator procedures for configuring file extracts to tape with the ECF. These procedures assume that you are logged on to the BASE24 system.

1. Enter ECF in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. ECF screen 1 is displayed, as shown below.

```
 BASE24-BASE  ECF MESSAGE FILE         LLLL        MM/DD/YY  HH:MM  01 OF 23

               TAG:                        SYMBOLIC NAME:
      EXTRACT DATE:        (YYMMDD)      EXTRACT TIME: 0000 (HHMM)
 LAST EXTRACT DATE:        (DISPLAY ONLY)     RESTART: N    (Y OR N)
   TAPE LABEL TYPE: IBM (IBM/ANS/BUR/NON  -  DEFAULT IS IBM)
   TAPE BLOCK SIZE:  4096 BYTES           RELEASE NUM: 60    CHARACTER SET: E
         TAPE NAME: $TAPE                               (DEFAULT IS $TAPE)
   REPORT LOCATION:
NUMERIC FLD FORMAT: B          READ PAST INITIAL EOF: N
 VOLUME IDENTIFIER: 000000       DATA SET IDENTIFIER: BASE24.SXT02122
         RETENTION: 00                 MOUNT MESSAGE: SUPER EXTRACT PROCESSING
           DENSITY: 6250   (800/1600/6250 - DEFAULT IS 6250)
                                  BASE FILES
        ILF: N (Y OR N)       DATE OFFSET:  +0  (+/- DAYS)
                              SWITCH FIID: ALL
                          REPORT LOCATION:
                              RPT-EXTRACT: N
                                   FORMAT: 00 (FIXED FORMAT)

 ********************************** BASE24 **********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                            F12-HELP
```

2. Determine a name for the ECF record and enter the name in the TAG field. The value entered in the TAG field is used to identify the extract session when manual extracts or reextracts are requested from a network control facility or the Extract screen. The value from this field is also placed in the tape header and trailer when the Super Extract process creates an extract tape.

3. Enter the symbolic name of the Super Extract process that will use this record in the SYMBOLIC NAME field.

4. Enter the date, in YYMMDD format, of the next automatic extract to be initiated using this record in the EXTRACT DATE field. If you do not want the Super Extract process to automatically start an extract using this record at some point in the future, you can enter a date in the past in this field. For example, if the current date is December 22, 2001, you could specify 011221 in this field and an automatic extract would not be run.

5. Enter the time, in hhmm format, at which automatic extracts using this record should be run in the EXTRACT TIME field.

6. Determine whether you want continuing, daily automatic extracts performed using this record and enter the appropriate value in the RESTART field. Valid values for this field are as follows:

   Y = Yes, perform daily automatic extracts.
   N = No, do not perform daily automatic extracts.

   Note: The value Y works only when a current or future date is entered in the EXTRACT DATE field. If a past date is entered in the EXTRACT DATE field, daily automatic extracts will not be performed.

7. Enter the tape label type in the TAPE LABEL TYPE field. Valid tape label type values depend on whether the Super Extract process or the HP NonStop Kernel TAPECOM product is performing tape label processing.

   If the HP NonStop Kernel TAPECOM product is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value N), valid tape label types are as follows:

   IBM = IBM/MVS
   ANS = ANSI
   NON = Unlabeled

   If the Super Extract process is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value Y), valid tape label types are as follows:

   IBM = IBM/MVS
   BUR = Burroughs
   NON = Unlabeled

8. Enter the block size to be used in the TAPE BLOCK SIZE field. Valid values are as follows:

   1–32 = Block size in kilobytes. The Super Extract process multiplies the value entered by 1024 to determine the block size in bytes (e.g., 2 kilobytes equals 2048 bytes, 10 kilobytes equals 10240 bytes, and 32 kilobytes equals 32768 bytes). BASE24 automatically displays a K following an entry in kilobytes (e.g., 2K, 10K, and 32K).

   33–32767 = Block size in bytes. The Super Extract process uses the value entered (e.g., 2048 equals 2048 bytes or 2 kilobytes, and 9216 equals 9216 bytes or 9 kilobytes).

9. If the extract session being defined includes extracts of shared files, enter the release number identifying the extract format to be used in the RELEASE NUM field. Valid values are as follows:

   50 = Use previous release extract formats for extracts of shared files.
   60 = Use current release extract formats for extracts of shared files.

   Note: The value in this field determines the format of the data on the extract tape. It does not indicate the release of the BASE24 database.

10. Enter the code identifying the character set to use in data records on the extract tape in the CHARACTER SET field. Valid codes are as follows:

    A = ASCII
    E = EBCDIC

11. Enter the SYSGEN device name of the tape drive in the TAPE NAME field.

12. Enter the name of the spooler location to which the Super Extract process should write its reports in the REPORT NAME field.

13. If the extract session will include an extract of the TLF, PTLF, ITLF, TTLF, ILF, or SAF, enter a code indicating the type of numeric field format to use in the extracted data records in the NUMERIC FLD FORMAT field. Valid values are as follows:

    A = The Super Extract process converts binary fields to ASCII display fields.
    B = The Super Extract process leaves binary fields in binary.

    Note: Only the fields in the TLF, PTLF, ITLF, TTLF, ILF, and SAF extract are affected by the setting in this field. Other files included in the extract are extracted in disk file format.

14. If the Super Extract process is to read past the initial end-of-file determined when the extract is first started, set the READ PAST INITIAL EOF field to a value of Y. Otherwise, any new records added to the file after the extract is started will not be included in the extract. In the ECF record used for the last extract performed on a transaction log file, this field must be set to a value of Y. Valid values are as follows:

    Y = Yes, the Super Extract process reads past the initially determined end-of-file.
    N = No, the Super Extract process does not read past the initially determined end-of-file.

    This field defaults to a value of N.

This field must be set to a value of Y in the following conditions to avoid records being missed:

- If the extract is the final one performed on a transaction log file.

  Note: If multiple extracts are performed each day, this field must be set to a value of N for each extract prior to the final one. However, it must be set to a value of Y for the final extract.

- If only one extract is performed each day.
- If the file is small enough to fit in one block (e.g., in testing situations).

15. If the HP NonStop Kernel TAPECOM product is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value N) and the tape used for extracts is a labeled tape, enter the volume identifier from the first tape to be used for the extract in the VOLUME IDENTIFIER field. The volume identifier can be assigned to the tape when the tape was labeled by TAPECOM.
16. If the HP NonStop Kernel TAPECOM product is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value N) and the tape used for extracts is a labeled tape, enter the file name to be used on the extract tape in the DATA SET IDENTIFIER field.
17. If the HP NonStop Kernel TAPECOM product is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value N) and the tape used for extracts is a labeled tape, enter the number of days to retain the extract file (the file on the tape) in the RETENTION field.

    When TAPECOM writes the label on the tape, the value in this field is used to calculate an expiration date for the tape. TAPECOM checks the expiration date when using labeled tapes to determine whether data on the tape can be overwritten. If the tape has not expired, TAPECOM rejects the tape and does not overwrite the existing data. Valid values are as follows:

00	=	No retention period. The tape expires immediately, meaning that new data can be written to the tape at any time.
01–99	=	The number of days to retain the data on the tape. This number is added to the current system date when TAPECOM writes the tape label to create an expiration date. New data cannot be written to the tape until after the expiration date (i.e., after the number of days specified to retain the data).

18. If the HP NonStop Kernel TAPECOM product is running on the system and additional information should be displayed with the system mount message, enter the additional information in the MOUNT MESSAGE field.

19. Enter the data density of the tape in bits per inch in the DENSITY field.
20. Identify the BASE24 files to be included in the extract. For more information on identifying the files to be included in the extract, refer to the topic “Specifying the Files to be Extracted” later in this section.

Using the ECF to Configure Extracts to Disk

The following are operator procedures for configuring file extracts to disk using the ECF. These procedures assume that you are logged on to BASE24.

1. Enter ECF in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. ECF screen 1 is displayed.
2. Determine a name for the ECF record and enter the name in the TAG field. The value entered in the TAG field is used to identify the extract session when manual extracts or reextracts are requested from a network control facility or the Extract screens. The value from this field is also placed in the tape header and trailer when the Super Extract process creates the extract file on disk.
3. Enter the symbolic name of the Super Extract process that will use this record in the SYMBOLIC NAME field.
4. Enter the date, in YYMMDD format, of the next automatic extract to be initiated using this record in the EXTRACT DATE field. If you do not want the Super Extract process to automatically start an extract using this record at some point in the future, you can enter a date in the past in this field. For example, if the current date is December 22, 2001, you could specify 011221 in this field and an automatic extract would not be run.
5. Enter the time, in hhmm format, at which automatic extracts using this record should be run in the EXTRACT TIME field.
6. Determine whether you want continuing, daily automatic extracts performed using this record and enter the appropriate value in the RESTART field. Valid values for this field are as follows:

   Y = Yes, perform daily automatic extracts.
   N = No, do not perform daily automatic extracts.

   Note: The value Y works only when a current or future date is entered in the EXTRACT DATE field. If a past date is entered in the EXTRACT DATE field, daily automatic extracts will not be performed.
7. Enter the value NON in the TAPE LABEL TYPE field. The value in the TAPE LABEL TYPE field is not used for extracts to disk; however, an entry is required in the field. The value NON indicates the tape labels are not used.

You have the option of leaving the default value of IBM in this field, if doing so will not cause confusion when someone reviews this ECF record at a later time.

8. Enter the block size to be used in the TAPE BLOCK SIZE field. For an extract to disk, the TAPE BLOCK SIZE field determines both the block size and the record size. Valid values are as follows:

1–3	=	Record/block size in kilobytes. The Super Extract process multiplies the value entered by 1024 to determine the record and block size in bytes (e.g., 2 kilobytes equals 2048 bytes, and 3 kilobytes equals 3072 bytes). BASE24 automatically displays a K following any entry in kilobytes (e.g., 2K and 3K). The Super Extract process sets the Enscribe record size equal to the value entered, and sets the Enscribe block size to the value entered plus 24 (the additional 24 bytes in each block is used for a 24-byte block header).
33–4072	=	Record/block size in bytes. The Super Extract process sets the Enscribe record size equal to the value entered, and sets the Enscribe block size to the value entered plus 24 (the additional 24 bytes in each block is used for a 24-byte block header).

Note: If an invalid value is entered (4 through 32 or 4073 through 32767), the Super Extract process uses a record size of 4072 bytes and a block size of 4096 bytes.

9. If the extract session being defined includes extracts of shared files, enter the release number identifying the extract format to be used in the RELEASE NUM field. Valid values are as follows:

50 = Use previous release extract formats for extracts of shared files.
60 = Use current release extract formats for extracts of shared files.

Note: The value in this field determines the format of the data in the extract disk file. It does not indicate the release of the BASE24 database.

10. Enter the code identifying the character set to use in data records on the extract disk file in the CHARACTER SET field. Valid codes are as follows:

A = ASCII
E = EBCDIC

11. Enter the name of the disk file where the extract is to be written in the TAPE NAME field.

    Enter at least one letter in this field. BASE24 products take the first letter of the file name (not including the volume or subvolume names) entered in the TAPE NAME field, add a month and day (in MMDD format), and add a 3-digit sequence number to the end of the name. For example, if EXTRA is the disk file name and the date is September 5, BASE24 products attempt to create a file for E0905000. If a complete disk name is not entered, BASE24 products expand this name to a disk file name based on a default volume and subvolume taken from where the network originated.

12. Enter the name of the spooler location to which the Super Extract process should write its reports in the REPORT NAME field.

13. If the extract session will include an extract of the TLF, PTLF, ITLF, TTLF, ILF, or SAF, enter a code indicating the type of numeric field format to use in the extracted data records in the NUMERIC FLD FORMAT field. Valid values are as follows:

    A = The Super Extract process converts binary fields to ASCII display fields.
    B = The Super Extract process leaves binary fields in binary.

    Note: Only the fields in the TLF, PTLF, ITLF, TTLF, ILF, and SAF extract are affected by the setting in this field. Other files included in the extract are extracted in disk file format.

14. If the Super Extract process is to read past the initial end-of-file determined when the extract is first started, set the READ PAST INITIAL EOF field to a value of Y. Otherwise, any new records added to the file after the extract is started will not be included in the extract. Valid values are as follows:

    Y = Yes, the Super Extract process reads past the initially determined end-of-file.
    N = No, the Super Extract process does not read past the initially determined end-of-file.

    This field defaults to a value of N.

    This field must be set to a value of Y in the following conditions to avoid records being missed:

    - If the extract is the final one performed on a transaction log file.

      Note: If multiple extracts are performed each day, this field must be set to a value of N for each extract prior to the final one. However, it must be set to a value of Y for the final extract.

- If only one extract is performed each day.
- If the file is small enough to fit in one block (e.g., in testing situations).

15. Tab to the NEW PAGE field at the bottom of the screen, enter the value 3 in the field, and press the **F7** key. ECF screen 3 will be displayed, as shown below.

```
BASE24-BASE  ECF MESSAGE FILE         LLLL        YY/MM/DD  HH:MM  03 OF 23

              TAG:                         SYMBOLIC NAME:

                          FILE CONFIGURATION

           PRIMARY EXTENT: 00001           FILE CODE: 0000
         SECONDARY EXTENT: 00001      MAXIMUM EXTENTS: 0016
              FILE FORMAT: 1                 BUFFERED: N

                       PARTITION FILE INFORMATION
    PARTITION 1 NAME:                PARTITION 2 NAME:
 PARTITION 1 PRI EXT: 00000     PARTITION 2 PRI EXT: 00000
 PARTITION 1 SEC EXT: 00000     PARTITION 2 SEC EXT: 00000

    PARTITION 3 NAME:                PARTITION 4 NAME:
 PARTITION 3 PRI EXT: 00000     PARTITION 4 PRI EXT: 00000
 PARTITION 3 SEC EXT: 00000     PARTITION 4 SEC EXT: 00000


*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                          F12-HELP
```

16. Enter the size, in pages, of the primary extents for the extract disk file in the PRIMARY EXTENT field. One page is equal to 2048 bytes.

    Note: If the EXTR-DISK-EXTENT-SIZE-MULT param is used, the value of this field is multiplied by the value of the param to obtain the number of primary extents created for the extract.

17. Enter a file code to be assigned to the disk file in the FILE CODE field. Any value can be used to identify and organize disk files except values 100 through 999, which are reserved for use by HP.

18. Enter the size, in pages, of the secondary extents for the extract disk file in the SECONDARY EXTENT field. One page is equal to 2048 bytes.

    Note: If the EXTR-DISK-EXTENT-SIZE-MULT param is used, the value of this field is multiplied by the value of the param to obtain the number of secondary extents created for the extract.

19. If the extract disk file is not to be partitioned, enter the maximum number of extents that can be allocated for the extract disk file in the MAXIMUM EXTENTS field. Together with the primary and secondary extent sizes, this field determines the maximum size of the extract disk file.

    Note: The value in the MAXIMUM EXTENTS field can be exceeded when the EXTR-DISK-EXTENT-SIZE-MULT param is used.

20. Enter a code indicating whether the extract file is created as a Format 1 or Format 2 file in the FILE FORMAT field. Valid values are as follows:

    1 = Create the extract file as a Format 1 file.
    2 = Create the extract file as a Format 2 file.

    This field defaults to a value of 1.

21. Enter a code indicating whether the extract disk file is created as a buffered file in the BUFFERED field. Valid values are as follows:

    Y = Yes, create the extract disk file as a buffered file.
    N = No, do not create the extract disk file as a buffered file.

    Note: When extracting to a buffered disk file, a CPU failure or disk process takeover can cause the loss of data to the extract file. If this occurs, you must re-extract the file.

22. If the extract disk file is to be partitioned, perform the following steps:

    a. For each partition to be used in the file, identify the volume on which to place the partition in a PARTITION *n* NAME field, where *n* ranges consecutively from 1 to 4. The name can be in $*volume* format.

    b. For each partition to be used in the file, specify the primary extent size, in pages, for the partition in the PARTITION *n* PRI EXT field, where *n* ranges consecutively from 1 to 4. One page is equal to 2048 bytes.

    c. For each partition to be used in the file, specify the secondary extent size, in pages, for the partition in the PARTITION *n* SEC EXT field, where *n* ranges consecutively from 1 to 4. One page is equal to 2048 bytes.

23. Press the **F3** key to add the record.

24. Tab to the NEW PAGE field at the bottom of the screen, enter the value 1 in the field, and press the **F7** key. ECF screen 1 will be displayed.

25. Identify the BASE24 files to be included in the extract, then press the **F5** key to update the record. For more information on identifying the files to be included in the extract, refer to the topic “Specifying the Files to be Extracted.”

Specifying the Files to be Extracted

The remaining fields on ECF screens identify the files to be included in the extract. The paragraphs that follow identify various types of fields that appear on the ECF screens and identify the valid values for the fields. For more information on the ECF and its fields, refer to the ***BASE24 Base Files Maintenance Manual***.

File Name Fields — The majority of the fields are flags indicating whether a specific file should be extracted. These fields are named according to the acronym for the file. For example, the ILF field is a flag indicating whether to include the Interchange Log File (ILF) in the extract. Valid values for these fields are as follows:

Y = Yes, extract the file.
N = No, do not extract the file.

DATE OFFSET fields — There is a DATE OFFSET field for each of the following files:

- Interchange Log File (ILF).
- Online Maintenance File (OMF).

  Note: The OMF has two date offset fields—one to indicate where to begin extracting and one to indicate where to end extracting.
- BASE24-atm Transaction Log File (TLF).
- BASE24-atm Hardware Status File (HSF). The HSF is only used with the IBM 3624 version 8 Device Handler when the Hardware Status Error Logging module (ELARG) has been purchased.
- POS Transaction Log File (PTLF).
- Teller Transaction Log File (TTLF).
- Host Mail Box File (HMBF).
- Mail Box File (MBF).
- Update Log File (ULF).
- ITS Transaction Log File (ITLF).

For each of these files, the value in the field specifies the time difference, in number of days, to add to or subtract from the current system date to obtain the date of the file to extract. For reextracts of each file except the ILF, HMBF, and MBF, this field contains the time difference, in number of days, to add to or subtract from the value in the LAST EXTRACT DATE field on ECF screen 1 when determining the date of the file to reextract. For reextracts of the ILF, HMBF, and MBF, the Super Extract process determines the date of the file to reextract in the same manner as for an extract, using the current system date and the DATE OFFSET field in the ECF.

The value entered in this field is added to or subtracted from the current system date to obtain the date of the file to extract. A minus sign (–) indicates the value is subtracted and a plus sign (+) or no sign indicates the value is added. Valid values are –99 to +99 or –99 to 999, depending on whether the plus sign is used.

To perform a partial extract of a current log file (i.e., ILF, TLF, PTLF, TTLF, or ITLF) after logical network cutover and before midnight, the value entered in this field is +1. This happens because the log file's date is increased at logical network cutover while the system date is not increased until midnight. For example, if the value entered in this field is +1 and the current system date is August 25, the date of the log file to extract is August 26.

RELEASE NUM fields — Each product has a product-specific RELEASE NUM field. This field specifies the format of the data placed on the extract tape (not the release of the BASE24 database from which data is being extracted). Valid values are as follows:

11 = Release 1.*x* extract format (BASE24-telebanking only)
50 = Release 5.*x* extract format
60 = Release 6.0 extract format

The Super Extract process supports extracts of all extractable files in release 1.*x* or 5.*x* (previous release format) and release 6.0 (current release format).

Since release number flags are at the product level (such as Base, BASE24-atm, BASE24-pos), all files being extracted in a single extract session for a given product are in the same format. However, files for different products can be extracted in different formats during a single extract session. For example, Base files can be extracted in release 6.0 format in the same session that BASE24-atm files are being extracted in release 5.*x* format.

FORMAT fields — A code that specifies the format of records in the log file extract Each of the following log files has a FORMAT field:

- Interchange Log File (ILF)
- BASE24-atm Transaction Log File (TLF)

- POS Transaction Log File (PTLF)
- Teller Transaction Log File (TTLF)

The ITLF does not have a FORMAT field because variable-length fields in ITLF records cannot be extracted as fixed-length fields. Valid values are as follows:

00 = Fixed format
01 = Variable format

Fixed format means that each record extracted from the transaction log file is extracted for the same length, not including token data. For example, TLF extracts can include four different types of records: financial transaction records (485 bytes), terminal balancing records (856 bytes), terminal cash adjustment records (150 bytes), and terminal settlement records (182 bytes). When the FORMAT field is set to 00 (fixed), all records are extracted for the length of the longest record type—in this case, for 856 bytes, the length of financial transaction records. Other record types are padded with blanks so that the total record length of each record is 856 bytes.

In addition, each record extracted of a given type and subtype contains the same tokens. If a token configured to be extracted from the transaction log file is not present in the transaction log file record, the Super Extract process creates an empty token for the maximum length of the token and includes the empty token in the extract. The Super Extract process also pads variable-length tokens included in the extract to their maximum length. Tokens to be extracted are configured through the Token File (TKN).

Note: If this field contains the value 00 (fixed) and the ORDER FLAG field on TKN screen 3 contains the value N (do not extract in any particular order), the Super Extract process creates an extract record that includes all tokens named in the TKN record in the order that they appear on TKN screen 3.

GROUP NAME fields — The TLF, HMBF, MBF, PTLF, TTLF, ULF, and ITLF each have a GROUP or GROUP NAME field. This field identifies the refresh group for which records are to be extracted. The Super Extract process matches this against the name in the REFRESH GROUP field on screen 1 of the IDF.

For the transaction log files (i.e., the TLF, PTLF, TTLF, and ITLF), the value specified in this field affects not only what records are extracted, but also the way in which extracts and reextracts are performed.

When the value ALL is specified in this field, the Super Extract process can perform multiple file extracts during a single day without extracting all of the records in the file every time. Reextracts extract a specified subset of the records in the file.

When any other value (i.e., any other refresh group) is specified in this field, the Super Extract process extracts only those records associated with the card-issuing or account-owning financial institutions belonging to the refresh group. A subsequent extract or reextract extracts the same records, plus any new records for institutions in the refresh group logged to the file since the first extract was performed.

For the TLF, TTLF, and ITLF, the Super Extract process uses one FIID value when determining whether to extract a record. For the PTLF, however, the Super Extract process uses two FIID values to make this determination. The Super Extract process compares the card-issuing financial institution FIID and terminal-owning financial institution FIID from each PTLF record with the FIID in the IDF record that is associated with the refresh group. A PTLF record is extracted if either FIID it contains matches the FIID in the IDF record associated with the refresh group. If a BASE24-pos Terminal Data Files (PTD) record is available for the terminal, the terminal-owning financial institution FIID in the PTLF record is taken from the FIID field on PTD screen 1. If a PTD record is unavailable (i.e., CRT Authorization is used), the terminal-owning financial institution FIID in the PTLF record is taken from the FIID field on POS Retailer Definition File (PRDF) screen 1.

Token File

A token is a collection of related data required to perform a specific function. As transactions are processed and additional information is required, BASE24 processes add tokens to the internal message. The Token File (TKN) is used to configure the following token processing:

- The tokens that get logged to the transaction log files. The transaction log files affected are the following:
  - BASE24-atm Transaction Log File (TLF)
  - POS Transaction Log File (PTLF)
  - Interchange Log Files (ILFs)
  - ITS Transaction Log File (ITLF)
  - Teller Transaction Log File (TTLF)

- The tokens that get extracted from the transaction log files by the Super Extract process. Extracting allows the token information to be processed by the host.
- The tokens that get sent to the host with each transaction in the ISO-based external message. The ANSI-based external message does not support message tokens.

For more information on configuring the tokens that get logged, extracted, and sent in the external message, refer to the ***BASE24 Tokens Manual***.

Network Control Commands

An extract is started by one of two events: the Super Extract process determining that an extract timer has expired or the network operator issuing a command to the Super Extract process. When the network operator initiates an extract, the command can be issued either from the Extract screens or from a network control facility. The extract message identifies the ECF record to use in performing the extract, and includes some configuration fields that can be used to override information in the ECF.

The network operator can issue the following commands to the Super Extract process:

- 9501
- EXTR
- REEX
- STATUS
- TLFRSET
- TRACE

For more information on issuing these commands to the Super Extract process, refer to the procedures in section 4. For more information on the processing the Super Extract process performs when it receives one of these commands, refer to the ***BASE24 Text Command Reference Manual***.

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3: Refresh Operator Procedures

Network operators can control all Refresh processes from a set of function-key driven screens or with commands entered from a network control facility. The following Refresh processes can be controlled with these screens and commands:

- The Base National Negative Refresh process is an Enscribe process that is used to refresh the Base national negative files used by the BASE24-pos product.
- The INAS National Negative Refresh process is an Enscribe process that is used to refresh the INAS national negative files used by the BASE24-pos product.
- The Enscribe Refresh process is used to refresh Enscribe files other than national negative files.
- The SQL Refresh process is used to refresh SQL tables.

This section provides an overview of the text commands that can be sent to the Refresh processes, and provides procedures for using the Refresh screens.

Caution: BASE24 Remote Banking applications currently require system downtime in order to perform the full-file refresh of an SQL table. This is because the name of an SQL table cannot be changed without recompiling the processes that use the table. A full-file refresh is recommended only at system installation for SQL tables that can be refreshed, which include the following:

- Customer Table (CSTT)
- Customer/Account Relation Table (CACT)
- Customer/Personal ID Relation Table (CPIT)
- Personal Information Table (PIT)

Using Network Control Text Commands

A network control facility allows the network operator to monitor and control the network from a single screen. Operators enter commands from the network control facility using defined syntax. In most cases, the commands entered from the network control facility are processed by the XPNET process; however, one network control command allows messages to be sent to any application process running in the network.

The command you use depends on the release of the network under which the Refresh process is running. The command used to send a text string to an application process is the DELIVER PROCESS command. The command syntax illustrations and examples provided in this manual use quotation marks as text delimiters for consistency and readability. Refer to the ***BASE24-XPNET Network Control Command Reference Manual*** for the procedures involved in sending DELIVER PROCESS commands, including the other delimiter options that are available.

Like all network control commands, DELIVER PROCESS commands must be entered using defined syntax. Each process recognizes and can act upon a different set of DELIVER PROCESS commands. The DELIVER PROCESS commands that can be sent to the Refresh process are defined on the following pages.

Refer to the “Conventions Used in this Manual” topic that follows the preface of this manual for the notation conventions necessary to understand the network control commands.

CONTINUE Command

The CONTINUE command directs an Enscribe Refresh process to continue processing, even though at least one parity error has been encountered. Depending on the type of refresh being performed, the Refresh process logs a message at various points indicating the number of parity errors encountered and asking whether to continue processing. This command is used in response to those event messages, to direct the Refresh process to continue processing. The Base National Negative, INAS National Negative, and SQL Refresh processes do not support this command.

Syntax

DELIVER PROCESS *name,* “CONTINUE”

name — The symbolic name assigned to the Refresh process.

REFRESH Command

A general use operator command that starts a Base National Negative Refresh process, Enscribe Refresh process, INAS National Negative Refresh process, or SQL Refresh process.

Note: The *label-type* and *character-set* fields are required.

Syntax

DELIVER PROCESS *name*, "REFRESH *label-type***,** *input-file-name*,
[*partition-number*], [*bypass-number*], [*mail-name*], [*mail-ID*],
[*mail-group*], [*atm-impact-start-date/time*],
[*pos-impact-start-date/time*], [*teller-impact-start-date/time*],
[*telebanking-impact-start-date/time*], [*character-set*],
[*mount-message*], [*file-ID*], [[*volume-ID*],]..."

name — The symbolic name assigned to the Refresh process.

label-type — Specifies the tape label type for the refresh tape. If this field and the actual tape label do not match, the Refresh process stops processing. Valid values are as follows:

IBM	=	IBM OS
DOS	=	IBM DOS
BLS	=	Burroughs Large Scale
BUR	=	Burroughs 5500
ANS	=	ANSI
NON	=	No label processing

input-file-name — The HP NonStop file name of the tape device on which the refresh tape is mounted.

This field can also contain the name of a disk file, if disk file input is being used. Disk file input is used primarily when disk files can be transferred from the host to the HP NonStop processor.

[*partition-number*] — The specific partition of a partitioned file to be refreshed during this operation. Valid values are 0 through 15. Spaces or AL signify all partitions.

[*bypass-number*] — The number of tape parity errors that can be encountered without causing the Refresh process to abort. The default is 0. Valid values are as follows:

-1	=	Bypass all tape parity errors.
0–32767	=	Bypass the number of tape parity errors specified.

An entry in this field overrides the value in the REF-BYPASS-PARITY-ERRORS param in the Logical Network Configuration File (LCONF). This field is only used for tape input.

[*mail-name*] — The name of the Mail process, for MBF refreshes. During an MBF refresh, the Refresh process creates mail messages from the information on the refresh tape, and sends the messages to the process named in this field. The Mail process writes the messages to the MBF.

[*mail-ID*] — The mail FIID. This field is not used.

[*mail-group*] — The Mail refresh group. This field is only used for previous-release MBF refreshes.

[*atm-impact-start-date/time*] — The impact starting date and time for BASE24-atm transactions, in the format YYYYMMDDHH mmssmmmmmm (milliseconds are optional). An entry in this field overrides the values in the ATM.IMP-STRT-DAT and ATM.IMP-STRT-TIM fields in the Refresh File Header (RFH).

[*pos-impact-start-date/time*] — The impact starting date and time for BASE24-pos transactions, in the format YYYYMMDDHH mmssmmmmmm (milliseconds are optional). An entry in this field overrides the values in the POS.IMP-STRT-DAT and POS.IMP-STRT-TIM fields in the Refresh File Header (RFH).

[*teller-impact-start-date/time*] — The impact starting date and time for BASE24-teller transactions, in the format YYYYMMDDHH mmssmmmmmm (milliseconds are optional). An entry in this field overrides the values in the TLR.IMP-STRT-DAT and TLR.IMP-STRT-TIM fields in the Refresh File Header (RFH).

[*telebanking-impact-start-date/time*] — The impact starting date and time for BASE24-telebanking transactions, in the format YYYYMMDDHH mmssmmmmmm (milliseconds are optional). An entry in this field overrides the values in the TB.IMP-STRT-DAT and TB.IMP-STRT-TIM fields in the Refresh File Header (RFH).

[*character-set*] — Indicates whether the form of the data on the refresh tape is ASCII or EBCDIC. Valid values are as follows:

A = ASCII
E = EBCDIC (default)

[*mount-message*] — A message to be displayed along with the system mount message when the Refresh process requests use of a tape drive. This field can be any alphanumeric message of 25 characters or less surrounded by single quotes. The default message is MOUNT REFRESH TAPE. This field is not used for refreshes from disk.

[*file-ID*] — The name of the file on the refresh tape. This field is not used for refreshes from disk.

This field should contain the value of the Data Set ID field in the IBM header labels. This field is only used when the tape label type is IBM and the BASE24-TAPE-LBL-PROC-USED param in the LCONF is set to N (Not used with disk input, TAPECOM is processing the labels).

[[*volume-ID*],]... — The volume identifiers for each reel to be used in the refresh. Volume identifiers are required when the tape label type is IBM or ANSI. Up to 35 volume identifiers can be entered, each separated from the previous volume ID by a comma. This field is not used for refreshes from disk.

Note: The bypass number, mount message, file ID, and volume IDs are only used with tape input. The file ID and volume IDs are only used with TAPECOM processing the tape labels.

When optional items are included in the REFRESH command, the command might be too long to be entered directly in the NCS COMMAND field on the Network Control Supervisor (NCS) screen. If the REFRESH command exceeds 139 characters, it can be placed in an Obey file and then issued to the Refresh process using the OBEY command or from the Refresh screen. For more information on the OBEY command, refer to the command-specific documentation in the ***NET24-XPNET Network Control Command Reference Manual***.

RESUME Command

The RESUME command directs the Enscribe Refresh process to resume processing a previously suspended refresh. This command can also be used to resume processing of an abended refresh of the MBF or SDF. The Base National Negative, INAS National Negative, and SQL Refresh processes do not support this command.

Syntax

DELIVER PROCESS *name*, "RESUME *tape-date*, *tape-time*, *refresh-group*, *application-id*, [*partition-number*]"

name — The symbolic name assigned to the Refresh process.

tape-date — The date that the refresh input file was created, in YYYYMMDD format.

tape-time — The time that the refresh input file was created, in hhmm format.

refresh-group — The refresh group.

application-id — A code indicating the type of file being refreshed at the time the Refresh process aborted or suspended processing.

Valid values are as follows:

CC = PBF for credit accounts only
CF = Cardholder Authorization File (CAF)
CH = Card History File (CHF)
CI = Customer/Card Information File (CCIF)
CM = Customer/Card Memo File (CCMF)
CO = Corporate Check File (CCF)
CS = Check Status File (CSF)
DA = PBF for checking or NOW accounts only
MB = Mail Box File (MBF)
NB = No Book File (NBF)
NF = Negative Card File (NEG)
PF = PBF for all account types
SD = Switch Dispute File (SDF)
SM = Scoring Engine Master File (SEMF)
SP = Stop Payment File (SPF)
SV = PBF for savings accounts only
TH = Transaction History File (THF)
WH = Warning/Hold/Float File (WHFF)

partition-number — The specific partition of a partitioned file being refreshed at the time the refresh was suspended. Valid values are 0 through 15 and ALL.

STATUS Command

The STATUS command requests status information from the Base National Negative, INAS National Negative, Enscribe, or SQL Refresh process.

Syntax

DELIVER PROCESS *name,* “STATUS”

name — The symbolic name assigned to the Refresh process.

For more information on the STATUS command and for a description of the STATUS command output, refer to the ***BASE24 Text Command Reference Manual.***

STOP Command

The STOP command directs the Base National Negative, INAS National Negative, Enscribe, or SQL Refresh process to abort and log a message that processing has been terminated.

Syntax

DELIVER PROCESS *name,* “STOP”

name — The symbolic name assigned to the Refresh process.

SUSPEND Command

The SUSPEND command directs the Enscribe Refresh process to suspend processing at the next logical point. The Base National Negative, INAS National Negative, and SQL Enscribe Refresh processes do not support this command.

Syntax

DELIVER PROCESS *name,* “SUSPEND”

name — The symbolic name assigned to the Refresh process.

Using the Refresh Screens

The Refresh screens provide for direct initiation of file refreshes from a series of function key-driven screens. There are three Refresh screens, accessible from the Operator Product Menu in the BASE24 CRT access system. Currently, screens 1 and 3 are used to initiate refreshes. The remaining screen is reserved for future use.

When the operator presses certain function keys from one of the Refresh screens, the Refresh Requester process builds a command message from the information entered on the screen and sends the message to the Refresh process. Depending on the function key pressed, the message created by the Refresh Requester process matches one of the text commands that the operator can issue from the network control facility. The text commands, and their corresponding Refresh screen function keys, are as follows:

Text Command Name	Function Key	Supported by Refresh Processes
REFRESH	**F2**	Base National Negative, INAS National Negative, Enscribe, SQL
STATUS	**F3**	Base National Negative, INAS National Negative, Enscribe, SQL
SUSPEND	**F4**	Enscribe
RESUME	**F5**	Enscribe

Other function keys can also be used on the Refresh screens. These function keys perform standard CRT access functions, as described below.

Key	Description
F1	**Validate Data —** Checks the data entered on the screen for errors, and enters the data into the system.
F7	**Go to New Page —** Displays a different screen in the same file in which the operator is working. The screen to be displayed must be identified in the NEW PAGE field at the bottom of the screen.

Key	Description
F8	**Clear Screen —** Clears any values on the Refresh screens to spaces or replaces them with default values. This key impacts all Refresh screens. This includes the current screen, any previous screens, and any remaining screens.
F9	**Display Next Screen —** Displays the next Refresh screen. If the operator is on the last Refresh screen, this function key takes the operator to the first screen.
F10	**Print Screen —** Sends the screen currently being displayed to the spooler location indicated on the Logon screen. The Logon screen is explained in the ***BASE24 CRT Access Manual***.
F11	**Display Previous Screen —** Displays the previous Refresh screen. If the operator is on the first Refresh screen, this function key takes the operator to the last Refresh screen.
F12	**Display Help Screen —** Displays the Help screen. The Help screen displayed depends on the screen currently being viewed. The Help screen contains information about BASE24 function keys or menu options.
F13	**Change Logical Network —** Changes the current logical network while on the Refresh screens.
F16	**Exit or Go to File —** This key has several purposes. It exits BASE24 or the screens currently being accessed. It also allows the operator to move between logical networks and files. This functionality is explained in the ***BASE24 CRT Access Manual***. This function key is operable on the Help screen for files to enable the operator to change files or logical networks.
Shift-F16	**Log Off BASE24 —** Logs the operator off BASE24. When this function is used while an operator is accessing BASE24, a blank Logon screen is displayed. If the Logon screen is displayed when this function is used, the Logon screen continues to be displayed or a TACL prompt is displayed, depending upon how the terminal is set up.

Refresh Screen 1

Refresh screen 1 is used to initiate file refreshes. Refresh screen 1 is shown below, followed by descriptions of its fields.

```
 BASE24-BASE  REFRESH SCREENS          LLLL        YY/MM/DD  HH:MM  01 OF 03
  SYMBOLIC NAME:
           USER:                     PASSWORD:

   TAPE LABEL TYPE:  IBM  (IBM/DOS/BUR/BLS/ANS/NON)
   INPUT FILE NAME:  $TAPE
     CHARACTER SET:  E  (E/A)     PARTITION NUM:  __     BYPASS NUM: _____
     MOUNT MESSAGE:  _________________________   FILE ID:  _________________
VOLUME IDENTIFIERS:  ______  ______  ______  ______  ______  ______  ______
                     ______  ______  ______  ______  ______  ______  ______
                     ______  ______  ______  ______  ______  ______  ______
                     ______  ______  ______  ______  ______  ______  ______
                     ______  ______  ______  ______  ______  ______  ______
           IMPACT START DATE AND TIME (YYYYMMDD HHMMSSMMMMMM)
      ATM: ________ ____________     POS: ________ ____________
      TLR: ________ ____________      TB: ________ ____________
         THE FOLLOWING FIELDS ARE SPECIFIC TO THE RESUME COMMAND
    GROUP:  ________________________________  APPLICATION ID:  __
        TAPE DATE:  ________ (YYYYMMDD)        TAPE TIME:  ____ (HHMM)

 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
 F1-VALIDATE  F2-REFRESH  F3-STATUS  F4-SUSPEND  F5-RESUME  F12-HELP
```

SYMBOLIC NAME — The symbolic name identifying the Refresh process to be used. Symbolic names are defined as each process is added to the XPNET node.

USER — The name that uniquely identifies you to the BASE24 CRT access system. This name must match your alias in the Security File (SEC) or access to the refresh screens is denied.

Normally, your supervisor gives you your user name. You must have security clearance under an identical user name for all logical networks you want to access, if you want to access a different logical network without logging on again.

Field Length:	1–16 alphanumeric characters
Required Field:	Yes, if network control security is enabled for your installation (i.e., the ENABLE-SECURITY param for the NCPI Server process is set to a value of ON).
Default Value:	None

PASSWORD — Your password. The BASE24 CRT access system requires each user to select a confidential password to ensure that user names are not employed by unauthorized individuals. Each user name and password combination must be unique within the BASE24 CRT access system.

Field Length:	1–16 alphanumeric characters
Required Field:	Yes, if network control security is enabled for your installation (i.e., the ENABLE-SECURITY param for the NCPI Server process is set to a value of ON).
Default Value:	None

TAPE LABEL TYPE — A code that specifies the tape label type for the refresh tape. If this field and the actual tape label do not match, the refresh will not run. Valid values for this field depend on whether the HP NonStop Kernel TAPECOM product or the Refresh process is processing tape labels. Valid values are as follows:

ANS = ANSI
BLS = Burroughs Large Scale
BUR = Burroughs 5500
DOS = IBM DOS
IBM = IBM/MVS
NON = Unlabeled

If a tape label type is not specified, the field defaults to IBM.

Note: If the HP NonStop Kernel TAPECOM product is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value N) and this field indicates a tape label type not supported by TAPECOM (DOS, BUR, or BLS), the Refresh process performs tape label processing. If this field contains the value NON, tape labels are not processed.

INPUT FILE NAME — The HP NonStop file name of the tape device on which the refresh tape is mounted. If not entered, the value in this field defaults to $TAPE.

This field can also contain the name of a disk file if disk file input is being used. Disk file input is used primarily when disk files can be transferred from the host to the HP NonStop processor.

CHARACTER SET — Indicates whether the data in the refresh input file is ASCII or EBCDIC. Valid values are as follows:

A = ASCII
E = EBCDIC

The default value for this field is E.

PARTITION NUM — The specific partition of a partitioned file to be refreshed during this operation. Valid values are as follows:

AL = All partitions for Enscribe files or SQL tables
0–15 = Number of partitions for Enscribe files
0–99 = Number of partitions for SQL tables

BYPASS NUM — The number of tape parity errors that can be encountered without causing the Refresh process to abort. Valid entries are as follows:

-1 = Bypass all tape parity errors
0–32767 = Bypass the number of tape parity errors indicated

An entry in this field overrides the value in the REF-BYPASS-PARITY-ERRORS param in the LCONF.

MOUNT MESSAGE — A message to be displayed along with the system mount message when the Refresh process requests use of the tape drive. This field is used only for refreshes from tape when the HP NonStop Kernel TAPECOM product is running on the system.

FILE ID — The name of the file on the refresh tape. This field is only used if the HP NonStop Kernel TAPECOM product is performing tape label processing (i.e., when the LCONF param BASE24-TAPE-LBL-PROC-USED is set to a value of N and the label type used is IBM). This field is not used for refreshes from disk.

VOLUME IDENTIFIERS — The volume identifiers for each reel to be used in the refresh. This field is only used for refreshes from tape when the HP NonStop Kernel TAPECOM product is processing tape labels. Volume identifiers are required when the tape label type is IBM or ANSI. Up to 35 volume identifiers can be entered.

IMPACT START DATE AND TIME

The following fields contain the starting date and time to be used for impacting. Only the Enscribe Refresh process performs impacting.

Normally, the Refresh process uses all records in the transaction log file for impacting. When institutions perform a partial extract of the transaction log file, however, there may be records in the transaction log file that should not be used for impacting. These records can be bypassed by specifying an impact start date and time in the refresh file header (RFH) or in these fields.

If a start date and time is specified, the Refresh process locates the first transaction log file record that has an entry timestamp greater than the specified start date and time. This timestamp is the time the record was logged to the transaction log file. This record and all subsequent records are used for impacting.

ATM — The starting date and time to be used for CAF or PBF impacting for the BASE24-atm product. The Refresh process uses this field to select Transaction Log File (TLF) records for CAF or PBF impacting. Impacting is restricted to those transactions logged to the TLF after the date and time specified in this field.

An entry in this field overrides the values in the ATM.IMP-STRT-DAT and ATM.IMP-STRT-TIM fields in the RFH. If the refresh input file contains records for multiple files, the value in this field overrides the RFH for the first file in the refresh input file only.

POS — The starting date and time to be used for PBF impacting for the BASE24-pos product. The Refresh process uses this field to select POS Transaction Log File (PTLF) records for PBF impacting. Impacting is restricted to those transactions logged to the PTLF after the date and time specified in this field.

An entry in this field overrides the values in the POS.IMP-STRT-DAT and POS.IMP-STRT-TIM fields in the RFH. If the refresh input file contains records for multiple files, the value in this field overrides the RFH for the first file in the refresh input file only.

TLR — The starting date and time to be used for NBF, PBF, SPF, or WHFF impacting for the BASE24-teller product. The Refresh process uses this field to select Teller Transaction Log File (TTLF) records for impacting. Impacting is restricted to those transactions logged to the TTLF after the date and time specified in this field.

An entry in this field overrides the values in the TLR.IMP-STRT-DAT and TLR.IMP-STRT-TIM fields in the RFH. If the refresh input file contains records for multiple files, the value in this field overrides the RFH for the first file in the refresh input file only.

TB — The starting date and time to be used for PBF impacting for the BASE24-telebanking product. The Refresh process uses this field to select ITS Transaction Log File (ITLF) records for impacting. Impacting is restricted to those transactions logged to the ITLF after the date and time specified in this field.

An entry in this field overrides the values in the TB.IMP-STRT-DAT and TB.IMP-STRT-TIM fields in the RFH. If the refresh input file contains records for multiple files, the value in this field overrides the RFH for the first file in the refresh input file only.

THE FOLLOWING FIELDS ARE SPECIFIC TO THE RESUME COMMAND

The following fields are required to resume a previously suspended refresh. The Enscribe Refresh process uses these fields. The Base National Negative, INAS National Negative, and SQL Refresh processes do not use these fields.

REFRESH GROUP — The refresh group being processed at the time the Refresh process aborted or suspended processing. This field is used to read the Refresh Group Control File (RGCF) record when resuming a refresh. The value entered in this field should match the value from the refresh input file.

APPLICATION ID — A code indicating the type of file being refreshed at the time the Refresh process aborted or suspended processing. Valid values are as follows:

CC = PBF for credit accounts only
CF = Cardholder Authorization File (CAF)
CH = Card History File (CHF)
CI = Customer/Card Information File (CCIF)
CM = Customer/Card Memo File (CCMF)
CO = Corporate Check File (CCF)
CS = Check Status File (CSF)
DA = PBF for checking or NOW accounts only
MB = Mail Box File (MBF)
NB = No Book File (NBF)

NF = Negative Card File (NEG)
PF = PBF for all account types
SD = Switch Dispute File (SDF)
SM = Scoring Engine Master File (SEMF)
SP = Stop Payment File (SPF)
SV = PBF for savings accounts only
TH = Transaction History File (THF)
WH = Warning/Hold/Float File (WHFF)

TAPE DATE — The date, in YYYYMMDD format, that the refresh input file was created.

TAPE TIME — The time, in hhmm format, that the refresh input file was created.

Refresh Screen 3

Refresh screen 3 is used to initiate BASE24-mail refreshes. Refresh screen 3 is shown below, followed by descriptions of its fields.

```
BASE24-MAIL  REFRESH SCREENS          LLLL        YY/MM/DD  HH:MM  03 OF 03
 SYMBOLIC NAME:
          USER:                     PASSWORD:



                          B A S E 2 4 - M A I L
                        MAILBOX:
                           FIID:
                  REFRESH GROUP:







*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F1-VALIDATE  F2-REFRESH  F3-STATUS  F4-SUSPEND  F5-RESUME  F12-HELP
```

MAILBOX — The symbolic name of the Mail process to be used. Symbolic names are defined as each process is added to the XPNET node.

FIID — The FIID of the financial institution whose Host Mail Box File (HMBF) or Mail Box File (MBF) is to be refreshed.

REFRESH GROUP — An identifier specifying a group of institutions for the refresh.

Allowing Unsorted Data to be Loaded by Refresh

The Refresh process allows unsorted data input from the information provided by the CAF or PBF only during a full-file or partial-file refresh.

Within the LCONF, two params are used to accept and load unsorted data.

- REFR-INPUT-DATA-UNSORTED
- REFR-MAX-RECS-UNSORTED

The REFR-INPUT-DATA-UNSORTED param allows unsorted data input. If this param is not present or contains invalid values, the default is for the option to be off as the Refresh process expects the input data to be sorted. Although the Refresh process runs slower if this param is present and set to yes, flexibility is provided for allowing unsorted data to be loaded from a file.

The REFR-MAX-RECS-UNSORTED param provides the number of unsorted data records passed from the Refresh process to FUP in the LOAD command. When FUP is used to load unsorted data records to a file, it creates a temporary scratch file. To ensure that the file is large enough for the data, the Refresh process sets a maximum number of records option in the FUP startup command so that it can calculate the correct size scratch file. The number of unsorted records refreshed must be monitored and updated as required.

Performing a Full-File Refresh of an SQL Table

BASE24 Remote Banking applications currently require system downtime in order to perform a full-file refresh of an SQL table. This is because the name of an SQL table cannot be changed without recompiling the processes that use the table. A full-file refresh is recommended only at system installation for SQL tables that can be refreshed, which include the following:

- Customer Table (CSTT)
- Customer/Account Relation Table (CACT)
- Customer/Personal ID Relation Table (CPIT)
- Personal Information Table (PIT)

When you perform a full-file refresh of an SQL table, you must perform the following steps in addition to the steps for performing a refresh from tape or disk that are presented later in this section. These limits and additional processing steps do not apply to partial-file refreshes of SQL tables.

Before Performing the Full-File Refresh

You must perform the following steps to create empty SQL tables without indexes and re-SQL compile the SQL Refresh process before you perform a full-file refresh. The SQL Refresh process can be compiled prior to creating these tables, but must be re-SQL compiled after the new tables have been created.

Note: The user ID of the operator who started the network must own the SQL catalog and tables being refreshed at this time.

1. Verify the following Logical Network Configuration File (LCONF) assign and param settings for accuracy, based on the table being refreshed:

Table	Assigns	Params
All tables	REF-SOURCEDICT	REFR-INPUT-BLK-DISK or REFR-INPUT-BLK-TAPE
CACT	CACT-PART*nn*	CACT-PRI-PARTONLY
CPIT	CPIT-PART*nn*	CPIT-PRI-PARTONLY

Table	Assigns	Params
CSTT	CSTT-PART*nn*	CSTT-PRI-PARTONLY
PIT	PIT-PART*nn*	PIT-PRI-PARTONLY

When a table is not partitioned, the PART00 assign identifies the location of the table. A value of Y should be used for the PRI-PARTONLY param when a table is not partitioned.

When a table is partitioned and all partitions are being refreshed, the PART00 assign identifies the location of the primary partition. A value of N for the PRI-PARTONLY param indicates that all partitions are being refreshed.

When a table is partitioned and a single partition is refreshed separately, the PART*nn* assign identifies the location of the partition being refreshed. Valid values are 00 through 99 for a single partition. A value of Y for the PRI-PARTONLY param is required when only partition 0 is being refreshed. A value of N for the PRI-PARTONLY param is required when a partition other than partition 0 is being refreshed. The value in the PART-NUM field in the refresh file header (RFH) or the PARTITION NUM field on Refresh screen 1 must match the value in the *nn* variable of the PART*nn* assign.

2. Verify that an SQL catalog exists for the logical network. The SQL catalog, which is required to perform the refresh, is a subvolume identified by the CTLG macro in the MAKE files listed in step 4. Use the SQL CREATE CATALOG command to create an SQL catalog at this time if one does not exist.

3. Verify that an ALLMACS define exists and identifies the correct macro overrides file. This define is required before MAKE files can be executed. Enter the following command at a TACL prompt to display current ALLMACS define details:

   ```
   INFO DEFINE =ALLMACS
   ```

   If an ALLMACS define does not exist or contains an invalid value, enter the following commands at a TACL prompt:

   ```
   PARAM MAX-RADIX-TBL-SIZE 50000000
   DELETE DEFINE =ALLMACS
   ADD DEFINE =ALLMACS, FILE $volume.ACILIBI.CUSTMACS
   ```

   For detailed information on the ACI MAKE utility, refer to the ***ACI Make Release 5 User Guide***.

4. Verify that the MAKE files you are going to use contain the correct values for each of the macros for the table being refreshed, as shown below. These MAKE files are located on the OC*xx*CMP subvolume, where *xx* is the number of the current release.

Table Being Refreshed	MAKE File	Macros
Customer Table (CSTT)	CSTTRM or ALLTBRM*	data_cstt ctlg
Customer/Account Relation Table (CACT)	CACTRM or ALLTBRM*	data_cact data_xcact01 data_xcact02 ctlg
Customer/Personal ID Relation Table (CPIT)	CPITRM or ALLTBRM*	data_cpit data_xcpit01 ctlg
Personal Information Table (PIT)	PITRM or ALLTBRM*	data_pit data_xpit01 data_xpit02 ctlg

* The ALLTBRM MAKE file is a combination of the CSTTRM, CACTRM, CPITRM, and PITRM MAKE files, as well as MAKE files for the Customer Allowed Transaction Table (CATT) and the Transaction Type By Source Code Table (TTSC). You can use the ALLTBRM MAKE file or the MAKE file for each table that you plan to refresh.

5. Enter the following command at a TACL prompt to create the new tables that you plan to refresh:

```
MAKEV202/OUT $S.#MAKE.x,NAME $MAKE,NOWAIT/ x -FORCE
-LISTREPLACE
```

   The *x* variable in the MAKE command is the name of the MAKE file being used (i.e., ALLTBRM, CSTTRM, CACTRM, CPITRM, or PITRM).

6. Log on as the operator of the network.

7. Enter the following command at the TACL prompt to start an SQL conversational interface (SQLCI) session:

```
SQLCI
```

8. Enter the commands from the following list at the SQLCI prompt that correspond to the indexes for the SQL tables that you created in step 5. You must delete the indexes that were created with the tables in step 5 before the tables can be refreshed, then replace the indexes after the tables have been refreshed.

    ```
    DROP INDEX $volume.subvolume.XCACT01;
    DROP INDEX $volume.subvolume.XCACT02;
    DROP INDEX $volume.subvolume.XCPIT01;
    DROP INDEX $volume.subvolume.XPIT01;
    DROP INDEX $volume.subvolume.XPIT02;
    ```

9. Stop the SQL conversational interface session by entering the following command at the SQLCI prompt:

    ```
    EXIT;
    ```

10. Verify that the REFRCMP and DEFADD files on the OC*xx*CMP subvolume contain the correct values for your system. The REFRCMP file is an obey file that SQL compiles the SQL Refresh process. The DEFADD file sets defines and is executed by the REFRCMP file.

11. Re-SQL compile the SQL Refresh process by entering the following command at the TACL prompt:

    ```
    O REFRCMP
    ```

12. Perform the full-file refresh.

After Performing the Refresh

You must perform the following steps to add indexes to SQL tables immediately after you perform the full-file refresh:

1. Log on as the operator of the network.

2. Enter the following command at the TACL prompt to start an SQL conversational interface (SQLCI) session:

    ```
    SQLCI
    ```

3. Enter the SQL commands for the indexes that you need to add. These commands appear at the end of the OC*xx*CUST.CACTRS, OC*xx*PSNL.CPITRS, and OC*xx*PNSL.PITRS files, where *xx* is the number of the current release. For example, the SQL commands in the CACTRS file for index XCACT01 are as follows:

```
CREATE UNIQUE INDEX =DATA_XCACT01 ON =DATA_CACT
(
ACCT ASCENDING,
ACCT_TYP ASCENDING
)

KEYTAG "AN"
EXTENT (100,100)
MAXEXTENTS 100;
```

4. If a significant number of rows were added or deleted from a refreshed table, you may want to enter the UPDATE STATISTICS command at the SQLCI prompt to optimize query performance on the table. If you do enter this command, you must then re-SQL compile all BASE24-telebanking and BASE24-billpay modules that access the table. Obey files and TACL macros for SQL compiling these modules are located on the OC*xx*CMP subvolume, where *xx* is the number of the current release.

   For more information on using the UPDATE STATISTICS command, refer to HP's ***NonStop SQL Reference Manual***.

5. Stop the SQL conversational interface session by entering the following command at the SQLCI prompt:

```
EXIT;
```

Performing a Refresh from Tape

The following are operator procedures for performing a file refresh from tape using the Refresh screen. These instructions assume that you are logged on to BASE24 and apply to all Refresh processes. Refer to the "Performing a Full-File Refresh of an SQL Table" topic in this section for additional steps necessary for such a refresh.

1. Mount the tape from the host on the HP NonStop processor tape drive.
2. Enter REFR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Refresh screen 1 will be displayed.
3. Enter the symbolic name of the Refresh process in the SYMBOLIC NAME field.
4. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
5. If the refresh is from tape, enter the tape label type in the TAPE LABEL TYPE field. Valid values for this field depend on whether the HP NonStop Kernel TAPECOM product or the Refresh process is processing tape labels.

   If the HP NonStop Kernel TAPECOM product is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value N), valid tape label types are as follows:

   IBM = IBM/MVS
   ANS = ANSI
   NON = Unlabeled

   If the Refresh process is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value Y), valid tape label types are as follows:

   IBM = IBM/MVS
   DOS = IBM DOS
   BUR = Burroughs 5500
   BLS = Burroughs Large Scale
   NON = Unlabeled

   The default is IBM if no tape label type is entered.
6. Enter the tape drive name in the INPUT FILE NAME field.

   The default is $TAPE if no tape drive name is entered.

7. Identify the character set of the data on the tape in the CHARACTER SET field. Valid values are as follows:

   E = EBCDIC
   A = ASCII

8. Enter the number of parity errors to bypass in the BYPASS NUM field (if applicable).

   This field can be used to override the number of parity errors to bypass specified in REF-BYPASS-PARITY-ERRORS param in the LCONF. Use of this field is optional.

   If no value is entered in this field or in the LCONF, the default is 1.

9. If the HP NonStop Kernel TAPECOM product is running on the system and an additional message is desired with the system mount message, enter the additional message in the MOUNT MESSAGE field. The information in this field is displayed along with the system mount message when the Refresh process requests use of the tape drive.

10. If the HP NonStop Kernel TAPECOM product is processing tape labels (i.e., the LCONF param BASE24-TAPE-LBL-PROC-USED is set to a value of N) and the tape label type is IBM, enter the tape file name in the FILE ID field.

11. If the HP NonStop Kernel TAPECOM product is processing tape labels (i.e., the LCONF param BASE24-TAPE-LBL-PROC-USED is set to a value of N) and the tape label type is IBM or ANSI, enter the volume identifier for each reel to be used in the extract in the VOLUME IDENTIFIERS fields. When there is more than one volume identifier, enter the volume IDs in order from left to right, with no blank fields between volume IDs.

12. Press the **F2** key to initiate the refresh.

13. Check the network log.

    Any error messages are displayed on the network log (using routing code 0). If there were no error conditions, a “REFRESH COMPLETE” message is generated; a message stating the percentage of file full is also displayed.

14. Review the refresh report.

    After the refresh completes, a hard copy report is sent to the spooler location identified in the appropriate LCONF assign (i.e., BP-RPTR, RPTR, or REF-RPT-SQL). For more information on the refresh report, refer to the “Verifying Reports” topic at the end of this section.

Performing a Refresh from Disk

The procedures for performing a refresh from disk are basically the same as those for refreshing from tape. The following are operator procedures for performing file refreshes from disk using the Refresh screen. These instructions assume that you are logged on to BASE24 and apply to all Refresh processes. Refer to the “Performing a Full-File Refresh of an SQL Table” topic earlier in this section for additional steps necessary for such a refresh.

1. Verify that the refresh input file is present on the disk. The input file must meet the requirements presented in the “Disk Input File Organization and Content” discussion in appendix A.
2. Enter REFR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Refresh screen 1 will be displayed.
3. Enter the symbolic name of the Refresh process in the SYMBOLIC NAME field.
4. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
5. Enter the disk file name in the INPUT FILE NAME field.
6. Identify the character set of the data in the refresh input file in the CHARACTER SET field. Valid values are as follows:

   E = EBCDIC
   A = ASCII
7. Press the **F2** key to initiate the refresh.
8. Check the network log.

   Any error messages are displayed on the network log (using routing code 0). If there were no error conditions, a “REFRESH COMPLETE” message is generated; a message stating the percentage of file full is also displayed.
9. Review the refresh report.

   After the refresh completes, a hard copy report is sent to the spooler location identified in the appropriate LCONF assign (i.e., BP-RPTR, RPTR, or REF-RPT-SQL). For more information on the refresh report, refer to the topic “Verifying Reports” at the end of this section.

Performing a Refresh with Impacting

The Enscribe Refresh process can impact records in the CAF, PBF, SPF, NBF, and WHFF. The Base National Negative, INAS National Negative, and SQL Refresh processes do not perform impacting.

Impacting updates the refreshed file with the transactions that have occurred since the last extract. The Refresh process determines whether to impact the file based on the setting in the IMP-TYP field in the refresh file header (RFH).

If the refresh input file includes the correct impact starting dates and times, the procedures for performing a refresh with impacting are no different from performing any other refresh. If the impact starting dates and times are not correct in the RFH, additional steps must be performed to indicate the impact starting dates and times to refresh.

The following paragraphs identify which impact starting dates and times must be specified for each file the Refresh process can impact. These paragraphs are followed by procedures which include the steps to specify the impact start dates and times on the Refresh screen.

CAF Impacting

To update the CAF, the Refresh process impacts the newly refreshed CAF with records from the BASE24-atm Transaction Log File (TLF). Therefore, the date of the last TLF fully extracted and the timestamp of the last TLF record extracted must be specified for CAF refreshes with impacting.

NBF Impacting

To update the NBF, the Refresh process impacts the newly refreshed NBF with records from the TTLF. Therefore, the date of the last TTLF fully extracted and the timestamp of the last TTLF record extracted must be specified for NBF refreshes with impacting.

PBF Impacting

To update the PBF, the Refresh process impacts the newly refreshed PBF with records from transaction log files (BASE24-atm Transaction Log File (TLF), POS Transaction Log File (PTLF), Teller Transaction Log File (TTLF), and ITS Transaction Log File (ITLF)). Therefore, the date of the last TLF, PTLF, TTLF, and ITLF fully extracted and the timestamp of the last record extracted from each of these files must be specified for PBF refreshes with impacting.

SPF Impacting

To update the SPF, the Refresh process impacts the newly refreshed SPF with records from the TTLF. Therefore, the date of the last TTLF fully extracted and the timestamp of the last TTLF record extracted must be specified for SPF refreshes with impacting.

WHFF Impacting

To update the WHFF, the Refresh process impacts the newly refreshed WHFF with records from the TTLF. Therefore, the date of the last TTLF fully extracted and the timestamp of the last TTLF record extracted must be specified for WHFF refreshes with impacting.

Performing a Refresh from Tape

The following are operator procedures for performing file refreshes from tape with impacting using the Refresh screen. These instructions assume that you are logged on to BASE24 and apply to the Enscribe Refresh process.

1. Mount the tape from the host on the HP NonStop processor tape drive.
2. Enter REFR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Refresh screen 1 will be displayed.
3. Enter the symbolic name of the Refresh process in the SYMBOLIC NAME field.
4. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.

5. Enter the tape label type in the TAPE LABEL TYPE field. Valid values for this field depend on whether the HP NonStop Kernel TAPECOM product or the Refresh process is processing tape labels. The BASE24-TAPE-LBL-PROC-USED param in the LCONF identifies the tape label processor.

   Valid tape label types when the HP NonStop Kernel TAPECOM product is processing tape labels are as follows:

   IBM = IBM/MVS
   ANS = ANSI

   Valid tape label types when the BASE24 Refresh process is processing tape labels are as follows:

   ANS = ANSI
   IBM = IBM/MVS
   DOS = IBM DOS
   BUR = Burroughs 5500
   BLS = Burroughs Large Scale
   NON = Unlabeled

   The default is IBM if no tape label is entered.

6. Enter the tape drive name in the INPUT FILE NAME field.

   The default is $TAPE if no value is entered.

7. Identify the character set of the data in the CHARACTER SET field. Valid values are as follows:

   E = EBCDIC
   A = ASCII

8. Enter the number of parity errors to bypass in the BYPASS NUM field (if applicable).

   This field can be used to override the number of parity errors to bypass specified in the REF-BYPASS-PARITY-ERRORS param in the LCONF. Use of this field is optional.

   If no value is entered in this field or in the LCONF, the default is 1.

9. Perform the following steps if the HP NonStop Kernel TAPECOM product is running on the system:

   a. Enter an additional message in the MOUNT MESSAGE field. The information in this field is displayed along with the system mount message when the Refresh process requests use of the tape drive.

   b. Enter the tape file name in the FILE IDENTIFIER field if you are processing tape labels (i.e., the LCONF param BASE24-TAPE-LBL-PROC-USED is set to a value of N) and the tape label type is IBM.

c. Enter the volume identifier for each reel to be used in the extract in the VOLUME IDENTIFIERS fields if you are processing tape labels and the tape label type is IBM or ANSI. When there is more than one volume identifier, enter the volume IDs from left to right, with no blank fields between volume IDs.

10. Enter the timestamp of the last record extracted from the log file if the product is part of the network and the specified file is being refreshed with impacting.

BASE24 Product	Log File	File Refreshed
BASE24-atm	TLF	CAF or PBF
BASE24-pos	PTLF	PBF
BASE24-telebanking	ITLF	PBF
BASE24-teller	TTLF	NBF, PBF, SPF, or WHFF

11. Press the **F2** key to initiate the refresh.

12. Check the network log.

    Any error messages are displayed on the network log (using routing code 0). If there were no error conditions, a REFRESH COMPLETE message is generated; a message stating the percentage of file full is also displayed.

13. Review the refresh report.

    After the refresh completes, a hard copy report is sent to the spooler location identified in the appropriate LCONF assign (i.e., RPTR). For more information on the refresh report, refer to the “Verifying Reports” topic at the end of this section.

Performing a Refresh from Disk

The following are operator procedures for performing file refreshes from disk with impacting using the Refresh screen. These instructions assume that you are logged on to BASE24 and apply to the Enscribe Refresh process.

1. Verify that the disk file is present.
2. Enter REFR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Refresh screen 1 will be displayed.
3. Enter the symbolic name of the Refresh process in the SYMBOLIC NAME field.

4. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.

5. Enter the disk file name in the INPUT FILE NAME field.

   The default is $TAPE if no value is entered.

6. Identify the character set of the data in the CHARACTER SET field. Valid values are as follows:

   E = EBCDIC
   A = ASCII

7. Enter the timestamp of the last record extracted from the log file if the product is part of the network and the specified file is being refreshed with impacting.

BASE24 Product	Log File	File Refreshed
BASE24-atm	TLF	CAF or PBF
BASE24-pos	PTLF	PBF
BASE24-telebanking	ITLF	PBF
BASE24-teller	TTLF	NBF, PBF, SPF, or WHFF

8. Press the **F2** key to initiate the refresh.

9. Check the network log.

   Any error messages are displayed on the network log (using routing code 0). If there were no error conditions, a REFRESH COMPLETE message is generated; a message stating the percentage of file full is also displayed.

10. Review the refresh report.

    After the refresh completes, a hard copy report is sent to the spooler location identified in the appropriate LCONF assign (i.e., RPTR). For more information on the refresh report, refer to the “Verifying Reports” topic at the end of this section.

Refreshing Partitioned Files and Tables

Refreshing of partitioned files and tables varies by Refresh process, as follows:

- The Enscribe Refresh process can refresh a partitioned file in its entirety or by individual partitions.*
- The SQL Refresh process can refresh a partitioned table in its entirety or by individual partitions.
- The Base and INAS National Negative Refresh processes do not use partitioned files.

* The Enscribe Refresh process maintains a version indicator in the File Control File (FCF) for the THF, which cannot be renamed because it is audited by the HP NonStop Kernel Transaction Monitoring Facility (TMF). THF partitions cannot be refreshed individually because the Enscribe Refresh process maintains only one version indicator in the FCF for each file.

When all partitions are being refreshed, the procedures for refreshing a partitioned file are no different from performing any other refresh. The procedures for refreshing a partitioned table involve setting an LCONF assign and param, as shown in the following table. The variable *nn* in the second column identifies the partition number being refreshed. The valid value is 00 for a refresh of all partitions. A value of Y for the param in the third column indicates that all partitions are being refreshed.

Table	Partition Number Assign	Full-File Refresh Param
CACT	CACT-PART*nn*	CACT-PRI-PARTONLY
CPIT	CPIT-PART*nn*	CPIT-PRI-PARTONLY
CSTT	CSTT-PART*nn*	CSTT-PRI-PARTONLY
PIT	PIT-PART*nn*	PIT-PRI-PARTONLY

Note: If the full-file refresh param is set to a value of Y and the partition number of the corresponding table is 00, a full-file refresh of only the primary partition is performed. If the full-file refresh param is set to a value of N and the partition number of the corresponding table is 00, a full-file refresh of all partitions in the table is performed. If the partition number of the table to be refreshed is greater than 00, a full-file refresh of only the specified partition is performed, regardless of the value of the full-file refresh param.

When a single partition of a file or table is refreshed separately, either the PART-NUM field in the refresh file header (RFH) or the PARTITION NUM field on Refresh screen 1 must specify the partition to be refreshed. When a single partition of a table is refreshed separately, the LCONF assigns and params shown in the table above are also used to properly identify the partition. The variable *nn* in the second column identifies the partition number being refreshed. Valid values are 01 through 99 for a single partition. A value of N for the param in the third column indicates that only one partition is being refreshed.

Note: When a full refresh of a single table partition is performed, the input tape or disk must contain only records that are within the key limits of the partition being refreshed.

Performing a Refresh from Tape

The following are operator procedures for performing a tape refresh of a single partition using the Refresh screen. These instructions assume that you are logged on to BASE24.

1. Mount the tape from the host on the HP NonStop processor tape drive.
2. Enter REFR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Refresh screen 1 will be displayed.
3. Enter the symbolic name of the Refresh process in the SYMBOLIC NAME field.
4. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
5. Enter the tape label type in the TAPE LABEL TYPE field. Valid values for this field depend on whether the HP NonStop Kernel TAPECOM product or the Refresh process is processing tape labels. The BASE24-TAPE-LBL-PROC-USED param in the LCONF identifies the tape label processor.

   Valid tape label types when the HP NonStop Kernel TAPECOM product is processing tape labels are as follows:

   IBM = IBM/MVS
   ANS = ANSI

Valid tape label types when the Refresh process is processing tape labels are as follows:

ANS = ANSI
IBM = IBM/MVS
DOS = IBM DOS
BUR = Burroughs 5500
BLS = Burroughs Large Scale
NON = Unlabeled

The default is IBM if no tape label is entered.

6. Enter the tape drive name in the INPUT FILE NAME field.

   The default is $TAPE if no value is entered.

7. Identify the character set of the data in the CHARACTER SET field. Valid values are as follows:

   E = EBCDIC
   A = ASCII

8. Enter the number of parity errors to bypass in the BYPASS NUM field (if applicable).

   This field can be used to override the number of parity errors to bypass specified in REF-BYPASS-PARITY-ERRORS param in the LCONF. Use of this field is optional.

   If no value is entered in this field or in the LCONF, the default is 1.

9. Perform the following steps if the HP NonStop Kernel TAPECOM product is running on the system:

   a. Enter an additional message in the MOUNT MESSAGE field. The information in this field is displayed along with the system mount message when the Refresh process requests use of the tape drive.

   b. Enter the tape file name in the FILE IDENTIFIER field if you are processing tape labels (i.e., the LCONF param BASE24-TAPE-LBL-PROC-USED is set to a value of N) and the tape label type is IBM.

   c. Enter the volume identifier for each reel to be used in the extract in the VOLUME IDENTIFIERS fields if you are processing tape labels (i.e., the LCONF param BASE24-TAPE-LBL-PROC-USED is set to a value of N) and the tape label type is IBM or ANSI. When there is more than one volume identifier, enter the volume IDs from left to right, with no blank fields between volume IDs.

10. Enter the partition number of the partition to be refreshed in the PARTITION NUM field.

11. Press the **F2** key to initiate the refresh.
12. Check the network log.

    Any error messages are displayed on the network log (using routing code 0). If there were no error conditions, a REFRESH COMPLETE message is generated; a message stating the percentage of file full is also displayed.
13. Review the refresh report.

    After the refresh completes, a hard copy report is sent to the spooler location identified in the appropriate LCONF assign (i.e., RPTR or REF-RPT-SQL). For more information on the refresh report, refer to the “Verifying Reports” topic at the end of this section.

Performing a Refresh from Disk

The following are operator procedures for performing a disk refresh of a single partition using the Refresh screen. These instructions assume that you are logged on to BASE24.

1. Verify that the disk file is present.
2. Enter REFR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Refresh screen 1 will be displayed.
3. Enter the symbolic name of the Refresh process in the SYMBOLIC NAME field.
4. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
5. Enter the disk file name in the INPUT FILE NAME field.

   The default is $TAPE if no value is entered.
6. Identify the character set of the data in the CHARACTER SET field. Valid values are as follows:

   E = EBCDIC
   A = ASCII
7. Enter the partition number of the partition to be refreshed in the PARTITION NUM field.
8. Press the **F2** key to initiate the refresh.

9. Check the network log.

   Any error messages are displayed on the network log (using routing code 0). If there were no error conditions, a REFRESH COMPLETE message is generated; a message stating the percentage of file full is also displayed.

10. Review the refresh report.

    After the refresh completes, a hard copy report is sent to the spooler location identified in the appropriate LCONF assign (i.e., RPTR or REF-RPT-SQL). For more information on the refresh report, refer to the “Verifying Reports” topic at the end of this section.

Monitoring Refresh Processing

The progress of a refresh can be monitored through the network logging facility. As a Base National Negative, INAS National Negative, Enscribe, and SQL Refresh process processes the refresh input file, it logs event messages to the network logging facility indicating what is happening. The messages logged range from informational messages, such as a message indicating that the Refresh process has completed processing for a specific file, to critical messages, such as a message indicating that the Refresh process is aborting due to a missing assign in the LCONF.

Operators can request additional information on the status of a refresh by sending a STATUS command to the Refresh process. When the Refresh process receives the STATUS command, it sends a series of messages to the network logging facility that provide information about the refresh in progress. The following are operator procedures for requesting the status of a refresh using the Refresh screen. These instructions assume that you are logged on to BASE24.

1. Enter REFR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Refresh screen 1 will be displayed.
2. Enter the symbolic name of the Refresh process in the SYMBOLIC NAME field.
3. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
4. Press the **F3** key to request the status of the refresh.
5. Check the network log.

   Event messages displayed on the network log (using routing code 0) indicate the status of the refresh.

Bypassing Parity Errors

Occasionally, parity errors occur during refreshes. Parity errors denote inconsistent data in the records. Unless otherwise specified, the first record with a parity error causes the Refresh process to abort.

The REF-BYPASS-PARITY-ERRORS param in the LCONF specifies the maximum number of parity errors that a Refresh process can encounter without aborting. An operator can override the value in this param when starting the

Refresh process through the Refresh screens or a network control facility. If the REF-BYPASS-PARITY-ERRORS param does not exist, the Refresh process abends when it encounters the first parity error.

When the REF-BYPASS-PARITY-ERRORS param is set, the Refresh process bypasses that number of records having parity errors; therefore, the file is not updated with those records. Either a separate partial-file refresh input file must be created for the records that were bypassed, or a BASE24 operator must manually update the bypassed records.

The Refresh process logs event messages as parity errors occur. It also generates warning messages at the end of the refresh, indicating that the record counts are incorrect. Using a STATUS command entered from the network control facility while the Refresh process is running, an operator can determine whether parity errors have been encountered, the number of parity errors that have been encountered, and the maximum number of parity errors that can be encountered.

Parity Errors and Full-File Refreshes

If the maximum number of parity errors is reached during a full-file refresh, the Refresh process abends immediately. Otherwise, when a parity error is encountered, the Refresh process logs event messages indicating the primary keys of the last and next records that were successfully read. The Refresh process then continues to build the new file.

If the maximum number of parity errors is not reached by the time the Refresh process has finished building the new file, the Refresh process logs event messages indicating the number of parity errors encountered and the maximum number of parity errors allowed.

The Base and INAS National Negative Refresh processes then cause the new file or table to be used for processing.

The Enscribe Refresh process then pauses until a CONTINUE command is received from an operator using the network control facility, or until a stop message is received from the network. The CONTINUE command, which is one of the DELIVER PROCESS commands that the Enscribe Refresh process can receive, causes the new file to be used for processing. A stop message, which is issued from the network control facility using the DELIVER PROCESS STOP command, causes the new file to be rejected.

SQL processes must be re-SQL compiled before they begin to use a fully refreshed table, regardless of whether parity errors have occurred.

Parity Errors and Partial-File Refreshes

If the maximum number of parity errors is reached during a partial-file refresh, the Refresh process immediately abends. Otherwise, when a parity error is encountered, the Refresh process logs a message indicating the primary key of the last record that was successfully read. The INAS National Negative and SQL Refresh processes continue processing as long as the maximum number of parity errors is not reached. The Base National Negative Refresh process does not support partial-file refreshes.

Upon receipt of a parity error, the Enscribe Refresh process pauses until a CONTINUE command is received from an operator using the network control facility, or until a stop message is received from the network. The CONTINUE command, which is one of the DELIVER PROCESS commands that the Enscribe Refresh process can receive, causes the Refresh process to continue processing. A stop message, which is issued from the network control facility using the DELIVER PROCESS STOP command, causes the Refresh process to stop processing. Note that changes made prior to the process aborting are not backed out of the file.

Parity Errors and Partial-File Refreshes With Impacting

If the maximum number of parity errors is reached during a partial-file refresh with impacting, the Enscribe Refresh process abends immediately. Otherwise, when a parity error is encountered, the Enscribe Refresh process logs a message indicating the primary keys of the last and next records that were successfully read, and continues updating the file. The Base National Negative, INAS National Negative, and SQL Refresh processes do not support impacting.

Once the extended memory is full or the end of the tape is reached, the Enscribe Refresh process logs a message indicating the number of parity errors encountered and the maximum number of parity errors allowed. The Refresh process then pauses until a CONTINUE command is received from an operator using the network control facility, or until a stop message is received from the network. A CONTINUE command causes the Refresh process to update the file with the records in memory. A stop message causes the Refresh process to abort. The records in extended memory when the Refresh process abends are not applied to the file. However, changes made to the file prior to the process abending are applied to the file.

Suspending Refresh Processing

Occasionally, operators may want to suspend a refresh that is in progress. For example, a refresh may need to be suspended because of another urgent request for the tape drive or because of heavy transaction processing. The following are operator procedures for suspending a refresh in progress. These instructions assume that you are logged on to BASE24. The Enscribe Refresh process supports the DELIVER PROCESS SUSPEND command to suspend a refresh in process. The Base National Negative, INAS National Negative, and SQL Refresh processes do not support the SUSPEND command.

Note: Depending on where the Refresh process is in its processing, it may take some time before the Refresh process reaches the next logical stopping point and suspends processing.

1. Enter REFR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Refresh screen 1 will be displayed.
2. Enter the symbolic name of the Refresh process in the SYMBOLIC NAME field.
3. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
4. Press the **F4** key to suspend the refresh.
5. Check the network log.

   If the SUSPEND command is accepted, the Enscribe Refresh process generates event message 0315 indicating that the refresh was suspended. The Enscribe Refresh process also generates event messages 0312 and 0317, indicating the status of the refresh at the time that it was suspended, and may generate event messages 0313 and 0314 to indicate the progress of transaction impacting. The information in these messages is required to resume refresh processing, and should be noted for later use.

   If the SUSPEND command is rejected, the Enscribe Refresh process generates event message 0332 indicating that the process does not need to be suspended. The command is rejected if the refresh is already complete at the time the command is received or if the next logical stopping point for the refresh is when processing is complete.

Resuming Refresh Processing

After a refresh has been suspended, it can be restarted by sending the Refresh process a RESUME command. The following are operator procedures for resuming a suspended refresh. These instructions assume that you are logged on to BASE24. The Enscribe Refresh process supports the DELIVER PROCESS RESUME command to restart a refresh that has been suspended. The Base National Negative, INAS National Negative, and SQL Refresh processes do not support the RESUME command.

1. Enter REFR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Refresh screen 1 will be displayed.
2. Enter the symbolic name of the Enscribe Refresh process in the SYMBOLIC NAME field.
3. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
4. Using the information sent to the network logging facility at the time the refresh was suspended, perform the following steps to identify the refresh being resumed:
   a. Enter the refresh group identifying the financial institutions for which records were being refreshed at the time the refresh was suspended in the REFRESH GROUP field.
   b. Enter the code identifying the file being refreshed at the time the refresh was suspended in the APPLICATION ID field.
   c. Enter the number of the partition being refreshed at the time the refresh was suspended in the PARTITION NUM field.
   d. Enter the date the refresh input file was created in the TAPE DATE field.
   e. Enter the time the refresh input file was created in the TAPE TIME field.
5. Press the **F5** key to resume the refresh.
6. Check the network log.

   Messages are displayed on the network log (using routing code 0) that indicate the outcome of the RESUME command.

7. Review the refresh report.

   After the refresh completes, a hard copy report is sent to the Spooler location identified in the LCONF assign RPTR. For more information on the refresh report, refer to the topic "Verifying Reports" earlier in this section.

Special Case Refreshes

The following is a list of several special case refreshes which may be of interest to an operator:

- Multiple reel refreshes
- Multiple concurrent refreshes
- Multiple file refreshes
- Refreshes with no data records

The implications of the above refreshes are discussed below.

Performing Multiple Reel Refreshes

The Base National Negative, INAS National Negative, Enscribe, and SQL Refresh processes can perform refreshes where the refresh tape is composed of multiple reels. The procedures for performing a multiple reel refresh are no different from performing any other refresh.

Performing Multiple Concurrent Refreshes

Multiple concurrent refreshes can be performed using multiple copies of the Enscribe Refresh process. For each refresh to be run, your system must include the following:

- A separate Enscribe Refresh process definition. For example, to run three concurrent refreshes, you need to define three Enscribe Refresh processes.
- A tape drive for each refresh from tape.

The procedures for performing multiple concurrent refreshes are no different from performing any other refresh.

Note: Although it is possible for multiple copies of the Enscribe Refresh process to successfully perform refreshes with impacting of different partitions of the same file concurrently, caution should be taken. Possible errors include losing transactions or posting them multiple times, depending on the situation. For best results, perform a refresh with impacting of the whole file or of each partition individually (i.e., not concurrently).

Performing Multiple-File Refreshes

The Enscribe and SQL Refresh processes support refreshes where the refresh input file includes records for multiple application files or tables. These Refresh processes process each application file or table automatically when the refresh input file includes records for multiple application files or tables. The procedures for performing a multiple file refresh are no different from performing any other refresh. The Base and INAS National Negative Refresh processes each read one input file that contains records for two application files.

The following restrictions must be observed when you are performing multiple-file refreshes:

- If you are using previous-release refresh file formats, records for only one application file can be present in the refresh input file.
- Records for Enscribe files and SQL tables cannot be present in the same refresh input file.

Performing Refreshes with No Data Records

For most types of refreshes, if the Refresh process encounters a file header and trailer combination with no corresponding refresh data records, the process will log a message and stop. This is based on the premise that if there are no input data records, either the refresh should not have been run or there is a probable error in the creation of the refresh input (a refresh tape or disk file was created, but no actual input records were created).

In some situations, however, an operations center may be set up to perform the same refresh tasks each day regardless of whether there are records to be refreshed or not. Certain application file refreshes performed by the Enscribe Refresh process operate differently to allow for this. These refreshes are as follows:

- Negative Card File Refresh
- No Book File Refresh
- Stop Payment File Refresh
- Warning/Hold/Float Refresh

For the above types of refreshes, if the Enscribe Refresh process encounters a file header and trailer combination with no corresponding refresh data records, the process will simply log a message to note that no refresh records were found and continue processing. This continuation of processing allows operators to run the above refreshes, whether or not the refresh tape or disk file contains refresh records, without the Enscribe Refresh process stopping abnormally.

The procedures for performing a file refresh, for one of the specified files, with no refresh data records is no different from performing a refresh with data records. The only difference is that the operator will receive an extra event message noting the absence of data records.

Note: For any other type of refresh, if the tape or disk file does not contain corresponding refresh data records, the Refresh process will log a message and stop.

Verifying Reports

Base National Negative, INAS National Negative, Enscribe, and SQL Refresh processes produce reports each time they are run. All Refresh processes produce reports that provide processing totals. The Enscribe and SQL Refresh processes also produce reports that identify processing errors and totals for each application file and table.

Enscribe and SQL Refresh Processes

The Enscribe and SQL Refresh processes produce a report for each application file or table refreshed and a summary report for each refresh run. Each report contains general refresh information on the type of refresh performed, the refresh group, performance counts, when the refresh started, and when the refresh ended. The refresh report also identifies any errors the Refresh process encountered. These reports are shown and described on the following pages.

Application File Report

For each application file or table represented in the refresh input file, the Refresh process produces an individual report containing information about the refresh. A sample application file report for Enscribe and SQL refreshes is shown below, followed by descriptions of its fields.

```
                         BASE24 refresh-report-type          PAGE          001
                                                             DATE MON DD, YYYY
                           application-file-name             TIME HH:MM:SS.TT
                              FOR GROUP group


<error-information>



                        DISK READS  = 00000520
                        INPUT READS = 00000016
                        ADDED       = 00000000
                        DELETED     = 00000000
                        REFRESHED   = 00000010
                        BYPASSED    = 00000000
                file  REFRESH COMPLETE  HH:MM:SS.TT
```

BASE24 *refresh-report-type* — Indicates the type of refresh run. The *refresh-report-type* can be any of the following literals:

- PARTITION *number* REFRESH REPORT, where *number* is the number of the partition being refreshed
- FULL FILE REFRESH REPORT
- PARTIAL REFRESH REPORT
- REFRESH REPORT

***application-file-name* —** The name of the application file or table being refreshed. The *application-file-name* can be any of the following literals:

- CARD HISTORY FILE
- CARDHOLDER AUTHORIZATION FILE
- CHECK STATUS FILE
- CORPORATE CHECK FILE
- CUSTOMER TABLE
- CUSTOMER/ACCOUNT RELATION TABLE
- CUSTOMER/CARD INFORMATION FILE
- CUSTOMER/CARD MEMO FILE
- CUSTOMER/PERSONAL ID RELATION TABLE
- MAILBOX FILE
- NEGATIVE AUTHORIZATION FILE
- PBF ACCOUNT FILE / TYPE = *type*, where *type* is one of the following values:

  CC = Credit accounts only
  DA = Checking or negotiable order withdrawal (NOW) accounts only
  PF = All account types
  SV = Savings accounts only
- PERSONAL INFORMATION TABLE
- SCORING ENGINE MASTER FILE
- STOP PAYMENT FILE
- SWITCH DISPUTE FILE
- TELLER NO BOOK FILE

- TELLER WARNING/HOLD/FLOAT FILE
- TRANSACTION HISTORY FILE

FOR GROUP *group* — The name of the group or file being refreshed. This line is printed only for CAF, NEG, PBF, and THF refreshes. The *group* variable is the name of the THF being refreshed for THF refreshes and the name of the group being refreshed for CAF, NEG, and PBF refreshes.

<error-information> — Any errors that occurred while the specified application file was being refreshed. A list of possible error messages is included later in this section in the topic "Handling Problems."

DISK READS — The number of times the Refresh process accessed an application file to read records or an application table to read rows. This count includes every read made to the BASE24 database during the refresh.

INPUT READS — The number of times the Refresh process accessed the refresh input file to read records.

ADDED — The number of records added to the existing file or rows added to the existing table during a partial-file refresh or during a resumed full- or partial-file refresh.

DELETED — The number of records deleted from the existing file or rows deleted from the existing table during a partial-file refresh. This count is not used for full-file refreshes.

REFRESHED — The number of records created for the new file or rows created for the new table during a full-file refresh. For a partial-file refresh, this is the number of records updated or added to the file or number of rows updated or added to the table.

BYPASSED — The number of records read from the refresh input file that could not be added to the application file or table. The bypass count is reported only if records or rows could not be refreshed. If this count is present in the refresh report, other errors are also reported in the refresh report.

***file* REFRESH COMPLETE —** Indicates that the refresh completed and the type of refresh that was run generating this report (*file* is an acronym for the file or table, such as CAF or CSTT).

Totals Summary Report

After all of the application files represented in the refresh input file are processed, the Enscribe or SQL Refresh process produces a summary report containing information about the entire refresh run. A sample totals summary report is shown below, followed by descriptions of its fields.

```
                              BASE24 REFRESH REPORT             PAGE          002
                                                                DATE MON DD, YYYY
                                  TOTALS SUMMARY                TIME HH:MM:SS.TT



                           DISK READS  = 00000520
                           INPUT READS = 00000016
                           ADDED       = 00000000
                           DELETED     = 00000000
                           REFRESHED   = 00000010
                           BYPASSED    = 00000000
                           *****   REFRESH COMPLETE   *****
```

DISK READS — The number of times the Refresh process accessed an application file to read records or an application table to read rows. This count is the sum of the disk read counts from all application file reports in this refresh run.

INPUT READS — The number of times the Refresh process accessed the refresh input file to read records during the refresh run. This count is the sum of the input read counts from all application file reports in this refresh run.

ADDED — The number of records added to an existing file or rows added to an existing table during a partial-file refresh or during a resumed full- or partial-file refresh. This count is the sum of the added counts from all application file reports in this refresh run.

DELETED — The number of records deleted from the existing file or rows deleted from the existing table during partial-file refreshes. This count is the sum of the deleted counts from all application file reports in this refresh run.

REFRESHED — The number of records created for the new file or rows created for the new table during a full-file refresh. For a partial-file refresh, this is the number of records updated or added to the file or rows updated or added to the table. This count is the sum of the refreshed counts from all application file reports in this refresh run.

BYPASSED — The number of records read from the refresh input file that could not be added to the application file or table. The bypass count is reported only if records or rows could not be refreshed. This count is the sum of the bypassed counts from all application file reports in this refresh run.

Handling Problems

Processing errors that occur during a refresh are typically identified by an error number, field name, and primary key for the record where the error occurred. The errors that can appear on the refresh report are listed and described below. Each error description also identifies which Refresh process (i.e., Enscribe or SQL) can generate the message.

```
ERROR text
```

Where: *text* = The text of an error message generated by the SQL Command Interpreter process.

Cause: The Refresh process received an error message when it used the SQL Command Interpreter process to update a row in an SQL table.

Process Use: SQL

```
PARITY ERROR
KEY OF LAST SUCCESSFUL READ = primary-key
```

Where: *primary-key* = The primary key of the record from the refresh input file read immediately before the parity error was encountered.

Cause: A tape parity error was encountered.

Process Use: Enscribe

```
PARITY ERROR
KEY OF NEXT SUCCESSFUL READ = primary-key
```

Where: *primary-key* = The primary key of the record from the refresh input file read immediately after the parity error was encountered.

Cause: A tape parity error was encountered.

Process Use: Enscribe

```
PARTITION KEY ERROR
KEY = primary-key
```

Where: *primary-key* = The primary key of the record from the refresh input file.

Cause: During a full-file refresh of a single partition, a refresh input file record was read whose primary key did not fall within the given key range for the partition being refreshed. The Refresh process abended.

Process Use: Enscribe

```
PBF INPUT RECORD (PRIKEY primary-key)
FOUND IN BATCH FOR FIID fiid - RECORD NOT PROCESSED
```

Where: *primary-key* = The primary key of the PBF record.

fiid = The FIID currently being processed.

Cause: During a refresh of the PBF, a refresh input file record was read for which the FIID portion of the primary key did not match the FIID currently being refreshed. The record was skipped.

Process Use: Enscribe

```
PBF OVERDRAFT LMT VALUE GREATER THAN MAX ALLOWED FOR CRT ACCESS
KEY = primary-key
OVERDRAFT LIMIT = nnnnnnnnnn DEFAULTING TO 999999999
```

Where: *primary-key* = The primary key of the record from the refresh input file.

nnnnnnnnnn = The value from the overdraft field in the specified record.

999999999 = The new value to which the Refresh process set the overdraft limit field in the PBF.

Cause: The overdraft limit field contained more numeric data than can be displayed by the BASE24 CRT access system. The overdraft limit field in the record was refreshed with the default value specified.

Process Use: Enscribe

```
REAPPLY ADD ERROR - ROW LOCKED BY ANOTHER PROCESS
```

Cause: The SQL Command Interpreter process was unable to add a new row to the table because the row was locked by another process.

Process Use: SQL

```
REAPPLY DELETE ERROR - ROW LOCKED BY ANOTHER PROCESS
```

Cause: The SQL Command Interpreter process was unable to delete an existing row from the table because the row was locked by another process.

Process Use: SQL

```
RE-APPLY ERROR - PREAUTH SEGMENT FULL
KEY=primary-key ERR=Guardian error
PTLF TRAN DATE = date, TRAN TIME = time
```

Where: *date* = The transaction date, in YYMMDD format, from the PTLF record.

time = The transaction time, in hhmmss.tt format, from the PTLF record.

primary-key = The primary key of the record.

Guardian-error = The Guardian error returned at read.

Cause: During a refresh with impacting, the Preauthorized Holds segment of the PBF was full, and the Refresh process could not find any expired or unused entries. The Refresh process could not add the preauthorized hold indicated by the PTLF record.

Process Use: Enscribe

```
REAPPLY UPDATE ERROR - ROW LOCKED BY ANOTHER PROCESS
```

Cause: The SQL Command Interpreter process was unable to update an existing row in the table because the row was locked by another process.

Process Use: SQL

```
RECORD COUNT ERROR FOR INSTITUTION fiid
TAPE SAYS nnnnnnnnn - ACTUAL 9999999999
```

Where: *fiid* = The FIID of the institution.

nnnnnnnnn = The number of data records indicated in the BT.NUM-RECS field.

9999999999 = The internal count of data records kept by the Refresh process.

Cause: The number of data records indicated in the BT.NUM-RECS field of the organizational trailer did not match the internal count of data records for this particular institution. The Refresh process abends when it encounters this error.

Process Use: Enscribe

```
RECORD COUNT ERROR IN FILE TRAILER
TAPE SAYS nnnnnnnnn - ACTUAL 9999999999
```

Where: *nnnnnnnnn* = The total number of data records in the refresh input file as indicated in the FT.NUM-RECS field.

9999999999 = The internal count kept by the Refresh process for the total number of data records in the refresh input file.

Cause: The total number of data records indicated in the FT.NUM-RECS field of the file trailer did not match the internal count of all data records processed in the refresh input file.

Process Use: Enscribe

```
REFRESH ABORTED   HH:MM:SS.TT
```

Cause: The Refresh process encountered an unrecoverable error and abended.

Process Use: Enscribe

`REFRESH IS RESUMING FOR TAPE DATE` *`date`*`, TAPE TIME` *`time`*`,`
`GROUP` *`group`*`, APPL ID` *`id`*`, PARTITION NUM` *`num`*

Where: *date* = The date, in YYYYMMDD format, from the header of the tape refresh being resumed.

time = The time, in hhmm format, from the header of the tape refresh being resumed.

group = The refresh group that was being processed at the time the refresh was suspended or abended.

id = A code identifying the type of file being processed at the time the refresh was suspended or abended. Valid values are the same as those for the *application-id* variable in the RESUME command.

num = The partition being refreshed at the time the refresh was suspended or abended. If the file is not partitioned or if all partitions are being refreshed, this field contains the value AL.

Cause: The Refresh process received a RESUME command.

Process Use: Enscribe

`REFRESH RUN HAS BEEN SUSPENDED`

Cause: The Refresh process received a SUSPEND command.

Process Use: Enscribe

`TAPE KEY DATA IS GREATER THAN SQL FIRST KEY VALUE`

Cause: The primary key retrieved from the first record on the input tape for a full-file refresh was greater than or equal to the key of the first row in the SQL table.

Process Use: SQL

```
TAPE KEY DATA IS GREATER THAN SQL LAST KEY VALUE
```

Cause: The primary key retrieved from the current record on the input tape for a full-file refresh was greater than or equal to the key of the highest key defined for the current partition of the SQL table.

Process Use: SQL

TOTAL RECORDS BYPASSED DUE TO PARITY ERRORS = *nnnnnnnnn*

Where: *nnnnnnnnn* = The total number of records bypassed due to parity errors.

Cause: A tape parity error was encountered that resulted in the Refresh process being unable to apply one or more refresh record to the file.

Process Use: Enscribe

WARNING *text*

Where: *text* = The text of a warning message generated by the SQL Command Interpreter process.

Cause: The Refresh process received a warning message when it used the SQL Command Interpreter process to update a row in an SQL table.

Process Use: SQL

---

404 FIELD CONVERSION ERROR
FIELD = *field-name*
KEY = *primary-key*
RECORD NOT REFRESHED

Where: *field-name* = The name of the field that could not be converted (e.g., CAFXD.BASE.TTL^WDL^LMT).

primary-key = The primary key of the record from the refresh input file.

Cause: A record in the refresh input file contained nonnumeric data in a numeric field. The field could not be converted from the refresh input file or could not be written to the file or table. The last line of this message (i.e., RECORD NOT REFRESHED) is only printed if the record in the application file or row in the application table could not be refreshed because of the conversion error.

Process Use: Enscribe, SQL

`405 RE-APPLY ERROR - TOKEN ID` *`id`* `MISSING`
`KEY=`*`primary-key`*

Where: *id* = The ID number assigned to the token.

primary-key = The primary key of the record.

Cause: During a refresh with impacting, a token required to perform impacting was not present in the transaction log file.

Process Use: Enscribe

`407 RE-APPLY ERROR`
`KEY=`*`primary-key`* `ERR=`*`Guardian error`*

Where: *primary-key* = The primary key of the record.

Guardian-error = The Guardian error returned at read.

Cause: During a PBF partial-file refresh with impacting, a transaction was found in the transaction log file for an account that is no longer in the PBF.

Process Use: Enscribe

```
409 RE-APPLY ERROR
KEY=primary-key ERR=Guardian-error
```

Where: *primary-key* = The primary key of the record.

Guardian-error = The Guardian error returned at read.

Cause: During a PBF full-file refresh with impacting, a transaction was found on the transaction log file for an account that is no longer in the PBF.

Process Use: Enscribe

Base National Negative Refresh Reports

The Base National Negative Refresh process performs a full-file refresh of the two Visa national negative files and produces a summary report for the refresh run. Each report contains general refresh information on the refresh performed and performance counts. A sample report for a Visa national negative files refresh is shown below, followed by descriptions of its fields.

```
                    VISA NATIONAL NEG FILE REFRESH

             RUN DATE  MM/DD/YY       TAPE CREATION DATE: MM/DD/YY
                        EFFECTIVE DATE:  MM/DD/YY

                        RANGE RECORDS - 00000000020
                          NEG RECORDS - 00000000100
                  COUNTERFEIT RECORDS - 00000000010
                      TRAILER RECORDS - 00000000003
                        OTHER RECORDS - 00000000000
                                        -----------
                        TOTAL RECORDS - 00000000133
                    TAPE RECORD COUNT - 00000000133
```

RUN DATE — The date, in MM/DD/YY format, that the Base National Negative Refresh process is run.

TAPE CREATION DATE — The creation date, in MM/DD/YY format, obtained from the refresh tape header.

EFFECTIVE DATE — The effective date, in MM/DD/YY format, for the records on the refresh tape. This is the start date contained in the refresh tape header.

RANGE RECORDS — The number of range records (i.e., records with record types 13 and 14) on the refresh tape that are added to the Base National Range File (BNR).

NEG RECORDS — The number of negative records (i.e., records with record types 11 and 12) on the refresh tape that are added to the Base National Negative File (BNEG).

COUNTERFEIT RECORDS — The number of counterfeit records (i.e., records with record type 15) on the refresh tape that can be added to the BNEG. Only counterfeit records containing an account number with 16 or fewer digits are added to the BNEG.

TRAILER RECORDS — The number of trailer records (i.e., records with record type 20) on the refresh tape.

OTHER RECORDS — The number of counterfeit records (i.e., records with record type 15) on the refresh tape that cannot be added to the BNEG. Counterfeit records containing an account number with more than 16 digits are not added to the BNEG.

TOTAL RECORDS — The total number of records, excluding trailer records, read from the refresh tape.

TAPE RECORD COUNT — The sum of record count values obtained from trailer records on the refresh tape.

INAS National Negative Refresh Reports

The INAS National Negative Refresh process performs full-file and partial-file refreshes of the two INAS national negative files and produces a summary report for the refresh run. Each report contains general refresh information and performance counts.

INAS National Negative Full-File Refresh Report

A sample report for an INAS National Negative full-file refresh is shown below, followed by descriptions of its fields.

```
              INAS NATIONAL NEG FULL FILE REFRESH

        RUN DATE  MM/DD/YY       TAPE CREATION DATE: MM/DD/YY

                   RANGE RECORDS - 00000000020
                     NEG RECORDS - 00000000150
                   OTHER RECORDS - 00000000002
         OUT OF SEQUENCE RECORDS - 00000000003
           CARD TOO LONG RECORDS - 00000000005
                                   -----------
                   TOTAL RECORDS - 00000000180
           TAPE NEG RECORD COUNT - 00000000180
         TAPE RANGE RECORD COUNT - 00000000000
```

RUN DATE — The date, in MM/DD/YY format, that the INAS National Negative Refresh process is run.

TAPE CREATION DATE — The creation date, in MM/DD/YY format, obtained from the refresh tape header.

RANGE RECORDS — The number of range records (i.e., records with record type 0004) on the refresh tape that are added to the INAS National Range File (INR).

NEG RECORDS — The number of negative records (i.e., records with record type 0003) on the refresh tape that are added to the INAS National Negative File (INEG).

OTHER RECORDS — The number of records on the refresh tape that are not range, negative, header, or trailer records.

OUT OF SEQUENCE RECORDS — The number of negative records on the refresh tape that are not in ascending order by account number.

CARD TOO LONG RECORDS — The number of negative records on the refresh tape with account numbers that exceed 16 digits in length.

TOTAL RECORDS — The sum of range, negative, other, out-of-sequence, and card-too-long records read from the refresh tape.

TAPE NEG RECORD COUNT — The sum of record count values obtained from trailer records on the refresh tape.

TAPE RANGE RECORD COUNT — This field is not used at the present time.

INAS National Negative Partial-File Refresh Report

A sample report for an INAS National Negative partial-file refresh is shown below, followed by descriptions of its fields.

```
                INAS NATIONAL NEG PARTIAL FILE REFRESH

            RUN DATE  MM/DD/YY       TAPE CREATION DATE: MM/DD/YY

                   NEG RECORDS ADDED - 00000000010
                 RANGE RECORDS ADDED - 00000000003
                 NEG RECORDS DELETED - 00000000001
           RANGE RECORDS NOT DELETED - 00000000002
                       OTHER RECORDS - 00000000001
               CARD TOO LONG RECORDS - 00000000000
                                       -----------
                       TOTAL RECORDS - 00000000017
               TAPE NEG RECORD COUNT - 00000000017
```

RUN DATE — The date, in MM/DD/YY format, that the INAS National Negative Refresh process is run.

TAPE CREATION DATE — The creation date, in MM/DD/YY format, obtained from the refresh tape header.

NEG RECORDS ADDED — The number of negative records (i.e., records with record type 0003) on the refresh tape that are added to the INAS National Negative File (INEG).

RANGE RECORDS ADDED — The number of range records (i.e., records with record type 0004) on the refresh tape that are added to the INAS National Range File (INR).

NEG RECORDS DELETED — The number of negative records (i.e., records with record type 0003) on the refresh tape that are deleted from the INAS National Negative File (INEG).

RANGE RECORDS NOT DELETED — The number of range records (i.e., records with record type 0004) on the refresh tape that should be deleted from the INAS National Range File (INR). These records cannot be deleted by the INAS National Negative Refresh process because the INR is an entry-sequenced file. They must be deleted manually following the update.

OTHER RECORDS — The number of records on the refresh tape that are not range, negative, header, or trailer records. This total also includes negative records that cannot be added or deleted.

CARD TOO LONG RECORDS — The number of negative records on the refresh tape with account numbers that exceed 16 digits in length.

TOTAL RECORDS — The sum of negative, range, other, and card-too-long records read from the refresh tape.

TAPE NEG RECORD COUNT — The sum of record count values obtained from trailer records on the refresh tape.

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4: Extract Operator Procedures

File extract sessions can be performed either automatically or manually. Automatic file extracts are performed at set times specified in the Extract Configuration File (ECF). Manual file extracts are performed using a network control facility or the BASE24 Extract screens. This section provides the following:

- An overview of how automatic extracts are performed
- An overview of the text commands that can be sent to the Super Extract process
- Procedures for using the Extract screens

Performing Automatic Extracts

When the Super Extract process initializes, it reads into memory all of the Extract Configuration File (ECF) records that contain its symbolic name in the ALTKEY. SYM-NAME field. Each record contains all of the information needed to initiate one extract session.

For each record that it reads, the Super Extract process sets a timer to expire when the extract is to be performed. When the timer expires, the Super Extract process uses the information in the ECF record in memory to perform the extract. After all files identified in the record are extracted, the Super Extract process checks the RESTRT-FLG field in the ECF record in memory. If the RESTRT-FLG field contains the value Y, the Super Extract process sets another timer to expire when the extract is to be performed. This processing causes the extract to be automatically performed on a daily basis. If the RESTRT-FLG field contains the value N, future extracts using that ECF record must be requested manually.

Note: If the EXTRACT-DAT and EXTRACT-TIM fields combined indicate a time in the past, no extract timer is set for the record.

When extracts are performed automatically, the operator is responsible for the following items:

- Ensuring that a tape is in the tape drive for extracts to tape
- Ensuring that sufficient disk space is available for extracts to disk
- Verifying the reports produced by the Super Extract process to determine whether the extract completed successfully

Using Network Control Text Commands

A network control facility allows the network operator to monitor and control the network from a single screen. Operators enter commands from the network control facility using defined syntax. In most cases, the commands entered from the network control facility are processed by the XPNET process; however, one network control command allows messages to be sent to any application process running in the network.

The command you use depends on the release of the network under which the Super Extract process is running. The command used to send a text string to an application process is the DELIVER PROCESS command. The command syntax illustrations and examples provided in this manual use quotation marks as text delimiters for consistency and readability. Refer to the ***BASE24-XPNET Network Control Command Reference Manual*** for the procedures involved in sending DELIVER PROCESS commands, including the other delimiter options that are available.

Like all network control commands, DELIVER PROCESS commands must be entered using defined syntax. Each process recognizes and can act upon a different set of DELIVER PROCESS commands. The DELIVER PROCESS commands that can be sent to the Super Extract process are defined on the following pages.

Refer to the "Conventions Used in this Manual" topic that follows the preface of this manual for the notation conventions necessary to understand the network control commands.

9501 Command

The 9501 command directs the Super Extract process to reinitialize. When the Super Extract process is warmbooted, it deletes all the timers set in its timer table and then goes through all of its initialization steps.

Syntax

DELIVER PROCESS *name,* "9501"

name — The symbolic name assigned to the Super Extract process.

EXTR Command

A general use operator command that causes the files indicated in the command or in the Extract Configuration File (ECF) to be extracted to tape or disk.

Syntax

DELIVER PROCESS *name,* "EXTR*ecf-key* [, *override-flag*] [, *file-map*]
[, *volume-id*]"

name — The symbolic name assigned to the Extract process.

ecf-key — The tag to use when reading the ECF record for this extract. The value in this field must match the value in a TAG field in the ECF.

[*override-flag*] — Indicates whether information in the *file-map* variable is to override values in the ECF. Valid values are as follows:

Y = Yes, override the values in the ECF with the values in the *file-map* variable.
N = No, do not override the values in the ECF.

This variable is optional. The Super Extract process allows you to either extract a file already specified in the ECF or not extract a file already specified in the ECF.

[*file-map*] — Identifies the BASE24 files to be extracted if the *override-flag* variable is set to the value Y. There is a flag for each file supported by the Super Extract process (e.g., ILF, OMF, SAF, etc.). Valid values are as follows:

Y = Yes, extract this file.
N = No, do not extract this file.

[*volume-id*] — The volume identifier for the first tape the Super Extract process uses.

Processing

An operator sends this command from a network control facility or the Extract (EXTR) screen to start a manual extract. The files to be extracted are controlled by the ECF record and the optional *override-flag* and *file-map* variables in the command. The *override-flag* variable defaults to a value of N, indicating that the ECF fields should determine which files are to be extracted. If a value of Y is placed in the *override-flag* variable, the *file-map* variable is compared byte by byte with the FILE-MAP field in the ECF, using the following tests:

- If the FILE-MAP field in the ECF and in the command contains the value Y, the Super Extract process uses a value of Y.
- If the FILE-MAP field contains the value Y in the ECF and contains the value N or blank in the command, the Super Extract process uses a value of N.
- If the FILE-MAP field contains the value N in the ECF and the value Y in the command, the Super Extract process uses a value of N and logs an event message indicating that the ECF cannot be overridden in this circumstance.
- If the FILE-MAP field contains the value N in the ECF and in the command, the Super Extract process uses a value of N.

If the above tests result in the Super Extract process using a value of Y for a file, the Super Extract process extracts the file. If the above tests result in the Super Extract process using a value of N for a file, the Super Extract process does not extract the file.

The byte positions in the *file-map* variable are identified in the following table.

Byte Position	File Acronym	Byte Position	File Acronym
1	ILF	22	not assigned
2	OMF	23	not assigned
3	SAF	24	not assigned
4	ICF	25	not assigned
5	IDF	26	not assigned
6	TLF	27	not assigned

Byte Position	File Acronym	Byte Position	File Acronym
7	ITLF	28	MBF
8	ICFE	29	HMBF
9	not assigned	30	ULF
10	not assigned	31	not assigned
11	PTLF	32	not assigned
12	PRDF	33	not assigned
13	TTLF	34	not assigned
14	TTF	35	HSF
15	not assigned	36	not assigned
16	not assigned	37	not assigned
17	not assigned	38	not assigned
18	SVHF	39	not assigned
19	not assigned	40	not assigned
20	not assigned	41–42	user field
21	not assigned		

Example

The following is an example of the EXTR command. In this example, the Super Extract process is P1A^EXTR1, the *ecf-key* is PTLFEXTR, the *override-flag* is set to a value of N, no *file-map* value is entered, and the *volume-id* is 000001.

DELIVER PROCESS P1A^EXTR1, “EXTR PTLFEXTR, N, , 000001”

Note: When optional items are included in the EXTR command, the command might be too long to be entered directly in the NCS COMMAND field on the Network Control Supervisor (NCS) screen. If the EXTR command exceeds 139 characters, it can be placed in an Obey file and then issued to the Super Extract process using the OBEY command. For more information on the OBEY command, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

REEX Command

A general use operator command that directs the Super Extract process to reextract the files indicated in the command or in the ECF record to tape or disk. A reextract can only be performed after an initial extract has been performed.

Syntax

DELIVER PROCESS *name,* “REEX*ecf-key* [, *override-flag*] [, *file-map*]
[, *volume-id*][, *start-partial*] [, *end-partial*]”

name — The symbolic name assigned to the Refresh process.

ecf-key — The tag to use when reading the ECF record for this reextract. The value in this field must match the value in a TAG field in the ECF.

[*override-flag*] — Indicates whether information in the *file-map* variable is to override values in the ECF. Valid values are as follows:

Y = Yes, override the values in the ECF with the values in the *file-map* field.
N = No, do not override the values in the ECF.

This field is optional. If this field is not included in the message, the Super Extract process reextracts the files identified in the ECF record.

[*file-map*] — Identifies the BASE24 files to be reextracted if the *override-flag* variable is set to the value Y. Valid values are as follows:

Y = Yes, reextract this file.
N = No, do not reextract this file.

[*volume-id*] — The volume identifier for the first tape the Super Extract process will use.

[*start-partial*] — The number of the partial extract at which to begin when reextracting a transaction log file. Valid values are 1 through 100. The value in the *end-partial* variable must be greater than or equal to the value in the *start-partial* variable.

[*end-partial*] — The number of the partial extract at which to end when reextracting a transaction log file. Valid values are 1 through 100. The value in the *end-partial* variable must be greater than or equal to the value in the *start-partial* variable.

Processing

An operator sends this command from a network control facility or the Extract (EXTR) screen. The files to be reextracted are controlled by the ECF record and the optional *override-flag* and *file-map* variables in the command. The *override-flag* variable defaults to a value of N, indicating that the ECF fields should determine which files are to be extracted. If a value of Y is placed in the *override-flag* variable, the *file-map* variable is compared byte by byte with the FILE-MAP field in the ECF, using the following tests:

- If the FILE-MAP field in the ECF and in the command contain the value Y, the Super Extract process uses a value of Y.
- If the FILE-MAP field contains the value Y in the ECF and contains the value N or blank in the command, the Super Extract process uses a value of N.
- If the FILE-MAP field contains the value N in the ECF and the value Y in the command, the Super Extract process uses a value of N and logs an event message indicating that the ECF cannot be overridden in this circumstance.
- If the FILE-MAP field contains the value N in the ECF and in the command, the Super Extract process uses a value of N.

If the above tests result in the Super Extract process using a value of Y for a file, the Super Extract process extracts the file. If the above tests result in the Super Extract process using a value of N for a file, the Super Extract process does not reextract the file.

The byte positions in the *file-map* variable are identified in the following table.

Byte Position	File Acronym	Byte Position	File Acronym
1	ILF	22	not assigned
2	OMF	23	not assigned
3	SAF	24	not assigned
4	ICF	25	not assigned
5	IDF	26	not assigned
6	TLF	27	not assigned
7	ITLF	28	MBF
8	ICFE	29	HMBF
9	not assigned	30	ULF
10	not assigned	31	not assigned
11	PTLF	32	not assigned
12	PRDF	33	not assigned
13	TTLF	34	not assigned
14	TTF	35	HSF
15	not assigned	36	not assigned
16	not assigned	37	not assigned
17	not assigned	38	not assigned
18	SVHF	39	not assigned
19	not assigned	40	not assigned
20	not assigned	41–42	user field
21	not assigned		

Example

The following is an example of the REEX command. In this example, the Super Extract process is P1A^EXTR1, the *ecf-key* is PTLFEXTR, the *override-flag* is set to a value of N, no *file-map* value is entered, the *volume-id* is 000001, the *start-partial* variable is 5, and the *end-partial* variable is 7.

DELIVER PROCESS P1A^EXTR1, “REEX PTLFEXTR, N, , 000001, 5, 7”

Note: When optional items are included in the REEX command, the command might be too long to be entered directly in the NCS COMMAND field on the Network Control Supervisor (NCS) screen. If the REEX command exceeds 139 characters, it can be placed in an Obey file and then issued to the Super Extract process using the OBEY command. For more information on the OBEY command, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

TLFRSET Command

The TLFRSET command directs the Super Extract process to reset the extract position markers stored in record 0 of a transaction log file. There is a separate form of this command for each transaction log file that can be extracted.

Syntax

DELIVER PROCESS *name,* “*file-id*TLFRSET, *date* [, *start-partial*]”

name — The symbolic name assigned to the Super Extract process.

file-id — A letter identifying the transaction log file for which the extract position markers are being reset. Valid values are as follows:

A = BASE24-atm Transaction Log File (TLF)
B = ITS Transaction Log File (ITLF)
P = BASE24-pos Transaction Log File (PTLF)
T = BASE24-teller Transaction Log File (TTLF)

For example, to reset the position markers in the Teller Transaction Log File (TTLF), the string would be TTLFRSET.

date — The date of the transaction log file in which to reset the extract position markers. The date is part of the transaction log file name, in YYMMDD format.

start-partial — The number of the partial extract at which to begin resetting extract position marker information. Valid values are 1 through 100.

TLFSTAT Command

The TLFSTAT command directs the Super Extract process to log a message indicating how many partial extracts have been performed against a transaction log file. There is a separate form of this command for each transaction log file that can be extracted.

Syntax

DELIVER PROCESS *name,* "*file-id*TLFSTAT *date*"

name — The symbolic name assigned to the Super Extract process.

file-id — A letter identifying the transaction log file for which the extract status is being requested. Valid values are as follows:

A = BASE24-atm Transaction Log File (TLF)
I = ITS Transaction Log File (ITLF)
P = BASE24-pos Transaction Log File (PTLF)
T = BASE24-teller Transaction Log File (TTLF)

For example, to request the extract status for the POS Transaction Log File (PTLF), the string would be PTLFSTAT.

date — The date of the transaction log file for which extract status information is being requested. The date is part of the transaction log file name, in YYMMDD format.

TRACE Command

The TRACE command directs the Super Extract process to log event messages identifying the Token File (TKN) records used during transaction log file extracts. The Super Extract process accepts both a TRACE ON and TRACE OFF command to start and stop the trace function. This command should only be issued at the request of ACI personnel. For more information on the TRACE command, refer to the ***BASE24 Text Command Reference Manual.***

Extract Screen Descriptions

The Extract screens provide for direct initiation of file extracts and reextracts from a series of function key-driven screens. This subsection describes the Extract screens and function keys. These screens and function keys are referenced in the procedures that follow this subsection. There are 23 Extract screens, of which seven screens are currently used. Each screen contains a list of files that can be extracted. This section contains documentation for the following Extract screens:

- Screen 1 controls extracts of BASE24 shared files
- Screen 5 controls extracts of BASE24-atm files
- Screen 7 controls extracts of BASE24-pos files
- Screen 9 controls extracts of BASE24-teller files
- Screen 17 controls extracts of BASE24-mail files
- Screen 19 controls extracts of BASE24-from host maintenance files
- Screen 23 controls extracts of BASE24-telebanking files

The remaining screens are reserved for future use.

Extract Screen 1

Extract screen 1 is used to control manual extracts of shared files. Extract screen 1 is shown below, followed by descriptions of its fields.

```
BASE24-BASE  EXTRACT SCREENS          LLLL        YY/MM/DD  HH:MM  01 OF 23

 SYMBOLIC NAME:
          USER:                                PASSWORD:
       ECF KEY:                                       OVERRIDE ECF: N
     VOLUME ID:                     STARTING PARTIAL EXTRACT NUMBER:   0
                                      ENDING PARTIAL EXTRACT NUMBER:   0


                                BASE FILES
                                ILF:  N
                                OMF:  N
                                SAF:  N
                                ICF:  N
                                ICFE: N
                                IDF:  N



********************************** BASE24 **********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F1-VALIDATE   F2-EXTRACT   F3-REEXTRACT   F12-HELP
```

SYMBOLIC NAME — The symbolic name identifying the Super Extract process to be used. Symbolic names are defined as each process is added to the XPNET node.

USER — The name that uniquely identifies you to the BASE24 CRT access system. This name must match your alias in the Security File (SEC) or access to the refresh screens is denied.

Normally, your supervisor gives you your user name. You must have security clearance under an identical user name for all logical networks you want to access, if you want to access a different logical network without logging on again.

Field Length:	1–16 alphanumeric characters
Required Field:	Yes, if network control security is enabled for your installation (i.e., the ENABLE-SECURITY param for the NCPI Server process is set to a value of ON).
Default Value:	None

PASSWORD — Your password. The BASE24 CRT access system requires each user to select a confidential password to ensure that user names are not employed by unauthorized individuals. Each user name and password combination must be unique within the BASE24 CRT access system.

Field Length:	1–16 alphanumeric characters
Required Field:	Yes, if network control security is enabled for your installation (i.e., the ENABLE-SECURITY param for the NCPI Server process is set to a value of ON).
Default Value:	None

ECF KEY — The tag, or primary key, of the ECF record. Using this value and the appropriate symbolic name, the system searches for the ECF record to be used for this extract or reextract session.

Note: There must be a corresponding record in the ECF in order to perform a manual extract.

OVERRIDE ECF — A flag indicating whether the Super Extract process should extract (or reextract) the files indicated in the ECF record or only those files indicated on the Extract screens. Valid values are as follows:

Y = Yes, override the ECF.
N = No, do not override the ECF.

If this field contains the value Y, the settings on the Extract screen are used to determine which files are extracted when a manual extract or a reextract is requested. Super Extract determines which files to extract as described in the note below.

If this field contains the value N, the settings in the ECF record specify which files are extracted when a manual extract or a reextract is requested.

Note: Files not marked to be extracted in the ECF record cannot be extracted by setting the OVERRIDE ECF field to Y, and the corresponding file flag to Y on the Extract screens. That is, the OVERRIDE ECF option can only be used to prevent a file from being extracted that is marked for extraction in the ECF record. Regardless of how the OVERRIDE ECF field is set, the files with flags set to a value of N in the ECF are not extracted.

VOLUME ID — The volume ID of the first tape volume Super Extract uses for the extract. The volume ID is assigned at the time the tape is labeled. It is intended to uniquely identify the tape. This field is only used when the HP NonStop Kernel TAPECOM product is used to handle tape processing and the TAPE LABEL TYPE field on ECF screen 1 is set to the value IBM or ANS. TAPECOM does not allow the extract to start if the volume ID named in this field does not match the volume ID on the mounted tape.

STARTING PARTIAL EXTRACT NUMBER — The number of the partial extract at which the Super Extract process should begin when reextracting a transaction log file. For example, if this field is set to a value of 5, the Super Extract process begins reextracting the transaction log file at the point where the fifth partial extract began.

This field is only used for reextracts of transaction log files (TLF, PTLF, TTLF, or ITLF). If this field is left blank when requesting a reextract of a transaction log file, the Super Extract process begins reextracting at the beginning of the file.

For more information on how partial extracts and reextracts are performed by the Super Extract process, refer to section 1.

ENDING PARTIAL EXTRACT NUMBER — The number of the partial extract at which the Super Extract process should end when reextracting a transaction log file. For example, if this field is set to a value of 7, the Super Extract process stops extracting after processing the last record extracted in the seventh partial extract.

This field is only used for reextracts of transaction log files (TLF, PTLF, TTLF, or ITLF). If this field is left blank when requesting a reextract of a transaction log file, the Super Extract process continues to extract until it reaches the initial end-of-file that is determined when the extract is first started or the actual physical end-of-file. The READ PAST INITIAL EOF field on ECF screen 1 indicates whether the process stops extracting when it reaches the initial end-of-file or continues past the initially determined end-of-file to the actual physical end-of-file.

For more information on how partial extracts and reextracts are performed by the Super Extract process, refer to section 1.

BASE FILES

The following fields are used to indicate which shared files should be extracted. These fields are used when a manual extract or a reextract is requested and the OVERRIDE ECF field contains the value Y.

ILF — A flag indicating whether to extract the Interchange Log File (ILF). Valid values are as follows:

Y = Yes, extract the ILF.
N = No, do not extract the ILF.

OMF — A flag indicating whether to extract the Online Maintenance File (OMF). Valid values are as follows:

Y = Yes, extract the OMF.
N = No, do not extract the OMF.

SAF — A flag indicating whether to extract the Store-and-Forward File (SAF). Valid values are as follows:

Y = Yes, extract the SAF.
N = No, do not extract the SAF.

Note: This is the ANSI or ISO Host Interface SAF, not an Interchange Store-and-Forward File (ISAF) or Gateway Store-and-Forward File (GPSAF).

ICF — A flag indicating whether to extract the Interchange Configuration File (ICF). Valid values are as follows:

Y = Yes, extract the ICF.
N = No, do not extract the ICF.

ICFE — A flag indicating whether to extract the Enhanced Interchange Configuration File (ICFE). Valid values are as follows:

Y = Yes, extract the ICFE.
N = No, do not extract the ICFE.

IDF — A flag indicating whether to extract the Institution Definition File (IDF). Valid values are as follows:

Y = Yes, extract the IDF.
N = No, do not extract the IDF.

Extract Screen 5

Extract screen 5 is used to control manual extracts of files used solely by the BASE24-atm product. Extract screen 5 is shown below, followed by descriptions of its fields.

```
BASE24-ATM   EXTRACT SCREENS          LLLL        YY/MM/DD  HH:MM  05 OF 23

 SYMBOLIC NAME:
          USER:                                 PASSWORD:
       ECF KEY:                                        OVERRIDE ECF: N

                                  ATM FILES
                                  TLF:  N
                                  HSF:  N









*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F1-VALIDATE   F2-EXTRACT   F3-REEXTRACT   F12-HELP
```

ATM FILES

The following fields are used to indicate which BASE24-atm product-specific files should be extracted. These fields are used when a manual extract or a reextract is requested and the OVERRIDE ECF field on Extract screen 1 contains the value Y.

TLF — A flag indicating whether to extract the Transaction Log File (TLF). Valid values are as follows:

Y = Yes, extract the TLF.
N = No, do not extract the TLF.

HSF — A flag indicating whether to extract the Hardware Status File (HSF). Valid values are as follows:

Y = Yes, extract the HSF.
N = No, do not extract the HSF.

Extract Screen 7

Extract screen 7 is used to control manual extracts of files used solely by the BASE24-pos product. Extract screen 7 is shown below, followed by descriptions of its fields.

```
BASE24-POS   EXTRACT SCREENS          LLLL        YY/MM/DD  HH:MM  07 OF 23

 SYMBOLIC NAME:
          USER:                                 PASSWORD:
       ECF KEY:                                        OVERRIDE ECF: N


                                    POS FILES
                                   PTLF:  N
                                   PRDF:  N
                                   SVHF:  N








*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F1-VALIDATE   F2-EXTRACT   F3-REEXTRACT   F12-HELP
```

POS FILES

The following fields are used to indicate which BASE24-pos product-specific files should be extracted. These fields are used when a manual extract or a reextract is requested and the OVERRIDE ECF field on Extract screen 1 contains the value Y.

PTLF — A flag indicating whether to extract the POS Transaction Log File (PTLF). Valid values are as follows:

Y = Yes, extract the PTLF.
N = No, do not extract the PTLF.

PRDF — A flag indicating whether to extract the POS Retailer Definition File (PRDF). Valid values are as follows:

Y = Yes, extract the PRDF.
N = No, do not extract the PRDF.

SVHF — A flag indicating whether to extract the Stored Value History File (SVHF). Valid values are as follows:

Y = Yes, extract the SVHF.
N = No, do not extract the SVHF.

Extract Screen 9

Extract screen 9 is used to control manual extracts of files used solely by the BASE24-teller product. Extract screen 9 is shown below, followed by descriptions of its fields.

```
BASE24-TLR   EXTRACT SCREENS          LLLL        YY/MM/DD  HH:MM  09 OF 23

 SYMBOLIC NAME:
          USER:                                 PASSWORD:
       ECF KEY:                                        OVERRIDE ECF: N


                                TELLER FILES
                                 TTLF:  N
                                  TTF:  N









*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F1-VALIDATE   F2-EXTRACT   F3-REEXTRACT   F12-HELP
```

TELLER FILES

The following fields are used to indicate which BASE24-teller product-specific files should be extracted. These fields are used when a manual extract or a reextract is requested and the OVERRIDE ECF field on Extract screen 1 contains the value Y.

TTLF — A flag indicating whether to extract the Teller Transaction Log File (TTLF). Valid values are as follows:

Y = Yes, extract the TTLF.
N = No, do not extract the TTLF.

TTF — A flag indicating whether to extract the Teller Transaction File (TTF). Valid values are as follows:

Y = Yes, extract the TTF.
N = No, do not extract the TTF.

Extract Screen 17

Extract screen 17 is used to control manual extracts of files used solely by the BASE24-mail product. Extract screen 17 is shown below, followed by descriptions of its fields.

```
BASE24-MAIL  EXTRACT SCREENS          LLLL        YY/MM/DD  HH:MM  17 OF 23

 SYMBOLIC NAME:
          USER:                                PASSWORD:
       ECF KEY:                                       OVERRIDE ECF: N


                                 MAIL FILES
                                  HMBF:  N
                                   MBF:  N










*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:       NEW LOGICAL NETWORK ID:
F1-VALIDATE   F2-EXTRACT   F3-REEXTRACT   F12-HELP
```

MAIL FILES

The following fields are used to indicate which BASE24-mail product-specific files should be extracted. These fields are used when a manual extract or a reextract is requested and the OVERRIDE ECF field on Extract screen 1 contains the value Y.

HMBF — A flag indicating whether to extract the Host Mail Box File (HMBF). Valid values are as follows:

Y = Yes, extract the HMBF.
N = No, do not extract the HMBF.

MBF — A flag indicating whether to extract the Mail Box File (MBF). Valid values are as follows:

Y = Yes, extract the MBF.
N = No, do not extract the MBF.

Extract Screen 19

Extract screen 19 is used to control manual extracts of files used solely by the BASE24-from host maintenance product. Extract screen 19 is shown below, followed by a description of its field.

```
BASE24-FHM   EXTRACT SCREENS          LLLL        YY/MM/DD  HH:MM  19 OF 23

  SYMBOLIC NAME:
           USER:                                 PASSWORD:
        ECF KEY:                                        OVERRIDE ECF: N


                                   FHM FILES
                                    ULF:  N








 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
 F1-VALIDATE   F2-EXTRACT   F3-REEXTRACT   F12-HELP
```

FHM FILES

The following field is used to indicate whether the BASE24-from host maintenance Update Log File (ULF) should be extracted. This field is used when a manual extract or a reextract is requested and the OVERRIDE ECF field on Extract screen 1 contains the value Y.

ULF — A flag indicating whether to extract the ULF. Valid values are as follows:

Y = Yes, extract the ULF.
N = No, do not extract the ULF.

Extract Screen 23

Extract screen 23 is used to control manual extracts of files used solely by the BASE24-telebanking product. Screen 23 is shown below, followed by description of its field.

```
BASE24-TB    EXTRACT SCREENS          LLLL        YY/MM/DD  HH:MM  23 OF 23

 SYMBOLIC NAME:
          USER:                                 PASSWORD:
       ECF KEY:                                        OVERRIDE ECF: N

                            TELEBANKING FILES
                                 ITLF: N











*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
F1-VALIDATE   F2-EXTRACT   F3-REEXTRACT   F12-HELP
```

TELEBANKING FILES

The following field is used to indicate which BASE24-telebanking product-specific files should be extracted. These fields are used when a manual extract or a reextract is requested and the OVERRIDE ECF field on Extract screen 1 contains the value Y.

ITLF — A flag indicating whether to extract the ITS Transaction Log File (ITLF). Valid values are as follows:

Y = Yes, extract the ITLF.
N = No, do not extract the ITLF.

Function Keys

The function keys used on the Extract screens are described below. Throughout BASE24 manuals, references to these function keys include only BASE24 function keys. Specific keyboards may require the use of a combination of keys to achieve the functionality.

The first column of information below shows the BASE24 key. The second column describes the functions that can be performed using these function keys.

Key	Description
F1	**Validate Data —** Checks the data entered on the screen for errors.
F2	**Begin Extract —** Sends an EXTR command to the Super Extract process, using the values entered on the Extract screens.
F3	**Begin Reextract —** Sends a REEX command to the Super Extract process, using the values entered on the Extract screens.
F7	**Go to New Page —** Displays a different screen in the same file in which the operator is working. The screen to be displayed must be identified in the NEW PAGE field at the bottom of the screen.

Key	Description
F8	**Clear Screen —** Clears any values on the Extract screens to spaces or replaces them with default values. This key impacts all Extract screens. This includes the current screen, any previous screens, and any remaining screens.
F9	**Display Next Screen —** Displays the next Extract screen. If the operator is on the last Extract screen, this function key takes the operator to the first screen.
F10	**Print Screen —** Sends the screen currently being displayed to the spooler location indicated on the Logon screen. Refer to the ***BASE24 CRT Access Manual*** for an explanation of the Logon screen.
F11	**Display Previous Screen —** Displays the previous Extract screen. If the operator is on the first Extract screen, this function key takes the operator to the last screen.
F12	**Display Help Screen —** Displays the Help screen. The Help screen displayed depends on the screen currently being viewed. The Help screen contains information about BASE24 function keys or menu options.
F13	**Change Current Logical Network —** Changes the current logical network while in the Extract screens.
F16	**Exit or Go to File —** This key has several purposes. It exits BASE24 or the set of screens currently being accessed. It also allows the operator to move between logical networks and files. Refer to the ***BASE24 CRT Access Manual*** for more information on how to move between logical networks and files. This function key is operable on the Help screen for files to enable the operator to change files or logical networks.
Shift-F16	**Log off BASE24 —** Logs the operator off BASE24. When this function is used while an operator is accessing BASE24, a blank Logon screen is displayed. If the Logon screen is displayed when this function is used, the Logon screen continues to be displayed or a TACL prompt is displayed, depending on how the terminal is set up.

Performing Manual Extracts to Tape

The following are operator procedures for using the Extract screens to perform manual file extracts to tape.

1. Mount a tape with a write ring on the HP NonStop processor tape drive. The drive must be the drive specified in the ECF record for the extract to be performed.
2. Enter EXTR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Extract screen 1 is displayed.
3. Enter the symbolic name of the Super Extract process in the SYMBOLIC NAME field.
4. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
5. Enter the ECF tag identifying the ECF record to be used to control this extract in the ECF KEY field.
6. Enter the value Y or N in the OVERRIDE ECF field to indicate whether to override the files marked for extraction in the ECF with those marked on the Extract screens.

   Y = Yes, override the ECF.
   N = No, do not override the ECF.
7. If necessary, specify the volume ID of the first tape to be used for the extract in the VOLUME ID field. The volume ID is only used when the HP NonStop Kernel TAPECOM product is handling tape processing and the TAPE LABEL TYPE field on ECF screen 1 contains the value IBM or ANS.
8. If necessary, mark the files on each of the Extract screens to be extracted. When the OVERRIDE ECF field contains the value Y, the files to be extracted must be marked on these screens.

   Y = Yes, extract the file.
   N = No, do not extract the file.

   Note: Files not marked to be extracted in the ECF record cannot be extracted by setting the OVERRIDE ECF field to Y and the corresponding file flag to Y on the Extract screens. That is, the OVERRIDE ECF option can only be used to prevent a file from being extracted that is marked for extraction in the ECF record. Regardless of how the OVERRIDE ECF field is set, the files with flags set to a value of N in the ECF are not extracted.

9. Press the **F1** key to validate the data entered on the screens. If the data is okay, continue to the next step. If the data is invalid, make the required corrections before continuing with the next step.

10. Press the **F2** key to initiate the extract.

11. Check the message log and the extract report.

    Any error messages are displayed on the message log (using routing code 0). If an error causes the Super Extract process to abort, return to step 1 and begin operator extract procedures again.

    If there were no error conditions, an event message containing the number of records extracted is generated. The extract report contains additional information about the extract.

Performing Reextracts to Tape

The following are operator procedures for using the Extract screens to perform reextracts to tape.

1. Mount a tape with a write ring on the HP NonStop processor tape drive. The drive must be the drive specified in the ECF record for the extract to be performed.
2. Enter EXTR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Extract screen 1 is displayed.
3. Enter the symbolic name of the Super Extract process in the SYMBOLIC NAME field.
4. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
5. Enter the ECF tag identifying the ECF record to be used to control this reextract in the ECF KEY field.
6. Enter the value Y or N in the OVERRIDE ECF field to indicate whether to override the files marked for extraction in the ECF with those marked on the Extract screens.

   Y = Yes, override the ECF.
   N = No, do not override the ECF.
7. Enter the volume identifier for the tape mounted on the tape drive in the VOLUME ID field.
8. If necessary, enter the starting partial extract number in the STARTING PARTIAL EXTRACT NUMBER field. Valid values are 1 through 100. For more information on how partial extracts and reextracts are performed by the Super Extract process, refer to section 1.
9. If necessary, enter the ending partial extract number in the ENDING PARTIAL EXTRACT NUMBER field. Valid values are 1 through 100, but the value must be greater than the value in the STARTING PARTIAL EXTRACT NUMBER field. For more information on how partial extracts and reextracts are performed by the Super Extract process, refer to section 1.
10. If necessary, mark the files on each of the Extract screens to be reextracted. When the OVERRIDE ECF field contains the value Y, the files to be reextracted must be marked on these screens.

    Y = Yes, extract the file.
    N = No, do not extract the file.

Note: Files not marked to be extracted in the ECF record cannot be extracted by setting the OVERRIDE ECF field to Y and the corresponding file flag to Y on the Extract screens. That is, the OVERRIDE ECF option can only be used to prevent a file from being extracted that is marked for extraction in the ECF record. Regardless of how the OVERRIDE ECF field is set, the files with flags set to a value of N in the ECF are not extracted.

11. Press the **F1** key to validate the data entered on the screens. If the data is okay, continue to the next step. If the data is invalid, make the required corrections before continuing with the next step.
12. Press the **F3** key to initiate the reextract.
13. Check the message log and extract report.

    Any error messages are displayed on the message log (using routing code 0). If an error causes the Super Extract process to abort, return to step 1 and begin operator reextract procedures again.

    If there were no error conditions, an event message containing the number of records reextracted is generated. The extract report contains additional information about the reextract.

Extracting to Disk

The procedures for extracting or reextracting to disk are basically the same as those for extracting to tape; however, the configuration of an extract to disk is somewhat different from the configuration of an extract to tape. The following list identifies the differences in setting up the ECF record when configuring an extract to disk:

- The value in the TAPE NAME field on ECF screen 1 identifies the disk to which extract records are to be written. The disk name should include a volume name, subvolume name, and a single letter file name. The Super Extract process uses the file letter to create a file name in the format *file-letter month-day seq-no*. For example, if the file letter is B and the date is October 12, the Super Extract process would create a file named B1012000 for the first extract that it performs that day. For subsequent extracts to disk with the same file letter and on the same date, the Super Extract process increments the *seq-no* by 1. The volume name and subvolume name are optional; if a volume and subvolume are not indicated in the TAPE NAME field, the Super Extract process creates the disk file on the volume and subvolume from which the BASE24 network was started.
- The value in the TAPE LABEL TYPE field is not used for extracts to disk; however, BASE24 requires an entry in this field. The value NON is recommended, to indicate that tape labels are not used.
- The value in the TAPE BLOCK SIZE field determines both the record size and the block size. Valid values are as follows:

1–3	=	Record/block size in kilobytes. The Super Extract process multiplies the value entered by 1024 to determine the record and block size in bytes (e.g., 2 kilobytes equals 2048 bytes, and 3 kilobytes equals 3072 bytes). BASE24 automatically displays a K following any entry in kilobytes (e.g., 2K and 3K). The Super Extract process sets the Enscribe record size equal to the value entered, and sets the Enscribe block size to the value entered plus 24 (the additional 24 bytes in each block is used for a 24-byte block header).
33–4072	=	Record/block size in bytes. The Super Extract process sets the Enscribe record size equal to the value entered, and sets the Enscribe block size to the value entered plus 24 (the additional 24 bytes in each block is used for a 24-byte block header).

Note: If an invalid value is entered (4 through 32 or 4073 through 32767), the Super Extract process uses a record size of 4072 bytes and a block size of 4096 bytes.

- The values in the VOLUME IDENTIFIER, DATA SET IDENTIFIER, RETENTION, MOUNT MESSAGE, and DENSITY fields on ECF screen 1 are not used for extracts to disk.
- ECF screen 3 contains fields used to specify disk file configuration information. This information is used to create the disk file to which the Super Extract process writes the extracted records.

The following are operator procedures for performing using the Extract screens to perform manual file extracts to disk.

1. Enter EXTR in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu, press the **F1** key. From a file screen, press the **F16** key. Extract screen 1 is displayed.
2. Enter the symbolic name of the Super Extract process in the SYMBOLIC NAME field.
3. If network control security is enabled, enter your user name in the USER field and your user password in the PASSWORD field.
4. Enter the ECF tag identifying the ECF record to be used to control this extract in the ECF KEY field.
5. Enter the value Y or N in the OVERRIDE ECF field to indicate whether to override the files marked for extraction in the ECF with those marked on the Extract screens.

   Y = Yes, override the ECF.
   N = No, do not override the ECF.
6. If necessary, mark the files on each of the Extract screens to be extracted. When the OVERRIDE ECF field contains the value Y, the files to be extracted must be marked on these screens.

   Y = Yes, extract the file.
   N = No, do not extract the file.

   Note: Files not marked to be extracted in the ECF record cannot be extracted by setting the OVERRIDE ECF field to Y and the corresponding file flag to Y on the Extract screens. That is, the OVERRIDE ECF option can only be used to prevent a file from being extracted that is marked for extraction in the ECF record. Regardless of how the OVERRIDE ECF field is set, the files with flags set to a value of N in the ECF are not extracted.

7. Press the **F1** key to validate the data entered on the screens. If the data is correct, continue to the next step. If the data is invalid, make the required corrections before continuing with the next step.
8. Press the **F2** key to initiate the extract.
9. Check the message log and extract report.

   Any error messages are displayed on the message log (using routing code 0). If an error causes the Super Extract process to abort, return to step 1 and begin operator extract procedures again.

   If there were no error conditions, an event message containing the number of records extracted is generated. The extract report contains additional information about the extract.

Verifying Reports

Each time an extract or reextract is performed, whether automatically or manually, the Super Extract process generates a report file. The report file reflects what the Super Extract process did during the extract or reextract. The report includes such information as the name of the file extracted, the number of records extracted, and the number of blocks extracted. The report also identifies any problems that occurred during the extract.

Because the report indicates what happened during a particular extract session, it should be checked after the extract session completes to verify that the extract completed normally. If any problems are identified on the report, they should be checked or corrected and the extract repeated.

The extract report is sent to the location identified in the REPORT NAME field on ECF screen 1.

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A: Refresh Disk and Tape Specifications

This appendix contains the specifications for input tapes and disks used by the BASE24 Refresh processes. It includes general input tape and disk specifications, as well as layouts for refresh input file control data records.

The information in this appendix applies to Base National Negative, INAS National Negative, Enscribe, and SQL Refresh processes, unless noted otherwise.

Tape Input File Organization and Content

The illustration on the next page shows the organization of a typical host-generated refresh tape.

Each refresh tape contains tape label records (optional) and data records. The data records consist of control records (one file header and trailer for each file on the tape, and a variable number of organization headers and trailers) and application data records.

One organization header and trailer pair is required for each file on the refresh tape. Additional pairs may be included in the following situations:

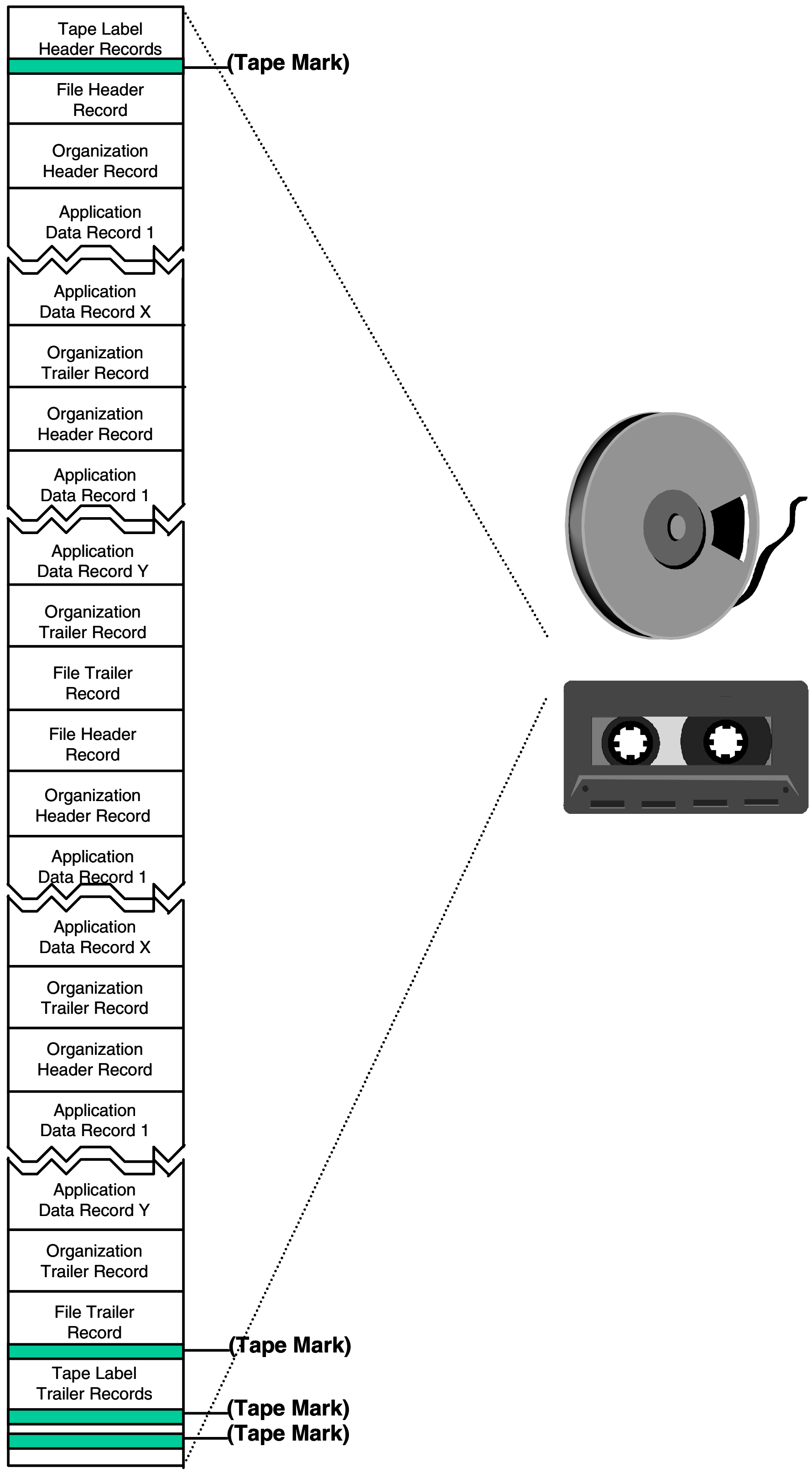
- If records need to be grouped by institution, as is the case for Positive Balance File (PBF) refreshes. All records for a particular institution are bracketed by an organization header and trailer for that institution. There are as many of these groupings as there are institutions to be refreshed. If records need not be grouped by institution, one organization header is required immediately following each file header, and one organization trailer is required immediately preceding each file trailer.
- If records need to be grouped by account number, as is the case for Transaction History File (THF) refreshes. THF refresh records must be organized according to the refresh group, FIID, and account number range combinations configured in the Transaction History Configuration File (THCF). Records for each of these refresh group, FIID, and account number range combinations must be bracketed by an organization header and trailer. If all institutions use a single THF, one organization header is required immediately following each file header, and one organization trailer is required immediately preceding each file trailer.

The last file trailer record on the refresh tape indicates in the NXT-FILE-IND field that there are no more application files to be refreshed.

Application data records must appear on the host-generated refresh tape in ascending order by primary key for the Enscribe file or SQL table being refreshed.

Note: Each Enscribe record or SQL row being refreshed can be impacted only by a single record in any one file contained on the host-generated tape.

Refresh tapes cannot contain records used by different types of Refresh processes (i.e., Base National Negative, INAS National Negative, Enscribe, and SQL) on the same tape.


Tape Label
Header Records
(Tape Mark)
File Header
Record
Organization
Header Record
Application
Data Record 1
Application
Data Record X
Organization
Trailer Record
Organization
Header Record
Application
Data Record 1
Application
Data Record Y
Organization
Trailer Record
File Trailer
Record
File Header
Record
Organization
Header Record
Application
Data Record 1
Application
Data Record X
Organization
Trailer Record
Organization
Header Record
Application
Data Record 1
Application
Data Record Y
Organization
Trailer Record
File Trailer
Record
(Tape Mark)
Tape Label
Trailer Records
(Tape Mark)
(Tape Mark)

Record Lengths

Data records on the refresh tape can be variable-length. Data records include file headers and trailers, organization headers and trailers, and application data records. The host must add block and record length fields to these data records for use by the Refresh process. The Refresh process uses information from block and record length fields for checking record validity and deblocking the records from the refresh file. The first record in each data block must start with a field indicating the block length and a field indicating the length of the data record. Each subsequent record in a data block must start with a record length field.

For example, the first record in each data block appears as follows:

Block Length	Record Length	Data Record

Each remaining record in the data block appears as follows:

Record Length	Data Record

If the data on the tape includes binary fields, the block length and record length fields are in binary. Each of these fields is 4 bytes long.

If the data on the tape is in display format, the block length and record length fields are in display format. Each of these fields is 6 bytes long.

The value in the record length field is the length of the data record plus the length of the record length field. The value in the block length field is the sum of the values in all record length fields in the block plus the length of the block length field itself.

Tape Blocking

The Refresh process supports variable length records and variable length blocks. Block lengths on refresh tapes can be up to 32,766 bytes. Each block must contain at least one record.

Using the largest block size possible is generally recommended because the larger the block size, the faster a refresh can be run. A larger block size also decreases the amount of overhead associated with the block length fields. For example, if a refresh tape contained twenty records and each record was in its own block, there would be twenty block length fields on the tape—one for each block. This amounts to 80 bytes of information for a binary format refresh, and 120 bytes of information for a display format refresh. If the same records were all placed in one block on the refresh tape, there would only be one block length field. The block length field is 4 bytes for a binary format refresh and 6 bytes for a display format refresh.

Because both the records and the blocks are variable length, there is no overhead associated with setting the block size too large. Each record is only the length that it needs to be—it is not padded so that all records are the same fixed length. Likewise, each block can be a different length, with the length being determined by the lengths of the records in the block.

Tape Density

The tape density for refresh tapes is tape drive dependent. A density that is supported by both the host and the HP NonStop processor tape drive should be selected when creating refresh tapes.

Tape Labels

The tape label types supported depend on whether the HP NonStop Kernel TAPECOM product or the Refresh process is processing tape labels.

If the HP NonStop Kernel TAPECOM product is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value N), the following tape labels are supported:

- IBM/MVS (IBM)
- ANSI (ANS)
- Unlabeled (NON)

If the BASE24 Refresh process is processing tape labels (i.e., the BASE24-TAPE-LBL-PROC-USED param in the LCONF contains the value Y), the following tape labels are supported:

- IBM/MVS (IBM)
- ANSI (ANS)
- IBM DOS (DOS)
- Burroughs 5500 (BUR)
- Burroughs Large Scale (BLS)
- Unlabeled (NON)

If the HP NonStop Kernel TAPECOM product is processing the tape labels and a tape label type is specified that is not supported by the HP NonStop Kernel TAPECOM product, BASE24 Refresh processes the tape labels.

Disk Input File Organization and Content

The organization and structure of a typical host-generated refresh disk file is basically the same as that for a tape input file, with the following exceptions:

- Disk input files should not include tape label records
- Block lengths (either binary or display format) should not be included in front of each block
- Record lengths (either binary or display format) should not be included in front of each record

The disk file can be entry-sequenced, key-sequenced, or relative, subject to the limitations identified below:

Structure	Comments
Entry-sequenced	Supported for all refreshes. Entry-sequenced files offer the best performance, since the Refresh process can use bulk I/O.
Key-sequenced	Supported for refreshes of nonsegmented files. If a key-sequenced structure is used, a key must be specified within the refresh input file record format.
Relative	Supported for all refreshes. Although the input file records may vary logically in length, each physical record is a fixed length in the disk file. This can result in wasted disk space.

Record Segments

Many BASE24 files contain segmented records that allow for the integration of various combinations of BASE24 products and features. This segmentation affects the refresh input file records in that different segments may be required in the refresh records depending on the combination of BASE24 products and features in use by each institution.

This discussion refers to the segmented files BASE24 uses for authorization purposes. This includes the Cardholder Authorization File (CAF), Negative Card File (NEG), and Positive Balance File (PBF). The Enscribe Refresh process handles all three of these files.

How the Refresh Process Determines Which Segments Are Required

The Enscribe Refresh process determines the segments that must be present in a particular institution's refresh record based on the FIID AUTH FILE SEGMENT INDICATORS fields on screens 5 and 6 of the Institution Definition File (IDF). When processing the Refresh File Header (RFH) for a file, the Refresh process reads the IDF record for each institution in the refresh group and builds a table containing the values from the FIID AUTH FILE SEGMENT INDICATORS fields.

During refresh processing, the Refresh process then uses the FIID in the refresh record—the value from the FIID field in the Base segment of the CAF or NEG refresh records, or for PBF refreshes, the value in the CRD-ISS field in the Refresh Bank Header—to select an entry from the table. From that entry, the Refresh process determines the segments to expect for that institution's refresh records and the appropriate disk segments to build for that institution's refreshed file records.

For example, if an institution uses BASE24-pos, but not the BASE24-pos Address Verification add-on product, the POS field on IDF screen 5 for that institution would be set to a value of Y, and the ADDR VERIF field would be set to a value of N. In this case, the Refresh process would require that all CAF refresh records for that institution include a BASE24-pos segment, but would not allow these records to include an Address Verification segment. On the other hand, if that institution were set up to use address verification, both the POS and ADDR VERIF fields would be set to a value of Y, and the Refresh process would require that all CAF refresh records for that institution include both a BASE24-pos segment and an Address Verification segment.

How the Refresh Process Handles Variable-Length Segments

Some segments store data that is not required for every disk record. For example, if an institution supports preauthorized holds and stores the hold information in the Cardholder Authorization File (CAF), not every CAF record for that institution will include a Preauthorized Holds segment—only those CAF records that actually have preauthorized holds associated with them.

However, if an institution supports a particular product or feature, the refresh records for that institution must include the corresponding segments, even if they have no data. In the above example, CAF refresh records that do not have preauthorized holds associated with them must still include a Preauthorized Holds segment. In this case, the segment should contain a length and count field only.

The Refresh process handles the above type of segments as variable-length segments in the refresh records. Variable-length segments might carry no data (just a length indicator and possibly a count) or they might carry varying amounts of data up to their maximum size, depending on the type of segment. If a variable-length segment does not carry any actual data, the Refresh process will not add the segment to the corresponding refreshed disk record.

Refresh Record Segment Summary

The following tables summarize the CAF, NEG, and PBF segments that can be present in a refresh record, along with the corresponding IDF screen 5 and 6 fields that control their required presence or absence. The tables also indicate whether the segments are variable length for the purposes of refresh and, for variable-length segments, how the LGTH fields in the segments are set for each record based on whether the segment contains data for the record.

Cardholder Authorization File (CAF) Segments				
			LGTH Field Setting	
Segment	**IDF Screen 5 and 6 Field**	**Variable Length**	**Without Data**	**With Data**
cafxbase-display	BASE	No		
atmcafx-display	ATM	No		
poscafx-display	POS	No		
emvcafx-display	EMV	No		
crdcafx-display	CMS	No		
ssbbcafx-display	SSB BASE	Yes	0004	0184
ssbccafx-display	SSB CHECK	No		
avcafx-display	ADDR VERIF	No		
preauthcafx-display	PREAUTH HOLD	Yes	0006*	Based on the number of holds, as shown below. **Holds** **Length** 00 0006 01 0072 02 0138 03 0204 04 0270 05 0336 06 0402 07 0468 08 0534 09 0600 10 0666
ncdcafx-display	NON-CURR DSP	No	0004	0162

Cardholder Authorization File (CAF) Segments				
			LGTH Field Setting	
Segment	IDF Screen 5 and 6 Field	Variable Length	Without Data	With Data
acctcafx-display	Always present—not controlled by the IDF	Yes	0006*	Based on the number of accounts, as shown below. **Accounts** **Length** 00 0006 01 0040 02 0074 03 0108 04 0142 05 0176 06 0210 07 0244 08 0278 09 0312 10 0346 11 0380 12 0414 13 0448 14 0482 15 0516 16 0550

* Includes a PRE-AUTH-CNT or ACCT-CNT value of 00.

Negative Card File (NEG) Segments				
			LGTH Field Setting	
Segment	IDF Screen 5 Field	Variable Length	Without Data	With Data
negxbase-display	BASE	No		
ssbbnegx-display	SSB BASE	Yes	0004	0184
ssbcnegx-display	SSB CHECK	Yes	0004	0014

Positive Balance File (PBF) Segments				
			LGTH Field Setting	
Segment	**IDF Screen 5 Field**	**Variable Length**	**Without Data**	**With Data**
pbfxbase-display	BASE	No		
pospbfx-display	POS	No		
tlrpbfx-display	TELLER	No		
tbpbfx-display	TELEBANKING	Yes	0004	0084
crlinepbfx-display	PBF CR LINE	Yes	0004	0026
nampbfx-display	PBF SHORT NM	No		
csfcpbfx-display	CUST SRVC	Yes	0004	Based on the number of cycles, as shown below. **Cycles** **Length** 00 0070 01 0100 02 0130 03 0160 04 0190 05 0220 06 0250 07 0280 08 0310 09 0340 10 0370 11 0400 12 0430

Positive Balance File (PBF) Segments				
			LGTH Field Setting	
Segment	**IDF Screen 5 Field**	**Variable Length**	**Without Data**	**With Data**
preauthpbfx-display	PREAUTH HOLD	Yes	0006*	Based on the number of holds, as shown below. **Holds** **Length** 00 0006 01 0050 02 0094 03 0138 04 0182 05 0226 06 0270 07 0314 08 0358 09 0402 10 0446

* Includes a PRE-AUTH-CNT value of 00.

Data Record Layouts

Layouts for the following refresh input file control data records can be found in this appendix:

- File Header Records
- Organization Header Records
- Organization Trailer Records
- File Trailer Records

Layouts for refresh input file application data records are located on the product code subvolumes shown below, where *xx* is the number of the current release, and can be printed using the DDLDOC program. The printed DDLDOC output for many of these files is also provided on the BASE24 Product Documentation CD-ROM.

File Acronym	**Subvolume**	**Definition Name**
CACT	OC*xx*CUST	RFCACTDS
CAF	BA*xx*DDL	CAFXD
CCF	AT*xx*CDDL	CCFX
CCIF	OC*xx*CS	CCIFXDS
CCMF	OC*xx*CS	CCMFXDS
CHF	BP*xx*CRTA	CHX
CPIT	OC*xx*PSNL	RFCPITDS
CSF	AT*xx*CDDL	CSFX
CSTT	OC*xx*CUST	RFCSTTDS
MBF	MA*xx*DDL	MBFX*
NBF	TR*xx*DDL	NBFX-DISPLAY
NEG	BA*xx*DDL	NEGXD
PBF	BA*xx*DDL	PBFXD
PIT	OC*xx*PSNL	RFPITDS
SDF	SW*xx*SDF	SDFX
SEMF	PI*xx*DDL	SEMFXD
SPF	BA*xx*DDL	SPFX-DISPLAY
THF	OC*xx*HIST	THFXDS
WHFF	TR*xx*DDL	WHFX-DISPLAY

* The Refresh process sends BASE24-mail request messages to the Mail process, which creates and stores the MBF record.

Data Format

BASE24 files and tables reside on disk in a standard format that includes binary and ASCII fields.

The data format of the input file can be either EBCDIC or ASCII. EBCDIC is the default format used by the Refresh process. If you want to use the ASCII format, you must specify it on Refresh screen 1 or in the REFRESH text command when the refresh is started.

Note: When the ASCII format is used, amount fields that can contain a negative value must be presented in the zoned data format. Refer to the “Zoned Data Format” topic in this appendix.

The following table shows the valid combinations of tape label type and data format. These options are specified in the TAPE LABEL TYPE and CHARACTER SET fields on Refresh screen 1.

Tape Label Type	Character Set	Description
ANSI	EBCDIC	Not supported.
ANSI	ASCII	Both labels and data are in ASCII.
IBM	EBCDIC	Both label and data are in EBCDIC.
IBM	ASCII	Labels are in EBCDIC and data is in ASCII.
DOS, BUR, or BLS	EBCDIC	Both label and data are in EBCDIC.
DOS, BUR, or BLS	ASCII	Both labels and data are in ASCII.
NON	EBCDIC	No labels are present; data is in EBCDIC.
NON	ASCII	No labels are present; data is in ASCII.

Zoned Data Format

In an ASCII input file, the Enscribe Refresh process requires the zoned data format for fields that can contain a negative value. The zoned data format replaces the least significant digit in an amount field with an alphanumeric character. This alphanumeric character serves two purposes—it represents the digit and whether the amount is positive or negative.

The following fields can have the zoned data format:

- Amount fields in the Positive Balance File (PBF)
- Amount fields in the No Book File (NBF)
- Amount fields in the Warning/Hold/Float File (WHFF)
- AMT field in the Organization Trailer

The following table shows the alphanumeric characters that can be used in place of the least significant digit. The positive or negative sign and numeric value are given for each alphanumeric value. Upper-case alphabetic characters must be used.

{	=	+ 0	}	=	– 0
A	=	+ 1	J	=	– 1
B	=	+ 2	K	=	– 2
C	=	+ 3	L	=	– 3
D	=	+ 4	M	=	– 4
E	=	+ 5	N	=	– 5
F	=	+ 6	O	=	– 6
G	=	+ 7	P	=	– 7
H	=	+ 8	Q	=	– 8
I	=	+ 9	R	=	– 9

As an example, a zoned-data entry of 12C is converted to an amount of + 123 in the BASE24 database and a zoned-data entry of 12L is converted to an amount of – 123 in the BASE24 database.

File Header Records

Record: File Header

Length: 150 bytes

Description: The file header identifies the file to be refreshed and whether a full-file or partial-file refresh is to be run. It also contains various processing parameters required by the Refresh process. One file header is required per application file in the refresh input file.

Refer to Record Length and Tape Blocking discussions presented earlier in this appendix for additional information that must be included on tapes.

Layout: The file header record required by the BASE24 Refresh process is described below.

Position	Level	Field Name and Description	Data Type
1–150		RFH	
1–9	02	REC-CNT The record number. If this is the first record in the refresh input file, the value in this field should be 000000001. If this is not the first record in the refresh input file, the value in this field should be one greater than that in the previous record, which must be a file trailer record.	PIC 9(9)
10–11	02	REC-TYP The record type. This field must contain the value FH (File Header).	PIC X(2)

Position	Level	Field Name and Description	Data Type
12	02	REF-TYP	PIC X(1)

The type of refresh for which the records in the refresh input file are to be used. Valid values are as follows:

0 = Full-file refresh. The Refresh process creates an entirely new file and adds to that file every record in the refresh input file.
1 = Partial-file refresh. The Refresh process updates the existing file using records in the refresh input file.

Position	Level	Field Name and Description	Data Type
13–14	02	APPL	PIC X(2)

Indicates the BASE24 file or table being refreshed. Valid values are as follows:

AC = Customer/Account Relation Table (CACT)
CC = Positive Balance File (PBF) for credit accounts only
CF = Cardholder Authorization File (CAF)
CH = Card History File (CHF)
CI = Customer/Card Information File (CCIF)
CM = Customer/Card Memo File (CCMF)
CO = Corporate Check File (CCF)
CP = Customer/Personal ID Relation Table (CPIT)
CS = Check Status File (CSF)
CT = Customer Table (CSTT)
DA = PBF for checking or NOW accounts only
MB = Mail Box File (MBF)
NB = No Book File (NBF)
NF = Negative Card File (NEG)
PF = PBF for all account types
PT = Personal Information Table (PIT)
SD = Switch Dispute File (SDF)
SM = Scoring Engine Master File (SEMF)
SP = Stop Payment File (SPF)
SV = PBF for savings accounts only
TH = Transaction History File (THF)
WH = Warning/Hold/Float File (WHFF)

A single PBF can contain multiple account types, or a PBF can contain a single account type.

Position	Level	Field Name and Description	Data Type
15–18	02	GRP The refresh group to be refreshed. Refresh groups identify groupings of institutions that share the same physical files. Institutions are assigned to refresh groups using the value in the REFR-GRP field in the Base segment of the Institution Definition File (IDF).	PIC X(4)
19–26	02	TAPE-DAT The date, in YYYYMMDD format, that this input file was created.	PIC X(8)
27–30	02	TAPE-TIM The time, in hhmm format, that this input file was created.	PIC 9(4)
31–34	02	LN The identifier assigned to the logical network in which the records being refreshed reside.	PIC X(4)
35–36	02	REL-NUM The ACI-assigned release number identifying the BASE24 software release with which the file being refreshed is compatible. The valid values are as follows: 11 = Release 1.*x*. This value is used for previous release refreshes of the THF. 50 = Release 5.*x*. This value is used for previous release refreshes of files other than the THF. 60 = Release 6.0 and newer. This value is used for all current release refreshes. **Note**: The Refresh process supports previous release (i.e., BASE24-atm release 5.*x*, BASE24-mail release 5.*x*, BASE24-pos release 5.*x*, BASE24-telebanking release 1.*x*, and BASE24-teller release 5.*x*) file formats for all files that can be refreshed. When the previous release file format is used, so are previous release file header and file trailer record formats.	PIC 9(2)

Position	Level	Field Name and Description	Data Type
37–38	02	PART-NUM Indicates the file partition to be refreshed, if the file is partitioned. The value in this field is only used for full-file refreshes; it should be set to blanks for partial-file refreshes. Valid values are as follows: ƀƀ = The input file records represent the entire file, where ƀ is a blank. 00 = The input file records represent those records to be placed in the primary file partition. 01–16 = The input file records represent those records to be placed in the Base National Negative, INAS National Negative, or Enscribe file partition indicated. 01–99 = The input file records represent those records to be placed in the SQL table partition indicated.	PIC X(2)
39–64	02	ATM The following fields contain information about the last extract of the BASE24-atm Transaction Log File (TLF). These fields are used for CAF and PBF refreshes with impacting when the BASE24-atm product is part of the BASE24 system.	
39–44	04	LST-EXTR-DAT The date, in YYMMDD format, of the last TLF that was fully extracted. A value is required in this field when performing a CAF or PBF refresh with impacting and the BASE24-atm product is installed. The value in this field cannot be overridden.	PIC X(6)
45–52	04	IMP-STRT-DAT The date, in YYYYMMDD format, of the last TLF record extracted. The Refresh process can use the timestamp specified in this field and the IMP-STRT-TIM field to calculate where impacting should begin in the initial TLF.	PIC X(8)

Position	Level	Field Name and Description	Data Type
		A value is required in this field when performing a CAF or PBF refresh with impacting and the BASE24-atm product is installed. The value in this field can be overridden when the Refresh process is started by specifying a date in the ATM field on Refresh screen 1, or by specifying an atm-impact-start date/time parameter in the network control command. When performing multiple partial extracts of the TLF, the Super Extract process places the timestamp value of the last TLF record extracted in the first 20 bytes of the USER-FLD2 field in the TLF extract file trailer record. The value for this field can be echoed from the first eight bytes of the USER-FLD2 field in the TLF extract file trailer record. All blanks or all nines in this field and in the IMP-STRT-TIM field indicates that impacting should start at the beginning of the TLF.	
53–64	04	IMP-STRT-TIM	PIC X(12)
		The time, in hhmmssmmmmmm format, of the last TLF record extracted. The Refresh process can use the timestamp specified in this field and the IMP-STRT-DAT field to determine where impacting should begin in the initial TLF. A value is required in this field when performing a CAF or PBF refresh with impacting and the BASE24-atm product is installed. The value in this field can be overridden when the Refresh process is started by specifying a date in the ATM field on Refresh screen 1, or by specifying an atm-impact-start date/time parameter in the network control command. When performing multiple partial extracts of the TLF, the Super Extract process places the timestamp value of the last TLF record extracted in the first 20 bytes of the USER-FLD2 field in the TLF extract file trailer record. The value for this field can be echoed from bytes 9 through 20 of the USER-FLD2 field in the TLF extract file trailer record. All blanks or all nines in this field and in the IMP-STRT-DAT field indicates that impacting should start at the beginning of the TLF.	

Position	Level	Field Name and Description	Data Type
65–90	02	POS The following fields contain information about the last extract of the POS Transaction Log File (PTLF). These fields are used for PBF refreshes with impacting when the BASE24-pos product is part of the BASE24 system.	
65–70	04	LST-EXTR-DAT The date, in YYMMDD format, of the last PTLF that was fully extracted. A value is required in this field when performing a PBF refresh with impacting and the BASE24-pos product is installed. The value in this field cannot be overridden.	PIC X(6)
71–78	04	IMP-STRT-DAT The date, in YYYYMMDD format, of the last PTLF record extracted. The Refresh process can use the timestamp specified in this field and the IMP-STRT-TIM field to determine where impacting should begin in the initial PTLF. A value is required in this field when performing a PBF refresh with impacting and the BASE24-pos product is installed. The value in this field can be overridden when the Refresh process is started by specifying a date in the POS field on Refresh screen 1, or by specifying a pos-impact-start date/time parameter in the network control command. When performing multiple partial extracts of the PTLF, the Super Extract process places the timestamp value of the last PTLF record extracted in the first 20 bytes of the USER-FLD2 field in the PTLF extract file trailer record. The value for this field can be echoed from the first eight bytes of the USER-FLD2 field in the PTLF extract file trailer record. All blanks or all nines in this field and in the IMP-STRT-TIM field indicates that impacting should start at the beginning of the PTLF.	PIC X(8)

Position	Level	Field Name and Description	Data Type
79–90	04	IMP-STRT-TIM The time, in hhmmssmmmmmm format, of the last PTLF record extracted. The Refresh process can use the timestamp specified in this field and the IMP-STRT-DAT field to determine where impacting should begin in the initial PTLF. A value is required in this field when performing a PBF refresh with impacting and the BASE24-pos product is installed. The value in this field can be overridden when the Refresh process is started by specifying a date in the POS field on Refresh screen 1, or by specifying a pos-impact-start date/time parameter in the network control command. When performing multiple partial extracts of the PTLF, the Super Extract process places the timestamp value of the last PTLF record extracted in the first 20 bytes of the USER-FLD2 field in the PTLF extract file trailer record. The value for this field can be echoed from bytes 9 through 20 of the USER-FLD2 field in the PTLF extract file trailer record. All blanks or all nines in this field and in the IMP-STRT-DAT field indicates that impacting should start at the beginning of the PTLF.	PIC X(12)
91–116	02	TLR The following fields contain information about the last extract of the Teller Transaction Log File (TTLF). These fields are used for NBF, PBF, SPF, or WHFF refreshes with impacting when the BASE24-teller product is part of the BASE24 system.	
91–96	04	LST-EXTR-DAT The date, in YYMMDD format, of the last TTLF that was fully extracted. A value is required in this field when performing an NBF, PBF, SPF, or WHFF refresh with impacting and the BASE24-teller product is installed. The value in this field cannot be overridden.	PIC X(6)

Position	Level	Field Name and Description	Data Type
97–104	04	IMP-STRT-DAT The date, in YYYYMMDD format, of the last TTLF record extracted. The Refresh process can use the timestamp specified in this field and the IMP-STRT-TIM field to determine where impacting should begin in the initial TTLF. A value is required in this field when performing an NBF, PBF, SPF, or WHFF refresh with impacting and the BASE24-teller product is installed. The value in this field can be overridden when the Refresh process is started by specifying a date in the TLR field on Refresh screen 1, or by specifying a teller-impact-start date/time parameter in the network control command. When performing multiple partial extracts of the TTLF, the Super Extract process places the timestamp value of the last TTLF record extracted in the first 20 bytes of the USER-FLD2 field in the TTLF extract file trailer record. The value for this field can be echoed from the first eight bytes of the USER-FLD2 field in the TTLF extract file trailer record. All blanks or all nines in this field and in the IMP-STRT-TIM field indicates that impacting should start at the beginning of the TTLF.	PIC X(8)
105–116	04	IMP-STRT-TIM The time, in hhmmssmmmmmm format, of the last TTLF record extracted. The Refresh process can use the timestamp specified in this field and the IMP-STRT-DAT field to determine where impacting should begin in the initial TTLF. A value is required in this field when performing an NBF, PBF, SPF, or WHFF refresh with impacting and the BASE24-teller product is installed. The value in this field can be overridden when the Refresh process is started by specifying a date in the TLR field on Refresh screen 1, or by specifying a teller-impact-start date/time parameter in the network control command. When performing multiple partial extracts of the TTLF, the Super Extract process places the timestamp value of the last TTLF record extracted in the first 20 bytes of the USER-FLD2	PIC X(12)

Position	Level	Field Name and Description	Data Type

field in the TTLF extract file trailer record. The value for this field can be echoed from bytes 9 through 20 of the USER-FLD2 field in the TTLF extract file trailer record.

All blanks or all nines in this field and in the IMP-STRT-DAT field indicates that impacting should start at the beginning of the TTLF.

117	02	IMP-TYP	PIC X(1)

Indicates whether CAF, NBF, PBF, SPF, or WHFF records are to be impacted. Valid values are as follows:

0 = Do not impact refreshed records.
1 = Impact refreshed records.

Impacting refers to the process by which transactions occurring since the last transaction log file extract are applied to the new records created or updated by the Refresh process.

The value in this field is used only for CAF, CUST, NBF, PBF, SPF, or WHFF refreshes. Otherwise, it should be set to a blank or zero.

118	02	CAF-EXPNT	PIC X(1)

The power of 10 to be used when applying the following CAF amount fields:

Base segment fields. GRP-LMT.TTL-WDL-LMT, GRP-LMT.OFFL-WDL-LMT, GRP-LMT.TTL-CCA-LMT, GRP-LMT.OFFL-CCA-LMT, GRP-LMT.AGGR-LMT, and GRP-LMT.OFFL-AGGR-LMT.

BASE24-atm segment fields. GRP-LMT.TTL-WDL-LMT, GRP-LMT.OFFL-WDL-LMT, GRP-LMT.TTL-CCA-LMT, and GRP-LMT.OFFL-CCA-LMT.

BASE24-pos segment fields. GRP-LMT.TTL-PUR-LMT, GRP-LMT.OFFL-PUR-LMT, GRP-LMT.TTL-CCA-LMT, GRP-LMT.OFFL-CCA-LMT, GRP-LMT.TTL-WDL-LMT, GRP-LMT.OFFL-WDL-LMT, GRP-LMT.TTL-RFND-CR-LMT, and GRP-LMT.OFFL-RFND-CR-LMT.

Position	Level	Field Name and Description	Data Type
		BASE24-atm self-service banking check segment fields. GRP-LMT.TTL-CHK-LMT, GRP-LMT.OFFL-CHK-LMT, GRP-LMT.TTL-CSF-CHK-LMT, and GRP-LMT.OFFL-CSF-CHK-LMT. The Refresh process multiplies the CAF amount fields on the refresh tape or disk by the appropriate power of 10 before applying them to the CAF. Valid values are as follows: b̸ = Not applicable. This is not a CAF refresh. (b̸ indicates a space). 0 = Use the CAF amount field as it appears in the message. 1 = Multiply the refresh amount fields by 10. 2 = Multiply the refresh amount fields by 100. 3 = Multiply the refresh amount fields by 1000.	
119	02	PRE-AUTH-SUPPORT Indicates whether the host provides preauthorization information. This field is used for partial refreshes of the CAF and PBF. It specifies whether the Refresh process retains the current preauthorization holds in the CAF or PBF, or replaces them with the information from the input file. Valid values are as follows: 0 = Host does not provide preauthorization holds. 1 = Host provides preauthorization holds.	PIC X(1)
120–124	02	USER-FLD2 Not used. This field should be left blank.	PIC X(5)
125–150	02	TB The following fields contain information about the last extract of the ITS Transaction Log File (ITLF). These fields are used for PBF refreshes with impacting when the BASE24-telebanking product is part of the BASE24 system.	

Position	Level	Field Name and Description	Data Type
125–130	04	LST-EXTR-DAT The date, in YYMMDD format, of the last ITLF that was fully extracted. A value is required in this field when performing a PBF refresh with impacting and the BASE24-telebanking product is installed. The value in this field cannot be overridden.	PIC X(6)
131–138	04	IMP-STRT-DAT The date, in YYYYMMDD format, of the last ITLF record extracted. The Refresh process can use the timestamp specified in this field and the IMP-STRT-TIM field to determine where impacting should begin in the initial ITLF. A value is required in this field when performing a PBF refresh with impacting and the BASE24-telebanking product is installed. The value in this field can be overridden when the Refresh process is started by specifying a date in the TB field on Refresh screen 1, or by specifying a telebanking-impact-start date/time parameter in the network control command. When performing multiple partial extracts of the ITLF, the Super Extract process places the timestamp value of the last ITLF record extracted in the first 20 bytes of the USER-FLD2 field in the ITLF extract file trailer record. The value for this field can be echoed from the first eight bytes of the USER-FLD2 field in the ITLF extract file trailer record. All blanks or all nines in this field and in the IMP-STRT-TIM field indicates that impacting should start at the beginning of the ITLF.	PIC X(8)
139–150	04	IMP-STRT-TIM The time, in hhmmssmmmmmm format, of the last ITLF record extracted. The Refresh process can use the timestamp specified in this field and the IMP-STRT-DAT field to determine where impacting should begin in the initial ITLF. A value is required in this field when performing a PBF refresh with impacting and the BASE24-telebanking product is installed. The value in this field can be overridden when the	PIC X(12)

Position	Level	Field Name and Description	Data Type
		Refresh process is started by specifying a date in the TB field on Refresh screen 1, or by specifying a telebanking-impact-start date/time parameter in the network control command. When performing multiple partial extracts of the ITLF, the Super Extract process places the timestamp value of the last ITLF record extracted in the first 20 bytes of the USER-FLD2 field in the ITLF extract file trailer record. The value for this field can be echoed from bytes 9 through 20 of the USER-FLD2 field in the ITLF extract file trailer record. All blanks or all nines in this field and in the IMP-STRT-DAT field indicates that impacting should start at the beginning of the ITLF.	

Organization Header Records

Record: Organization Header

Length: 44 bytes

Description: The organization header identifies, by FIID, the institution whose files are to be refreshed. One organization header is required for each set of institution records in the refresh input file. If records are not grouped by institution, only one organization header is required in the refresh input file.

The organization header also identifies the end range of account numbers in a Transaction History File (THF) refresh. When multiple THFs are used, one organization header is required for each THF being refreshed.

Refer to the Record Length and Tape Blocking discussions presented earlier in this appendix for additional information that must be included on tapes.

Layout: The organization header record required by the BASE24 Refresh process is described below.

Position	Level	Field Name and Description	Data Type
1–44		RBH	
1–9	02	REC-CNT The record number. The value in this field should be one greater than that in the previous record, which must be a file header or an organization trailer.	PIC 9(9)
10–11	02	REC-TYP The record type. This field must contain the value BH (organization header record).	PIC X(2)

Position	Level	Field Name and Description	Data Type
12–15	02	CRD-ISS The FIID of the institution whose refresh records follow. The entry in this field is left-justified and blank filled on the right. It is used only with PBF and THF refreshes. The FIID is used for PBF refreshes because it is part of the primary key used to locate records in the PBF. The FIID is used for THF refreshes because it is part of the primary key used to locate records in the Transaction History Configuration File (THCF). Records in the THCF identify which THF contains history records for a particular account.	PIC X(4)
16–43	02	END-RANGE The end of the range of account numbers that can appear in the refresh records follow. This value is used only with THF refreshes. The value in positions 1 through 19 of this field is right-justified and zero filled on the left. Positions 20 through 28 of this field are blank. The end of the range of account numbers is used for THF refreshes because it is part of the primary key used to locate records in the Transaction History Configuration File (THCF). Records in the THCF identify which THF contains history records for a particular account.	PIC X(28)
44	02	USER-FLD Not used. This field should be left blank.	PIC X(1)

Organization Trailer Records

Record: Organization Trailer

Length: 38 bytes

Description: The organization trailer signals the end of an institution's refresh records. It contains a verification count and amount for the institution's records, which the Refresh process uses to check its processing. One organization trailer is required for each set of institution records in the refresh input file. If records are not grouped by institution, only one organization trailer is required in the refresh input file.

Refer to Record Length and Tape Blocking discussions presented earlier in this appendix for additional information that must be included on tapes.

Layout: The organization trailer record required by BASE24 is described below.

Position	Level	Field Name and Description	Data Type
1–38		RBT	
1–9	02	REC-CNT The record number. The value in this field should be one greater than that in the previous record.	PIC 9(9)
10–11	02	REC-TYP The record type. This field must contain the value BT (organization trailer record).	PIC X(2)
12–29	02	AMT A verification amount used with NBF, PBF, SPF, and WHFF refreshes. This amount is calculated as described below for each type of file being refreshed.	PIC 9(18)

Position	Level	Field Name and Description	Data Type

- For NBF refreshes, this field must contain the sum of all NBF transaction amounts included in the refresh input file for the organization. To derive this sum, a host must add together the amounts in the TRAN-AMT field of each NBF record.
- For PBF refreshes, this field must contain the sum of all PBF ledger balances included in the refresh input file for the organization. To derive this sum, a host must add together the LEDG-BAL amounts in the Base segment of all PBF records.
- For SPF refreshes, this field must contain the sum of all SPF stop payment amounts included in the refresh input file for the organization. To derive this sum, a host must add together the amounts in the AMT field of each SPF record.
- For WHFF refreshes, this field must contain the sum of all WHFF warning, hold, or deposit float amounts included in the refresh input file for the organization. To derive this sum, a host must add together the amounts in the PRIKEY.AMT field of each WHFF record.

Position	Level	Field Name and Description	Data Type
30–38	02	NUM-RECS	PIC 9(9)

The number of application records for the organization, not including the organization header or trailer records.

File Trailer Records

Record: File Trailer

Length: 24 bytes

Description: The file trailer signals the end of the refresh records. It contains a verification count, which is used by the Refresh process to check its processing. One file trailer is required for each refresh input file.

Refer to the Record Length and Tape Blocking discussions presented earlier in this appendix for additional information that must be included on tapes.

Layout: The file trailer record required by BASE24 is described below.

Position	Level	Field Name and Description	Data Type
1–24		RFT	
1–9	02	REC-CNT The record number for this record. The value in this field should be one greater than that in the previous record, which must be an organization trailer record.	PIC 9(9)
10–11	02	REC-TYP The record type. This field must contain the value FT (file trailer record).	PIC X(2)
12–20	02	NUM-RECS The total number of refresh records for the current application file, excluding file headers and trailers and organization headers and trailers.	PIC 9(9)

Position	Level	Field Name and Description	Data Type
21	02	NXT-FILE-IND Indicates whether another input file follows. Valid values are as follows: 0 = This is the last file in the input file. 1 = A subsequent file is present in the input file.	PIC X(1)
22–24	02	USER-FLD1 Not used. This field should be left blank.	PIC X(3)

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B: Extract Tape and Disk Specifications

This appendix contains the specifications for extract tapes and disks created by the BASE24 Super Extract process. It includes general extract tape and disk specifications, as well as the record layouts for the tape and file header and trailer records generated by the Super Extract process.

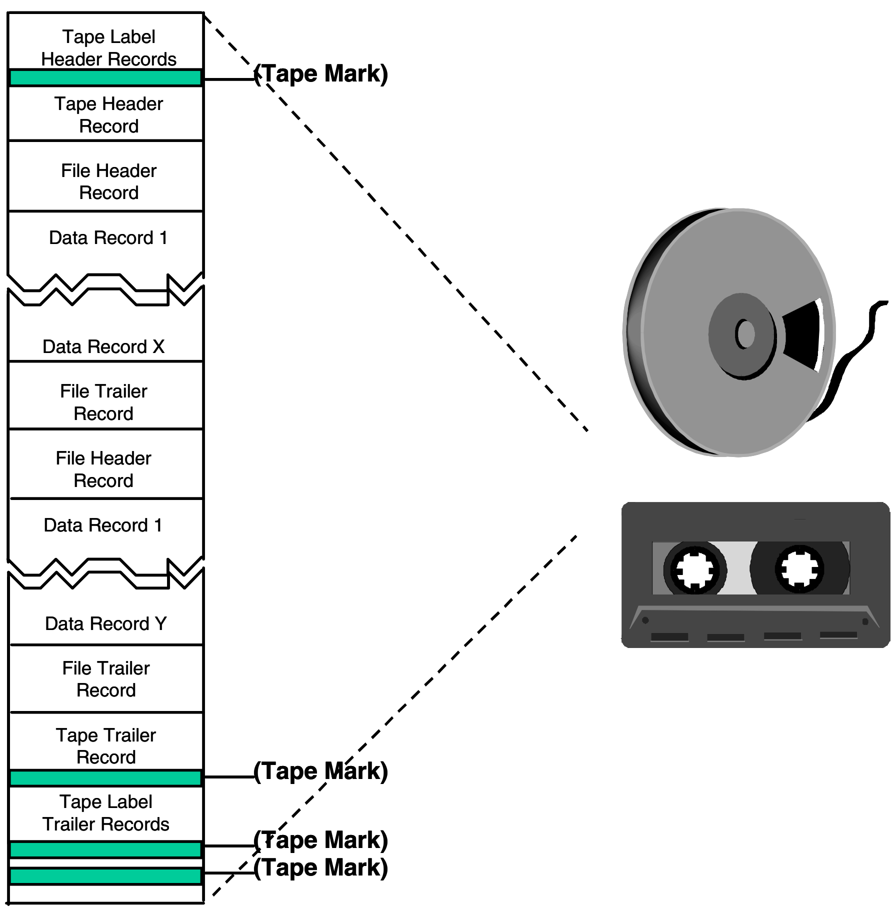
Tape Organization and Content

The organization of a BASE24 extract tape is illustrated below. Each extract tape contains data records and optional tape labels. The data records consist of control records (one tape header, one tape trailer, and a variable number of file headers and trailers) and extracted file records.

Extract records for multiple files can be contained on one BASE24 extract tape, depending on how the extract is set up. This is established through records in the Extract Configuration File (ECF).

Records from each extracted file are bracketed by a file header and trailer identifying the file. All extracted files on the tape are bracketed by a tape header and trailer.

Extract Tape Organization



Record Groupings

Except as noted in the following table, all records are extracted in the same order as they reside in the file. The first column of the table identifies the file, and the second column indicates the order in which records are extracted from that file.

File	Record Extraction Order
	BASE24 Shared Files
IDF	Sequenced by FIID.
ICF	Sequenced by Interchange Interface process name within the FIID (for the interchange).
ICFE	Sequenced by Interchange Interface process name within the FIID (for the interchange).
ILF	Extracted as they are encountered in the ILF for the interchange processes, for one FIID (optional), or for all FIIDs (optional).
SAF	Sequenced by date and time within product (optional) within DPC number.
	BASE24-atm Files
TLF	Grouped by FIID within refresh group or extracted as they are encountered in the TLF.
	BASE24-pos Files
PTLF	Grouped by FIID within refresh group or extracted as they are encountered in the PTLF.
PRDF	Sequenced by retailer ID within region within group number within FIID.
SVHF	Sequenced by date and time within FIID and card number.
	BASE24-telebanking Files
ITLF	Grouped by FIID within refresh group or extracted as they are encountered in the ITLF.

Record Format Conversions

The Super Extract process normally extracts records in a binary/ASCII format that matches the standard disk record format. These record formats are documented in DDL files located on product code subvolumes, and can be printed using the DDLDOC utility. For more information on using the DDLDOC utility, refer to the ***BASE24 Files and Record Structures Reference Manual***. The printed DDLDOC output for extract record formats is also provided on the BASE24 Product Documentation CD-ROM.

The Super Extract process also supports two types of record conversions which can have some specific applications for certain hosts. The first is ASCII-to-EBCDIC and the second is binary-to-ASCII. Each is described below.

ASCII-to-EBCDIC Conversion

The Super Extract process can convert values in ASCII fields in extract records to EBCDIC. The value in the CHARACTER SET field on Extract Configuration File (ECF) screen 1 controls this option. When this option is used, the Super Extract process converts the values in all ASCII fields to EBCDIC while leaving the binary fields untouched. Tape labels are always in EBCDIC. Converting the values in ASCII fields to EBCDIC does not affect the field lengths documented in the DDLs or this manual.

Note: When tokens are converted to EBCDIC, the eye catcher (!) is converted to a vertical bar (|).

Currently, a limitation exists on the ASCII-to-EBCDIC conversion for OMF extracts. OMF records are used in tracking file maintenance operations and, by design, contain images of other file records. The OMF extract converts OMF records accurately from ASCII to EBCDIC for file maintenance operations involving only the following files:

- Account Routing File (ARF)
- Account Type Table File (ATT)
- Acquirer Processing Code File (APCF)
- Address Table File (ADRT)
- Administrative Card File (ADMN)
- Bank Table (BANK)

- Bank Branch Table (BKBR)
- Bank Notice Table (BKNT)
- Bank Transaction Advice Table (BKAV)
- BASE24-atm Terminal Data Dynamic File—general data (ATDD1)
- BASE24-atm Terminal Data Static File—general data (ATDS1)
- BASE24-pos Terminal Data Dynamic File—general data (PTDD1)
- BASE24-pos Terminal Data Static File—general data (PTDS1)
- Billing Group Table (BLG)
- Billing Rate Table (BLRT)
- Billing Type Table (BLTY)
- Bin Sort File (BSF)
- Card Authorization Parameters File (CAPF)
- Card Prefix File (CPF)
- Cardholder Authorization File (CAF)
- Check Authorization Routing Table File (CART)
- Check Status File (CSF)
- Cleanup Configuration Table (CLPT)
- Corporate Check File (CCF)
- Customer Table (CSTT)
- Customer/Account Relation Table (CACT)
- Customer Allowed Transaction Table (CATT)
- Customer Billpay Table (CBPT)
- Customer/Card Information File (CCIF)
- Customer/Card Memo File (CCMF)
- Customer/Personal ID Relation Table (CPIT)
- Customer Vendor Table (CVND)
- Derivation Key File (KEYD)
- Enhanced Interchange Configuration File (ICFE)
- Exchange Rate File (ERF)
- External Message File (EMF)

- Extract Configuration File (ECF)
- File Control File (FCF)
- Host Configuration File (HCF)
- ICC Key File (KEYI)
- Institution Definition File (IDF)
- Institution Routing Configuration File (IRCF)
- Interchange Configuration File (ICF)
- Issuer Processing Code File (IPCF)
- Interface Security File (ISF)
- Interface Station Table File (IST)
- Internal Message Translation File (IMTF)
- ITS Interface Configuration File (IFCF)
- Key Authorization File (KEYA)
- Key File (KEYF)
- Key 6 File (KEY6)
- Log Configuration File (LCF)
- Logical Network Configuration File (LCONF)
- National Negative Card File (NNF)
- Negative Card File (NEG)
- No Book File (NBF)
- Override Response File (ORF)
- Personal Information Table (PIT)
- Positive Balance File (PBF)
- POS Retailer Definition File (PRDF)
- POS Terminal Data File (PTDF)
- Processing Code Definition File (PCDF)
- Processing Code Description File (PDF)
- Rate Table (RATE)
- Rate Group Table (RATG)
- Response Text Table (RTXT)

- Retry Table (RTRY)
- Routing Table File (RTBL)
- Scoring Engine Master File (SEMF)
- Security File (SEC)
- Service Provider Table (SVPT)
- Stop Payment File (SPF)
- Surcharge File (SURF)
- Switch Terminal File (STF)
- Teller Terminal Data File (TTDF)
- Teller Transaction File (TTF)
- Terminal Data File (TDF)
- Terminal Receipt File (TRF)
- Token File (TKN)
- Transaction Code File (TCF)
- Transaction History Configuration File (THCF)
- Transaction Type by Source Code Table (TTSC)
- Usage Accumulation File (UAF)
- User/FIID Relation File (UFIR)
- User/Product Files Relation File (UPFR)
- Vendor Table (VNDR)
- Vendor Branch Table (VNDB)
- Vendor Budget Category Table (VBUD)
- Vendor Mask Table (VNDM)
- Warning/Hold/Float File (WHFF)

Binary-to-ASCII Conversion

Some Burroughs host mainframes cannot accept binary fields on their tapes. The Super Extract process can convert the values in all binary fields on the tape to ASCII character format. The value in the NUMERIC FLD FORMAT field on ECF screen 1 controls this option. If this option is used, values in all binary fields are converted to ASCII. This includes tape and file headers and trailers, extract records, tokens, record headers, and block headers. The only data on the tape not included in the conversion are the IBM volume or other IBM labels. The ASCII-only format is available only for the files listed below.

- Interchange Log File (ILF)
- ITS Transaction Log File (ITLF)
- POS Transaction Log File (PTLF)
- Store-and-Forward File (SAF)
- Stored Value History File (SVHF)
- Transaction Log File (TLF)
- Teller Transaction Log File (TTLF)

Record structures for ASCII-only extract formats are also documented in DDL files located on product code subvolumes, and can be printed using the DDLDOC utility. For more information on using the DDLDOC utility, refer to the ***BASE24 Files and Record Structures Reference Manual***.

Block Size

The Super Extract process writes extract records to disk or tape in blocks. Each block contains one or more extract records, depending on the size of the block and the size of each record being extracted. During processing, the Super Extract process fills the block with extract records until no more can fit and then writes the block to tape or disk. The larger the block size, the more records each block can contain. This results in fewer blocks being written to disk or tape when extracting for a given number of records.

The maximum block size is configurable using the Extract Configuration File (ECF). This flexibility allows users to set the block size to take advantage of the processing capabilities of the host receiving the extract tape or disk file. Reducing the number of blocks written generally improves processing efficiency.

The value in the TAPE BLOCK SIZE field on ECF screen 1 determines the maximum size of each block written to tape or disk. There are two ranges of values for the TAPE BLOCK SIZE field—bytes and kilobytes (1 kilobyte equals 1024 bytes).

For extracts to tape, the Super Extract process processes the TAPE BLOCK SIZE field values as follows:

1–32	=	Block size in kilobytes. The Super Extract process multiplies the value entered by 1024 to determine the block size in bytes (e.g., 2 kilobytes equals 2048 bytes, 10 kilobytes equals 10240 bytes, and 32 kilobytes equals 32768 bytes).
33–32767	=	Block size in bytes. The Super Extract process uses the value entered (e.g., 2048 equals 2048 bytes or 2 kilobytes, and 9216 equals 9216 bytes or 9 kilobytes).

For extracts to disk, the value in the TAPE BLOCK SIZE field determines both the record size and the block size. For extracts to disk, the Super Extract process processes the TAPE BLOCK SIZE field values as follows:

1–3	=	Record/block size in kilobytes. The Super Extract process multiplies the value entered by 1024 to determine the record and block size in bytes (e.g., 2 kilobytes equals 2048 bytes, and 3 kilobytes equals 3072 bytes). The Super Extract process sets the Enscribe record size equal to the value entered, and sets the Enscribe block size to the value entered plus 24 (the additional 24 bytes in each block is used for a 24-byte block header).
4–32	=	Invalid values. The Super Extract process sets the Enscribe record size to 4072 bytes and the Enscribe block size to 4096 bytes.
33–4072	=	Record/block size in bytes. The Super Extract process sets the Enscribe record size equal to the value entered, and sets the Enscribe block size to the value entered plus 24 (the additional 24 bytes in each block is used for a 24-byte block header).
4073–32767	=	Invalid values. The Super Extract process sets the Enscribe record size to 4072 bytes and the Enscribe block size to 4096 bytes.

Block Headers

The tables below describe the block headers added to the front of each block placed on the extract tape or disk for both binary/ASCII and ASCII-only formats.

Binary/ASCII Format

For extracts in binary/ASCII format, four bytes are added to the front of each block to indicate the size of the block (in bytes). The following is a description of the four-byte block header:

Bytes	Description
1 and 2	A binary representation of the block length, which is the sum of the lengths of all the records and headers that fit in the block.
3 and 4	Null values (reserved).

ASCII-Only Format

For extracts in ASCII-only format, six bytes are added to the front of each block to indicate the size of the block (in bytes). The following is a description of the six-byte block header:

Bytes	Description
1–6	A right-justified, numeric display character representation of the block length, which is the sum of the lengths of all the records and headers that fit in the block. The value in this field is zero filled on the left.

Record Headers

The tables below describe the record headers added to the front of each record placed on the extract tape or disk (excluding tape label records on tape extracts) for binary/ASCII and ASCII-only formats.

Binary/ASCII Format

For extracts in the binary/ASCII format, six bytes are added to the front of each record. The following is a description of the six-byte record header:

Bytes	Description
1 and 2	A binary representation of the length of the record being extracted plus the length of this header.
3 and 4	Null values (reserved).
5 and 6	The type of record. Values are as follows: TH = Tape header FH = File header CF = Continuation record—first CL = Continuation record—last CR = Continuation record—intermediate DR = Data record ER = Error record (not used) TR = Truncation record FT = File trailer TT = Tape trailer

ASCII-Only Format

For extracts in the ASCII-only format, eight bytes are added to the front of each record. The following is a description of the eight-byte record header:

Bytes	Description
1–6	An ASCII representation of the length of the record being extracted plus the length of this header. The value in this field is right justified and zero filled on the left.

Bytes	Description
7 and 8	The type of record. Values are as follows: TH = Tape header FH = File header CF = Continuation record—first CL = Continuation record—last CR = Continuation record—intermediate DR = Data record ER = Error record (not used) TR = Truncation record FT = File trailer TT = Tape trailer

Record Continuation and Truncation

The Super Extract process uses a maximum extract block size determined by the value in the TAPE BLOCK SIZE field on ECF screen 1. Depending on the file being extracted, the size of a single record from the file (or single record plus record header) may exceed the maximum extract block size. When this occurs, the Super Extract process can truncate the data from the oversized record that does not fit in the extract block, or it can create the additional extract records necessary to extract the oversized record in its entirety. The setting of the EXTR-DISPOSITION param in the Logical Network Configuration File (LCONF) determines how the Super Extract process processes the oversized record.

The Super Extract process includes a record header with each record being extracted to identify its length and type. The record type is identified by a two-character code that is also known as the prefix. Normally, the Super Extract process writes a data record with a DR prefix. However, when a single record exceeds the maximum block size, the EXTR-DISPOSITION param setting determines which record type the Super Extract process writes. The Super Extract process can use the following prefixes and record types when extracting an oversized record:

CF = Continuation record—first
CL = Continuation record—last
CR = Continuation record—intermediate
DR = Data record
TR = Truncation record

The following is a description of the available EXTR-DISPOSITION param values and the resulting prefixes that the Super Extract process uses in each case:

Value	Description
C	The Super Extract process writes the number of extract records necessary to extract the oversized record in its entirety. The first extract record is identified with a CF prefix. The last extract record is identified with a CL prefix. If the disk record requires more than two extract records, one or more intermediate extract records are identified with a CR prefix.

Value	Description
N	The Super Extract process writes a single extract record identified with the usual DR prefix, but truncates the data from the oversized record that does not fit in the extract block. The host must compare the block length and record length to identify an oversized extract record. This is the default value for the param.
T	The Super Extract process writes a single extract record and truncates the excess data from the oversized record as it does with the N option; however, the Super Extract process changes the prefix from DR to TR. The TR prefix uniquely identifies an oversized extract record, eliminating the need for the host to compare the block and record lengths.

The length specified in each extract record header is always the length of the entire disk record plus the length of the header itself. The length specified in each extract block header is always the actual length of the block when it is written, including all headers. Normally, the length contained in the block header is greater than the length contained in any of the record headers in that block. When an oversized record is extracted, however, the length in the record header is greater than the length in the block header because the block does not contain the entire record.

When continuation occurs, each continuation record header (CF, CL, and CR prefix) carries the length of the oversized record even though each continuation record only contains a portion of the data. When truncation occurs, data and truncation record headers (DR and TR prefix) carry the length of the oversized record even though part of the data is lost.

Tape Labels

The tape labels that can be processed depend on whether the Super Extract process or the HP NonStop Kernel TAPECOM product is used to process tape labels. The BASE24-TAPE-LBL-PROC-USED param in the LCONF indicates whether BASE24 or the HP NonStop Kernel TAPECOM product performs tape label processing.

The HP NonStop Kernel TAPECOM product can process IBM or ANSI standard label tapes or unlabeled tapes.

The Super Extract process can process IBM, ANSI, and Burroughs standard label tapes or unlabeled tapes.

The tape label type is set using the TAPE LABEL TYPE field on ECF screen 1.

Tape Header Records

Record: Tape Header (Binary/ASCII and ASCII-only Formats)

Length: 64–82 bytes (variable-length)

Description: The tape header contains information identifying the extract. Since the tape header record contains no binary fields, the binary/ASCII and ASCII-only formats are the same.

Refer to the Block Headers and Record Headers discussions presented earlier in this appendix for additional information that are included on disks or tapes.

Layout: The tape header record generated by BASE24 is described below.

Position	Level	Field Name and Description	Data Type
1	02	CHAR-SET A code indicating the character set used to create the tape. Valid values are as follows: A = ASCII E = EBCDIC	PIC X
2–7	02	EXTRACT-DAT The calendar date, in YYMMDD format, the extract tape was created.	PIC 9(6)
8–15	02	EXTRACT-TIM The time, in hhmmsshh format, the extract tape was created.	PIC 9(8)
16–19	02	LN The identifier assigned to the logical network in which the files being extracted reside.	PIC X(4)

Position	Level	Field Name and Description	Data Type
20–21	02	REL-NUM The BASE24 software release number indicating the release of the software with which the extracted file is compatible. The value in this field is set from the value in the RELEASE NUM field on Extract Configuration File (ECF) screen 1.	PIC 9(2)
22–29	02	FILE-ID Not used. This field contains blanks.	PIC X(8)
30–64	02	NAME The tag identifying the ECF record used for the extract (for 10 bytes).	PIC X(35)
(65–82)	02	USER-FLD1 Not used. Currently, this field is not included in the tape header record. It has been reserved here for future use.	PIC X(18)

File Header Records

Record: File Header (Binary/ASCII and ASCII-only Formats)

Length: 64–82 bytes (variable-length)

Description: The file header identifies the file whose extracted records follow. Since the file header contains no binary fields, the binary/ASCII and ASCII-only formats are the same.

Refer to the Block Headers and Record Headers discussions presented earlier in this appendix for additional information that are included on disks or tapes.

Layout: The file header record generated by BASE24 is described below.

Position	Level	Field Name and Description	Data Type
1	02	CHAR-SET Not used. This field contains a blank.	PIC X
2–7	02	EXTRACT-DAT The date, in YYMMDD format, of the extracted file. If the extracted file is not a dated file, such as the Institution Definition File (IDF), this field contains the date the file header record was created.	PIC 9(6)
8–15	02	EXTRACT-TIM The time, in hhmmsshh format, the file header record was created.	PIC 9(8)
16–19	02	LN The identifier assigned to the logical network in which the file being extracted resides.	PIC X(4)

Position	Level	Field Name and Description	Data Type
20–21	02	REL-NUM The BASE24 software release number indicating the release of the software with which the extracted file is compatible. The value in this field is set from the value in the RELEASE NUM fields on the respective ECF product screens.	PIC 9(2)
22–29	02	FILE-ID A file mnemonic indicating the file that was extracted. The entry in this field is left-justified and blank filled. Valid values are as follows: HMBF = Host Mail Box File HSF = Hardware Status File ICF = Interchange Configuration File ICFE = Enhanced Interchange Configuration File IDF = Institution Definition File ILF = Interchange Log File ITLF = ITS Transaction Log File MBF = Mail Box File OMF = Online Maintenance File PRDF = POS Retailer Definition File PTLF = POS Transaction Log File SAF = Store-and-Forward File SVHF = Stored Value History File TLF = Transaction Log File TTF = Teller Transaction File TTLF = Teller Transaction Log File ULF = Update Log File	PIC X(8)
30–64	02	NAME The fully expanded file name of the extracted file for files other than the Interchange Log File (ILF). This field contains blanks for ILF extracts because records from more than one ILF can be placed between a single pair of extract file header and trailer records. As a result, the name of a single ILF would not be accurate.	PIC X(35)

Position	Level	Field Name and Description	Data Type
(65–82)	02	USER-FLD1	PIC X(18)

Miscellaneous information, depending on the file being extracted. This field is not present in the file header only as noted below.

ILF/TLF/TTLF. The USER-FLD1 field contains the values of the following LCONF params, in order:

File format:

1 = Format 1 files (default)
2 = Format 2 files

Byte 1 = ACCT-NUM-TYP param
Byte 2 = DATE-DISPLAY-TYPE param
Byte 3 = Blank
Byte 4 = FILE-FRMT (Default = 1)

PRDF/PTLF. The USER-FLD1 contains the values of the following LCONF params, in order:

Byte 1 = ACCT-NUM-TYP param
Byte 2 = DATE-DISPLAY-TYPE param
Byte 3 = Set to 2
Byte 4 = FILE-FRMT (Default = 1)

Note: If the above information is present on the tape, it is only four bytes in length (the entire 18 bytes is not used). If the above information is not required for the extract, this field is not included in the file header record unless it is a Format 2 file.

File Trailer Records (Binary/ASCII Format)

Record: File Trailer (Binary/ASCII Format)

Length: 58–388 bytes (variable-length)

Description: The file trailer signals the end of the extract records for a particular file. It contains a verification count and amount that can be used by hosts to check their processing. There are two formats for extract records: binary/ASCII and ASCII-only. The format used is specified by the value in the NUMERIC FLD FORMAT field on Extract Configuration File (ECF) screen 1. The ASCII-only option allows all binary fields on the tape to be converted to ASCII character format. If the value in the NUMERIC FLD FORMAT field is set to B, the Super Extract process uses the binary format. If the value in the NUMERIC FLD FORMAT field is set to A, the Super Extract process uses the ASCII character format.

Layout: The file trailer record (binary/ASCII format) generated by BASE24 is described below.

Position	Level	Field Name and Description	Data Type
1–8	02	FILE-ID The file mnemonic of the file being extracted. The entry in this field is left-justified and blank filled. Valid values for this field are the same as those for the FILE-ID field in the file header record.	PIC X(8)
9–43	02	NAME The fully expanded file name of the extracted file.	PIC X(35)
44	02	USER-FLD1 Not used. This field contains a blank.	PIC X

Position	Level	Field Name and Description	Data Type
45–46	02	STAT	BINARY 16

A code indicating processing errors encountered by the Super Extract process. A zero indicates that no error has occurred; any other value indicates that an error has occurred. Refer to the Extract Report in the spooler for additional information regarding the error.

Position	Level	Field Name and Description	Data Type
47–54	02	AMT	BINARY 64

A total kept by the Super Extract process during the processing of the records. The value in this field is normally set to 0. The table below identifies the files for which this field contains a total and indicates the values that are included in the total:

File	Included Value
ILF	Sum of the values in ATM.RQST.AMT-1 fields for BASE24-atm records (BASE24-atm records are identified by a value of 1 in the REC-TYP field). Sum of the values in POS.TRAN.AMT-1 fields for BASE24-pos records (BASE24-pos records are identified by a value of 2 in the REC-TYP field).
ITLF	Sum of the values in AMT_TXN.TXN.AMT fields for all records in which a transaction amount is logged.
PTLF	Sum of the values in AUTH.AMT-1 fields for all records with a value of 0210, 0220, 0412, or 0420 in the AUTH.TYP field. Note that if the HEAD.REC-TYP field contains the value 23, indicating the PTLF record contains invalid data, the AUTH.AMT-1 field for the record is not included in the total.
TLF	Sum of the values in AUTH.AMT-1 fields for all records with values of 01, 20, 21, or 22 in the HEAD.REC-TYP field.

Position	Level	Field Name and Description	Data Type

File	Included Value
TTLF	Sum of the values in AMT-1 fields of the Financial Token for all records with a value of 0210, 0420, or 0430 in the SYS.MSG-TYP field.

55–58 02 CNT BINARY 32

The number of file records extracted. This number does not include the file header and trailer records.

(59–388) 02 USER-FLD2 PIC X(330)

For extracts of the ITLF, TLF, PTLF, and TTLF, the first 20 bytes of this field contain the date and time, in YYYYMMDDhhmmssmmmmmm format, of the last record extracted from the file.

File Trailer Records (ASCII-Only Format)

Record: File Trailer (ASCII-only Format)

Length: 78–408 bytes (variable-length)

Description: The file trailer signals the end of the extract records for a particular file. It contains a verification count and amount that can be used by hosts to check their processing. There are two formats for extract records: binary/ASCII and ASCII-only. The format used is specified in the NUMERIC FLD FORMAT field on Extract Configuration File (ECF) screen 1. The ASCII-only option allows all binary fields on the tape to be converted to ASCII character format. If the value in the NUMERIC FLD FORMAT field is set to B, the Super Extract process uses the binary format. If the value in the NUMERIC FLD FORMAT field is set to A, the Super Extract process uses the ASCII character format.

Layout: The file trailer record (ASCII-only format) generated by BASE24 is described below.

Position	Level	Field Name and Description	Data Type
1–8	02	FILE-ID The file mnemonic of the file being extracted. The entry in this field is left-justified and blank filled. Valid values for this field are the same as those for the FILE-ID field in the file header record.	PIC X(8)
9–43	02	NAME The fully expanded file name of the extracted file.	PIC X(35)
44	02	USER-FLD1 Not used. This field contains a blank.	PIC X

Position	Level	Field Name and Description	Data Type
45–49	02	STAT	PIC 9(5)

A code indicating processing errors encountered by the Super Extract process. A zero indicates that no error has occurred; any other value indicates that an error has occurred. Refer to the Extract Report in the spooler for additional information regarding the error.

Position	Level	Field Name and Description	Data Type
50–68	02	AMT	PIC X(19)

A total kept by the Super Extract process during the processing of the records. The value in this field is normally set to 0. The table below identifies the files for which this field contains a total and indicates the values that are included in the total:

File	Included Values
ILF	Sum of the values in ATM.RQST.AMT-1 fields for BASE24-atm records (BASE24-atm records are identified by a value of 1 in the REC-TYP field). Sum of the values in POS.TRAN.AMT-1 fields for BASE24-pos records (BASE24-pos records are identified by a value of 2 in the REC-TYP field).
ITLF	Sum of the values in AMT_TXN.TXN.AMT fields for all records in which a transaction amount is logged.
PTLF	Sum of the values in AUTH.AMT-1 fields for all records with a value of 0210, 0220, 0412, or 0420 in the AUTH.TYP field. Note that if the HEAD. REC-TYP field contains the value 23, indicating the PTLF record contains invalid data, the AUTH.AMT-1 field for the record is not included in the total.
TLF	Sum of the values in AUTH.AMT-1 fields for all records with values of 01, 20, 21, or 22 in the HEAD.REC-TYP field.

Position	Level	Field Name and Description	Data Type

File	Included Values
TTLF	Sum of the values in AMT-1 fields of the Financial Token for all records with a value of 0210, 0420, or 0430 in the SYS.MSG-TYP field.

Position	Level	Field Name and Description	Data Type
69–78	02	CNT The number of file records extracted. This number does not include the file header and trailer records.	PIC 9(10)
(79–408)	02	USER-FLD2 For extracts of the ITLF, TLF, PTLF, and TTLF, the first 20 bytes of this field contain the date and time, in YYYYMMDDhhmmssmmmmmm format, of the last record extracted from the file.	PIC X(330)

Tape Trailer Records (Binary/ASCII Format)

Record: Tape Trailer (Binary/ASCII Format)

Length: 58–388 bytes (variable-length)

Description: The tape trailer signals the end of the extract records. It contains a record count that can be used by hosts to check their processing. There are two formats for extract records: binary/ASCII and ASCII-only. The format used is specified in the NUMERIC FLD FORMAT field on Extract Configuration File (ECF) screen 1. The ASCII-only option allows all binary fields on the tape to be converted to ASCII character format. If the value in the NUMERIC FLD FORMAT field is set to B, the Super Extract process uses the binary format. If the value in the NUMERIC FLD FORMAT field is set to A, the Super Extract process uses the ASCII character format.

Layout: The tape trailer record (binary/ASCII format) generated by BASE24 is described below.

Position	Level	Field Name and Description	Data Type
1–8	02	FILE-ID Not used. This field contains blanks.	PIC X(8)
9–43	02	NAME The tag identifying the ECF record used for the extract (for 10 bytes).	PIC X(35)
44	02	USER-FLD1 Not used. This field contains a blank.	PIC X
45–46	02	STAT Not used. This field contains zeros.	BINARY 16

Position	Level	Field Name and Description	Data Type
47–54	02	AMT Not used. This field contains zeros.	BINARY 64
55–58	02	CNT The count of all records on the tape, not including header and trailer records or tape label records.	BINARY 32
(59–388)	02	USER-FLD2 Not used. This field is currently not present in the tape trailer. It has been reserved here for future use.	PIC X(330)

Tape Trailer Records (ASCII-Only Format)

Record: Tape Trailer (ASCII-only Format)

Length: 78–408 bytes (variable-length)

Description: The tape trailer signals the end of the extract records. It contains a record count that can be used by hosts to check their processing. There are two formats for extract records: binary/ASCII and ASCII-only. The format used is specified in the NUMERIC FLD FORMAT field on Extract Configuration File (ECF) screen 1. The ASCII-only option allows all binary fields on the tape to be converted to ASCII character format. If the value in the NUMERIC FLD FORMAT field is set to B, the Super Extract process uses the binary format. If the value in the NUMERIC FLD FORMAT field is set to A, the Super Extract process uses the ASCII character format.

Layout: The tape trailer record (ASCII-only format) generated by BASE24 is described below.

Position	Level	Field Name and Description	Data Type
1–8	02	FILE-ID Not used. This field contains blanks.	PIC X(8)
9–43	02	NAME The tag identifying the ECF record used for the extract (for 10 bytes).	PIC X(35)
44	02	USER-FLD1 Not used. This field contains a blank.	PIC X
45–49	02	STAT Not used. This field contains zeros.	PIC 9(5)

Position	Level	Field Name and Description	Data Type
50–68	02	AMT Not used. This field contains zeros.	PIC X(19)
69–78	02	CNT The count of all records on the tape, not including header and trailer records or tape label records.	PIC 9(10)
(78–408)	02	USER-FLD2 Not used. This field is currently not present in the tape trailer. It has been reserved here for future use.	PIC X(330)

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C: Initialization and Processing

This appendix describes Refresh and Super Extract process initialization. Each Refresh process (i.e., Base National Negative, INAS National Negative, Enscribe, and SQL) has its own initialization procedure.

Base National Negative Refresh Initialization

When the Base National Negative Refresh process receives a REFRESH command, it performs the following steps:

1. Reads the name of the Logical Network Configuration File (LCONF) from the startup message into memory.

   The name of the LCONF is contained in the header attached to the startup message. This header contains standard information that is passed by the XPNET process whenever the XPNET process routes a message to a BASE24 process.

2. Initializes its global variables.
3. Retrieves the time from the HP NonStop processor.
4. Opens the LCONF and reads the following assigns and params:

Assigns	Params
BP-BNEG	BASE24-TAPE-LBL-PROC-USED
BP-BNR	LOGICAL-NET
BP-RPTR	POS-PREF-NOTIFY
LMCF	

   If these assigns and params cannot be read, the Refresh process abends.

5. Opens the network logging facility.
6. Reads the Log Message Configuration File (LMCF) for modified event messages.

   The LMCF contains a record for each modified event message in the system. The Base National Negative Refresh process opens the LMCF, retrieves its own modified event messages, and closes the file. When the Base National Negative Refresh process generates an event message, it checks these LMCF records in memory to determine whether the message needs to be modified.

7. Validates the REFRESH command. If the message format is incorrect, or contains invalid information, the Base National Negative Refresh process logs error messages and abends.
8. Opens the output location specified in the BP-RPTR assign. If the Base National Negative Refresh process cannot open the output location, the process logs a message and abends.

9. Opens the refresh input file. If the Base National Negative Refresh process cannot open the refresh input file, the process logs a message and abends.
10. Reads a record from the refresh input file and verifies that it is a refresh file header (RFH).

    If a parity error is encountered or a tape label is not the type specified in the REFRESH command, the Base National Negative Refresh process logs a message and abends. If the REFRESH command indicated the tape label is NON, the process assumes no tape labels are on the tape.
11. Begins a full-file refresh.

INAS National Negative Refresh Initialization

When the INAS National Negative Refresh process receives a REFRESH command, it performs the following steps:

1. Reads the name of the Logical Network Configuration File (LCONF) from the startup message into memory.

   The name of the LCONF is contained in the header attached to the startup message. This header contains standard information that is passed by the XPNET process whenever the XPNET process routes a message to a BASE24 process.

2. Initializes its global variables.
3. Retrieves the time from the HP NonStop processor.
4. Opens the LCONF and reads the following assigns and params:

Assigns	Params
BP-INEG	BASE24-TAPE-LBL-PROC-USED
BP-INR	LOGICAL-NET
BP-RPTR	POS-PREF-NOTIFY
LMCF	

   If these assigns and params cannot be read, the INAS National Negative Refresh process abends.

5. Opens the network logging facility.
6. Reads the Log Message Configuration File (LMCF) for modified event messages.

   The LMCF contains a record for each modified event message in the system. The INAS National Negative Refresh process opens the LMCF, retrieves its own modified event messages, and closes the file. When the INAS National Negative Refresh process generates an event message, it checks these LMCF records in memory to determine whether the message needs to be modified.

7. Validates the REFRESH command. If the message format is incorrect, or contains invalid information, the INAS National Negative Refresh process logs error messages and abends.
8. Opens the output location specified in the BP-RPTR assign. If the INAS National Negative Refresh process cannot open the output location, the process logs a message and abends.

9. Opens the refresh input file. If the INAS National Negative Refresh process cannot open the refresh input file, the process logs a message and abends.

10. Reads a record from the refresh input file and verifies that it is a refresh file header (RFH).

    If a parity error is encountered or a tape label is not the type specified in the REFRESH command, the INAS National Negative Refresh process logs a message and abends. If the REFRESH command indicated the tape label is NON, the process assumes no tape labels are on the tape.

11. Determines the type of refresh.

    If the IDENTIFICATION field in the refresh file header contains the entry MASTERCARD INTERNATIONAL RESTRICTED ACCOUNT UPDATES, the INAS National Negative Refresh process begins a partial-file refresh. If the IDENTIFICATION field in the refresh file header contains the entry MASTERCARD INTERNATIONAL ELECTRONIC WARNING BULLETIN, the process begins a full-file refresh.

Enscribe Refresh Initialization

When the Enscribe Refresh process receives a REFRESH command, it performs the following steps:

1. Reads the name of the Logical Network Configuration File (LCONF) from the startup message into memory.

   The name of the LCONF is contained in the header attached to the startup message. This header contains standard information that is passed by the XPNET process whenever the XPNET process routes a message to a BASE24 process.

2. Initializes its global variables.
3. Retrieves the time from the HP NonStop processor.
4. Opens the LCONF and reads the following assigns and params:

Assigns	Params
IDF	REF-BYPASS-PARITY-ERRORS
LMCF	REF-RGCF-RETN-PRD
RGCF	REF-MONITOR-COMMAND
RPTR	BASE24-TAPE-LBL-PROC-USED

   If these assigns and params cannot be read, the Enscribe Refresh process abends. Additional assigns and params are read later in the initialization procedures.

5. Opens the network logging facility.
6. Reads the Log Message Configuration File (LMCF) for modified event messages.

   The LMCF contains a record for each modified event message in the system. The Enscribe Refresh process opens the LMCF, retrieves its own modified event messages, and closes the file. When the Enscribe Refresh process generates an event message, it checks these LMCF records in memory to determine whether the message needs to be modified.

7. Validates the REFRESH command. If the message format is incorrect, or contains invalid information, the Enscribe Refresh process logs error messages and abends.
8. Opens the Refresh Group Control File (RGCF). If the Enscribe Refresh process cannot open the RGCF, the process logs a message and abends.

If the refresh is resuming, the Enscribe Refresh process reads the RGCF using information contained in the startup message. If the refresh is new, the process simply opens the RGCF at this time.

9. Opens the output location specified in the RPTR assign. If the Enscribe Refresh process cannot open the output location, the process logs a message and abends.

10. Opens the refresh input file. If the Enscribe Refresh process cannot open the refresh input file, the process logs a message and abends.

11. For a refresh that is beginning, the Enscribe Refresh process reads a record from the refresh input file and verifies that it is a refresh file header (RFH).

    If a parity error is encountered or a tape label is not the type specified in the REFRESH command, the Enscribe Refresh process logs a message and abends. If the REFRESH command indicated the tape label is NON, the process assumes no tape labels are on the tape.

12. Reads the LCONF for the remaining params required. The params that are read depend on the type of refresh and on the application file being refreshed.

13. If the current file being refreshed is not a BASE24-mail file, the Enscribe Refresh process identifies the institutions and products involved in the refresh.

    To do this, the Enscribe Refresh process compares the REFR-GRP field in the Base segment of each institution's Institution Definition File (IDF) record to the GRP field in the refresh file header as follows. Each IDF record that has a REFR-GRP field equal to the GRP field in the refresh file header is used in these steps.

    a. Identifies the products supported by each institution.

    b. Verifies that the following fields in the IDF for each institution involved in the refresh contain the same information.

       CAF-NAME
       NEG-NAME
       PBF1-NAME
       PBF2-NAME
       PBF3-NAME
       SPF-NAME
       POS.CHF-NAME
       TLR.NBF-NAME
       TLR.WHFF-NAME

       If the institutions' IDF records do not contain the same information in these fields, the Enscribe Refresh process logs an error message and abends.

Note: The Card History File (CHF), No Book File (NBF), Stop Payment File (SPF), and Warning/Hold/Float File (WHFF) do not have to be supported by all institutions in a refresh group. For these files, the Enscribe Refresh process allows either blanks (no support) or the same file name value used by other institutions in the refresh group.

14. Retrieves the corresponding REFR-IND field in the IDF. The REFR-IND field is used during impacting. The REFR-IND field that is accessed depends on the type of file being refreshed. Valid values are as follows:

REFR-IND.CAF1	=	Cardholder Account File (CAF)
REFR-IND.PBF1	=	Positive Balance File (PBF) for checking accounts or combined accounts
REFR-IND.PBF2	=	PBF for savings accounts
REFR-IND.PBF3	=	PBF for credit accounts
SPF-REFR-IND	=	Stop Payment File (SPF)

15. Reads the remaining assigns needed for processing, validates the information contained in each assign, and closes the LCONF. If an assign is not present or is not valid, the Enscribe Refresh process uses a default value.

16. For a refresh that is beginning, the Enscribe Refresh process validates the remaining fields in the refresh file header in the refresh input file. The refresh file header fields validated at this time are listed and described below:

Field	Description
LN	The logical network. This field must match the value of the LOGICAL-NET param in the LCONF.
product-IMP-STRT-DAT and *product*-IMP-STRT-TIM	The impact date and time. These fields identify the transaction log file record to start with when impacting files during refresh. If these fields are blank or contains all nines, impacting starts with the first record in the log file. An impact date and time specified in the REFRESH command overrides these fields. These fields are verified only when the Refresh process determines that impacting will take place.

Field	Description
PARTITION-NUM	The partition number, used for single-partition refreshes. This field specifies if a particular partition of a file is being refreshed. If this field is blank, the Enscribe Refresh process assumes an entire file is being refreshed. A partition number in the REFRESH command overrides this field. **Note:** A partition number cannot be specified for partial-file refreshes. The field in the refresh file header must be left blank

If any of the above fields contain incorrect or invalid information, the Enscribe Refresh process logs an error message and abends.

17. For a refresh that is beginning, the Enscribe Refresh process reads the next record and verifies that it is an organizational header.
18. For a refresh that is resuming, the Enscribe Refresh process positions to the correct record in the file, based on the information in the RGCF record, and resumes refresh processing. The RGCF indicates at what point the refresh was suspended.
19. Writes the headers of the refresh report to the spooler location.
20. If the refresh is for the Transaction History File (THF), the Enscribe Refresh process performs the following steps:
    a. Opens the File Control File (FCF).
    b. Reads the file indicator contained in the FCF for each THF.
    c. Creates the file name of the THF by appending the file indicator to the end of the file name defined in the LCONF.
21. Logs an event message indicating the refresh has begun or is resuming, depending on the circumstance.
22. If impacting is to occur, the Enscribe Refresh process opens the first log files to be used for impacting.
23. Determines the type of refresh.

    If the REC-TYP field in the refresh file header equals 0, the Enscribe Refresh process begins a full-file refresh. If the REC-TYP field in the refresh file header equals 1, the process begins a partial-file refresh.

SQL Refresh Initialization

When the SQL Refresh process receives a REFRESH command, it performs the following steps:

1. Reads the name of the Logical Network Configuration File (LCONF) from the startup message into memory.

   The name of the LCONF is contained in the header attached to the startup message. This header contains standard information that is passed by the XPNET process whenever the XPNET process routes a message to a BASE24 process.

2. Initializes its global variables.

3. Retrieves the time from the HP NonStop processor.

4. Initializes the HP NonStop Kernel Event Management Service (EMS), which includes opening the LCONF and reading the following assigns and params:

Assigns	Params
ALT-COLL	ALT-TIM-LMT
PRI-COLL	CONSOLE-PRT
REF-RPT-SQL	PRI-TIM-LMT
	REF-INPUT-BLK-DISK
	REF-INPUT-BLK-TAPE

   Of these assigns and params, the PRI-COLL and REF-RPT-SQL assigns are required. If these assigns cannot be read, the SQL Refresh process abends. Otherwise, initialization continues and the SQL Refresh process uses default values.

5. Reads the following required LCONF assign and param:

Assigns	Params
RGCF	LOGICAL-NET

   If this assign and param cannot be read, the SQL Refresh process abends.

6. Reads the following optional LCONF assign and params:

Assigns	Params
REF-SOURCEDICT	BASE24-TAPE-LBL-PROC-USED REF-BYPASS-PARITY-ERRORS REF-MONITOR-COMMAND REF-RGCF-RETN-PRD REF-SQLCI-CPU REF-SQLCI-DELAY

   If these assigns and params cannot be read or contain invalid values, initialization continues and the SQL Refresh process uses default values.

7. Validates the REFRESH command. If the message format is incorrect, or contains invalid information, the SQL Refresh process logs error messages and abends.

8. Opens the Refresh Group Control File (RGCF). If the SQL Refresh process cannot open the RGCF, the process logs a message and abends.

9. Opens the output location specified in the REF-RPT-SQL assign. If the SQL Refresh process cannot open the output location, the process logs a message and abends.

10. Opens the refresh input file. If the SQL Refresh process cannot open the refresh input file, the process logs a message and abends.

11. Performs the following steps:

    a. Reads a record from the refresh input file and verifies that it is a refresh file header (RFH).

       If a parity error is encountered or a tape label is not the type specified in the REFRESH command, the SQL Refresh process logs a message and abends. If the REFRESH command indicated the tape label is NON, the process assumes no tape labels are on the tape.

    b. Reads the LCONF for the remaining assigns and params required. The params that are read depend on the type of refresh and on the application file being refreshed.

c. Validates the remaining fields in the refresh file header in the refresh input file. The refresh file header fields validated at this time are listed and described below:

Field	Description
Logical network	The logical network. This field must match the value of the LOGICAL-NET param in the LCONF.
Refresh type	The type of refresh (i.e., full-or partial-file) being performed.
Partition number	The partition number, used for single-partition refreshes. This field specifies if a particular partition of a file is being refreshed. If this field is blank, the Refresh process assumes an entire file is being refreshed. A partition number in the REFRESH command overrides this field. **Note:** A partition number cannot be specified for partial-file refreshes. The field in the refresh file header must be left blank

If any of the above fields contain incorrect or invalid information, the SQL Refresh process logs an error message and abends.

d. Reads the next record and verifies that it is an organizational header.

12. Writes the headers of the refresh report to the spooler location.
13. Raises an event indicating the type of refresh being performed.
14. Verifies the SQL table is empty if a full-file refresh is being performed.
15. Starts the SQL conversational interface (SQLCI) process if a full-file refresh is being performed.
16. Performs the refresh.

Super Extract Initialization

When the Super Extract process is first started or initialized, it performs the following steps:

1. Reads the name of the LCONF and its own process name into memory.

   The startup message from the XPNET process contains the name of the LCONF and the Super Extract process name, which the Super Extract process reads into memory. The LCONF name allows the Super Extract process to access the LCONF for the assigns and params it uses in processing. The process name allows the Super Extract process to read the correct records from the ECF.

2. Opens the LCONF and reads the required assigns and params.
3. Initializes the network logging facility in order to write event messages.
4. Reads the LMCF for its modified event messages.

   The LMCF contains a record for each modified event message in the system.

5. Opens the Token File (TKN) and reads the extract configuration records into its extended memory.
6. Opens the ECF with read/write access. Any other process can read the ECF while the Super Extract process has the ECF open.
7. Reads the ECF for records that have its name in the ALTKEY.SYM-NAME field in the Base segment. For each record found, the Super Extract process checks the EXTRACT-DAT and the RESTRT-FLG fields.

   - If the RESTRT-FLG is set to a value of Y and the EXTRACT-DAT is less than the current day, the Super Extract process logs a warning message that shows the EXTRACT-DAT and TAG fields in the ECF and then changes the EXTRACT-DAT to the current date. The Super Extract process then starts a timer that expires at the time specified by the EXTRACT-DAT and EXTRACT-TIM fields.
   - If the RESTRT-FLG is set to a value of Y and the EXTRACT-DAT is equal to the current date or a future date, the Super Extract process starts a timer that will expire at the time specified by the EXTRACT-DAT and EXTRACT-TIM fields. When this timer expires, an extract is performed.

The RESTRT-FLG should be set to a value of N when extracts are to be completed manually on a general basis. Although a manual extract can be performed when the RESTRT-FLG is set to a value of Y, the Super Extract process sets a timer and performs an extract when the timer expires. Thus, two extracts would be performed on the same file(s).

8. If records were found, the Super Extract process issues a netread that times out when the first timer expires.

   If no records were found, the Super Extract process logs a message indicating this and abends.

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